



Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

e-Crafter 2019 ➤

e-TGE 2019 ➤

Brake systems

Edition 01.2019



List of Workshop Manual Repair Groups

Repair Group

- 00 - Technical data
- 45 - Anti-lock brake system
- 46 - Brakes - mechanism
- 47 - Brakes - hydraulics

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



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00 – Technical data

1 Safety information

(VRL012466; Edition 01.2019)

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⇒ ["1.2 Safety precautions when using testers and measuring instruments during a road test", page 1](#)

⇒ ["1.3 Safety precautions when working on high-voltage system", page 1](#)

⇒ ["1.4 Safety precautions when working in vicinity of high-voltage components", page 2](#)

⇒ ["1.5 Safety precautions when tow-starting or towing", page 2](#)

1.1 Safety measures when working on vehicles with a start/stop system

Risk of injury due to unexpected motor start

If the vehicle's start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the start/stop system is activated.

- Deactivate start/stop system by switching off the ignition.

1.2 Safety precautions when using testers and measuring instruments during a road test

Risk of injury caused by unsecured testing and measuring instruments

When the front passenger airbag is triggered in an accident, insufficiently secured testing and measuring instruments become dangerous projectiles.

- Secure testing and measuring instruments on the rear seat.
- or
- Have a second person operate the test and measuring equipment on the rear seat.

1.3 Safety precautions when working on high-voltage system

Danger to life from high voltage

The high-voltage system is under high voltage. Severe or fatal injury from electric shock.

- Persons with life-preserving or other electronic medical devices in or on their body must not perform any work on the high-voltage system. Such medical devices include internal analgesic pumps, implanted defibrillators, pacemakers, insulin pumps and hearing aids.
- The high-voltage system must be de-energised by a suitably qualified technician.



Risk of injury due to unexpected engine/motor start

On electric and hybrid vehicles, it can easily be overlooked that the vehicle is in "ready" mode. There is a risk of parts of the body becoming trapped or drawn in.

- Switch off ignition.
- Always store the ignition key outside the vehicle.

Risk of damage to high-voltage cables

Improper handling of high-voltage cables or high-voltage connectors can result in damage to their insulation.

- Never support body weight on high-voltage cables or high-voltage connectors.
- Never support any tools on high-voltage cables or high-voltage connectors.
- Never kink or severely bend high-voltage cables.
- Always observe the coding when connecting high-voltage connectors.

1.4 Safety precautions when working in vicinity of high-voltage components

Danger to life from high voltage

The high-voltage system is under high voltage. Damage to high-voltage components can result in severe or fatal injury from electric shock.

- Perform visual check of high-voltage components and high-voltage cables.
- Never use cutting or forming tools, or any other sharp-edged tools.
- Never use heat sources such as welding, brazing, soldering, hot air or thermal bonding equipment.

1.5 Safety precautions when tow-starting or towing

Risk of irreparable damage to electric drive motor

Risk of destroying electric drive motor when towing incorrectly.

- Position selector lever in "N" position when vehicle is being towed.
- 50 km/h must not be exceeded when towing.



2 Identification

⇒ [“2.1 Allocation of PR number - brakes”, page 3](#)

2.1 Allocation of PR number - brakes

⇒ [“2.1.1 Front brakes”, page 3](#)

⇒ [“2.1.2 Rear brakes”, page 3](#)

2.1.1 Front brakes

The type of brake system installed in the vehicle is indicated among other things by the corresponding PR number on the vehicle data sticker.

- Relevant information about the brake system installed in the vehicle are documented in the vehicle-specific notes in ELSA.

In this example, the front brake “1LJ” is installed in the vehicle.

- ◆ Allocation ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE

- ◆ The following table explains the PR numbers. These are important for the brake caliper/brake disc and brake pads.

Front brakes are differentiated according to PR numbers.

Fahrzeuginformationen			
Marke:	V	Fahrzeugidentifizierung:	WV0222542EW000129
Modelljahr:	2008	Bezeichnung:	Tiguan Trendline 2.0 TDI 140kW
Verkaufsdatum:	2007-04-13	Produktionsdatum:	2007-05-04
MSR:	CRAB	Auslieferungsdatum:	-
MSR:	200	Leasing:	-
Farbausstattung			
Typ	Farbcode	Lacknummer (Lack-ED-Nr.)	Beschreibung
außen (Chassis/Gelb)	BE/BE	LA7W	Reflexsilber Metallic/Reflexsilber Metallic
innen	YY		Contrast: ... (schwarz/Schwarz)
PR-Nummern			
PR-Nr.	Hersteller	PR-Nr.	PR-Nr.
8	015	L	HLZ
9	016	L	GMH
10	147	L	DRS
11	108	L	ANV
12	109	L	SZU
13	110	L	ANB
14	111	L	TV
15	112	L	SAU
16	113	L	SAV
17	114	L	SAV
18	115	L	SAV
19	116	L	SAV
20	117	L	SAV
21	118	L	SAV
22	119	L	SAV
23	120	L	SAV
24	121	L	SAV
25	122	L	SAV
26	123	L	SAV

Running gear size in inches	PR No.
16	1LA
16	1LB

2.1.2 Rear brakes

The type of brake system installed in the vehicle is indicated among other things by the corresponding PR number on the vehicle data sticker.

- The relevant information relating to which braking system is installed can also be called up via ELSA/vehicle-specific notes.

Allocation ⇒ Electronic parts catalogue (ETKA) or ⇒ webMAN-TIS for MAN TGE.

16" rear brakes (hydraulic disc brakes with mechanically actuated parking brake)



3 Repair instructions

⇒ [“3.1 General information”, page 4](#)

⇒ [“3.2 Rules for cleanliness”, page 4](#)

⇒ [“3.3 Rules for cleanliness when working on high-voltage system”, page 5](#)

⇒ [“3.4 General repair instructions”, page 5](#)

⇒ [“3.5 Checking vacuum system”, page 5](#)

⇒ [“3.6 Gaskets and seals”, page 6](#)

⇒ [“3.7 Nuts and bolts”, page 6](#)

⇒ [“3.8 Routing and attachment of lines”, page 6](#)

⇒ [“3.9 Contact corrosion”, page 6](#)

3.1 General information

- ◆ The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory.
- ◆ For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- ◆ Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and damage the lambda probe.
- ◆ Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- ◆ The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds to develop pressure in the fuel system.

3.2 Rules for cleanliness

- ◆ Absolute cleanliness is required when working on the anti-lock brake system. It is not permitted to use any products that contain mineral oil, such as oil, grease etc.
- ◆ Thoroughly clean all unions and adjacent areas before loosening. Do not use aggressive cleaning agents such as brake cleaner, petrol, thinners or similar.
- ◆ Place removed parts on a clean surface and cover them over.
- ◆ If the repair is not performed immediately, carefully cover or seal opened components.
- ◆ Only use lint-free cloths.
- ◆ Remove packing from replacement parts immediately prior to installation and not before.
- ◆ Only use genuine packed parts.
- ◆ When the system is open, do not work with compressed air or move the vehicle.
- ◆ Ensure that no brake fluid enters connectors.
- ◆ Make sure no brake fluid gets onto paintwork.
- ◆ Immediately remove any escaping brake fluid with water.



3.3 Rules for cleanliness when working on high-voltage system

When working on the high-voltage system, pay careful attention to the following rules for cleanliness:

- ◆ Thoroughly clean all connections/inspection holes and corresponding surrounding areas before disconnecting/opening them.
- ◆ Place removed parts on a clean surface and cover them over. Use lint-free cloths only.
- ◆ Carefully cover opened components or seal them if repairs cannot be carried out immediately.
- ◆ Install only clean components.
- ◆ Remove packing from replacement parts immediately prior to installation and not before.
- ◆ Do not use parts which have been stored unpackaged (e.g. in tool boxes, etc.).
- ◆ Transportation and protective packaging and sealing caps must be removed only immediately before fitting.
- ◆ If system is open, do not work with compressed air. Do not move the vehicle.

3.4 General repair instructions

- ◆ When performing repair work on brakes, hairline cracks are often found on the friction surface of the brake discs. Hairline cracks up to a length of 10 mm do not represent a technical problem and do not justify a renewal of brake discs.
- ◆ Brake discs and pads/linings with cracks going all the way across the friction surface must be renewed.

3.5 Checking vacuum system

The following test instructions are intended to help find the causes of problems effectively and objectively in the event of complaints about the brake servo or in the event of a so-called »hard brake pedal«.

This check relates to the following components:

- ◆ Brake servo
- ◆ Oil seal between brake master cylinder and brake servo.
- ◆ Non-return valve
- ◆ Vacuum hoses with connectors
- ◆ Vacuum pump (if included)

Measuring results will be influenced by the geographical location. The higher the location is above sea level, the lower the air pressure will be.

The average air pressure of the earth's atmosphere at sea level is 1013 mbar and decreases steeply as the altitude increases (approx. 100 mbar/1000 m altitude). Local and time fluctuations also influence vacuum generation.

A cold engine, a switched-on air-conditioning system as well as engine idling have an adverse influence on generation of a vacuum.

Take note of the following test requirements:



- ◆ Inspect all vacuum hoses for damage (e.g. cracks or marten bite) and to ensure that they are correctly and firmly attached
- ◆ Ensure cleanliness when working on vacuum system
- ◆ Before starting work, clean engine compartment if necessary

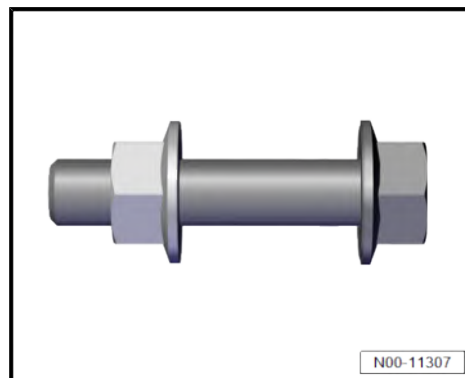
3.6 Gaskets and seals

- ◆ Always renew seals and gaskets.
- ◆ When seals have been removed, check contact surface on housings or shafts for burrs and damage and rectify as necessary.
- ◆ Completely remove all residue from liquid sealants from the sealing surfaces, making sure that no residual sealant gets into the steering rack housing.
- ◆ Open-side of seals point towards oil.



3.7 Nuts and bolts

- ◆ Avoid canting sensitive parts, such as servo motor with control unit. Loosen and tighten them diagonally in stages.
- ◆ Specified torques given are for unlubricated nuts, bolts and screws.
- ◆ Always renew self-locking bolts and nuts.
- ◆ Always replace nuts and bolts with turning further angle.



3.8 Routing and attachment of lines

Risk of damage to lines

Lines may become damaged by moving or hot components.

- Route lines in their original positions.
- Ensure there is sufficient clearance to moving or hot components.

3.9 Contact corrosion

Contact corrosion can occur if non-approved fasteners are used on the vehicle (bolts, nuts, washers etc.).

For this reason, only connecting elements with a special surface coating have been fitted.

In addition, rubber, plastic and adhesives are made of non-conductive materials.

If there are doubts about the suitability of fastening components, always use new parts ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.

Observe the following:

- ◆ It is recommended to use only genuine parts, since only genuine parts are approved.



- ◆ Damage resulting from contact corrosion is not covered by the warranty.



4 Technical data

⇒ [“4.1 Technical data for brakes”, page 8](#)

4.1 Technical data for brakes

⇒ [“4.1.1 Brake master cylinder and brake servo”, page 8](#)

⇒ [“4.1.2 Front brakes”, page 8](#)

⇒ [“4.1.3 Rear brakes”, page 8](#)

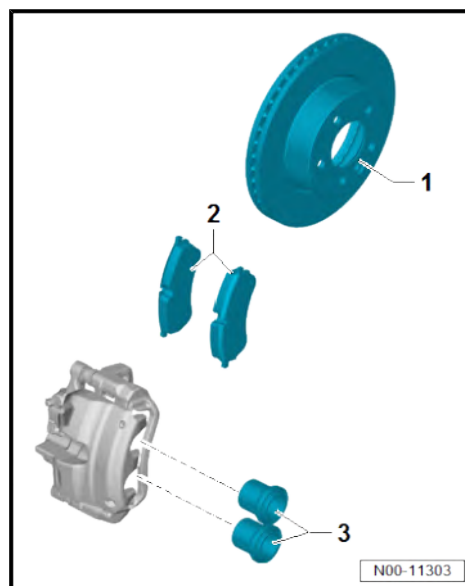
4.1.1 Brake master cylinder and brake servo

Description	Dimensions
Brake master cylinder, diameter	26.99 mm (tandem)
Brake servo, diameter	10"/10"

4.1.2 Front brakes

Front brake 16"

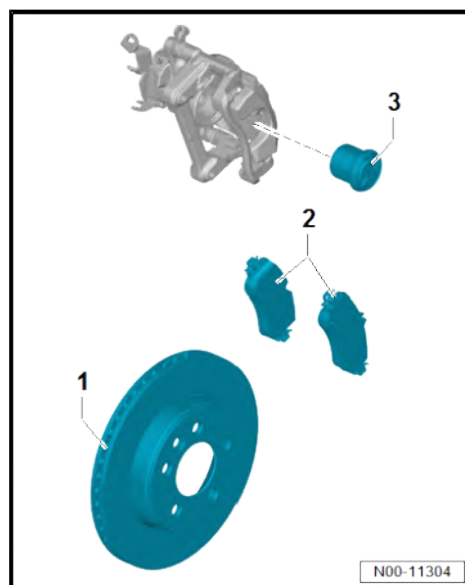
PR number		1LA	1LB
Item	Description	Dimensions [mm]	Dimensions [mm]
-1-	Brake disc, diameter	303	303
	Brake disc, thickness	28	28
	Brake disc, wear limit	25	25
-2-	Brake pad, thickness (without back-plate)	14	14
	Brake pad, wear limit (without back-plate)	2.5	2.5
-3-	Brake caliper piston, diameter	2 x 48	2 x 52



4.1.3 Rear brakes

Vehicles with single tyres

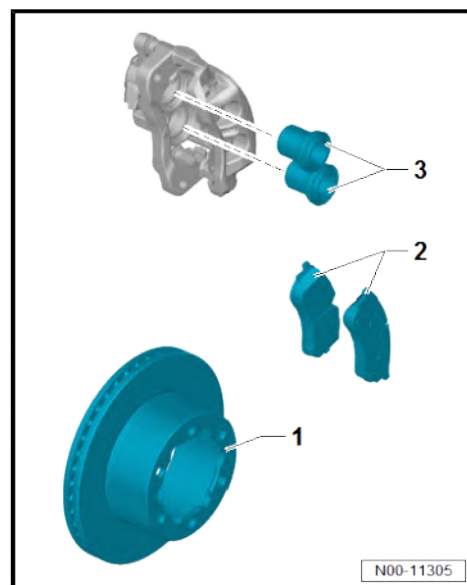
Item	Description	Dimensions [mm]
-1-	Brake disc, diameter	300
	Brake disc, thickness	22
	Brake disc, wear limit	19
-2-	Brake pad, thickness (without back-plate)	12.5
	Brake pad, wear limit (without back-plate)	2.5
-3-	Brake caliper piston, diameter	1 x 48





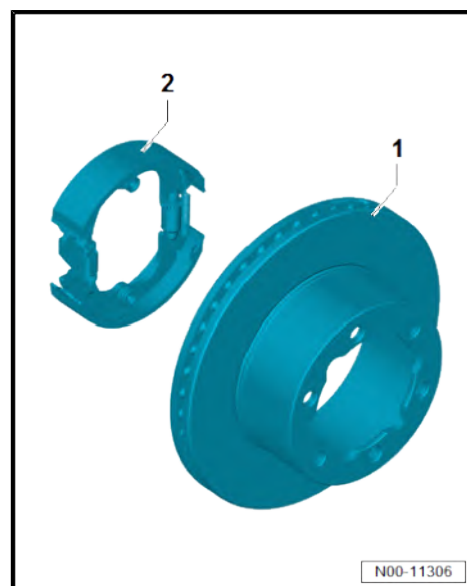
Vehicles with twin tyres

Item	Description	Dimensions [mm]
-1-	Brake disc, diameter	303
	Brake disc, thickness	28
	Brake disc, wear limit	25
-2-	Brake pad, thickness (without back-plate)	14
	Brake pad, wear limit (without back-plate)	2.5
-3-	Brake caliper piston, diameter	2 x 44



Parking brake, twin tyres

Item	Description	Dimensions [mm]
-1-	Brake disc, inner diameter	172
	Brake disc, wear limit	173
-2-	Brake lining, width	42
	Brake pad, thickness without brake shoe	3.9
	Brake pad, wear limit without brake shoe	1.5





5 Hazard classification for high-voltage system

DANGER

The vehicle's high-voltage system and the high-voltage battery are dangerous and can cause burns or other injuries and even lead to a fatal electric shock.

- Any work on the high-voltage system, or on systems which could be indirectly affected by it, must only be carried out by properly trained and qualified expert personnel.
- In the event of queries or uncertainties regarding the terms "high-voltage technician" or "high-voltage expert", or those concerning the high-voltage system, the responsible importer must be contacted prior to the start of any work.
- Any repair work must be performed in accordance with applicable laws and regulations, state-of-the-art technology, any relevant accident prevention regulations (in Germany, including but not limited to BGI/GUV-I 8686 – Training for work on vehicles with high voltage systems), as well as this workshop manual.

De-energise high-voltage system ⇒ Electric motor (270, LS2);
Rep. gr. 93 ; De-energising high-voltage system .

The types of work for which the high-voltage system has to be de-energised are indicated in the list entitled "Work on the high-voltage system" ⇒ [page 10](#) .

Work for which the high-voltage system has to be de-energised:

- ◆ Only the HVT ⇒ [page 11](#) is authorised to de-energise the high-voltage system so that it is certified.
- ◆ All work on an electric drive vehicle must be carried out only by those who are at least qualified as electrically instructed persons (EIP) ⇒ [page 11](#) .
- ◆ Regardless of the work to be performed, visually inspect high-voltage components in the work area.
- ◆ The high-voltage cables must not be extensively bent or kinked.
- ◆ In the event of conspicuous findings or uncertainties, the high-voltage technician (HVT) or the high-voltage expert (HVE) ⇒ [page 11](#) must be consulted.
- ◆ Any work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables is prohibited. In this case the high-voltage system must be de-energised and the respective component removed or sufficiently protected.
- ◆ All listed work refers to the removal and installation or the renewal of the individual components.
- ◆ For safety reasons, no further work should be performed in or on the vehicle during the charging process.
- ◆ For regular maintenance work, the high-voltage system does not have to be de-energised.



Explanation of qualifications

Working on high-voltage system

Qualification VW	Qualification MAN	Qualification DGUV-I 200-005	Area of application
EIP (electrically instructed person)	Person made aware of high-voltage issues in connection with MAN Truck & Bus AG	Chapter 2	Trained for non-electrical work (< 60 VDC)
HVT (high-voltage technician)	Qualified vehicle electrician of the MAN Truck & Bus AG (for vehicles not intrinsically safe in terms of high voltage)	Chapter 3.1 b	Work on production vehicles intrinsically safe in terms of high voltage ♦ Vehicles are de-energised exclusively with certification. ♦ Complete contact protection in place.
Level 1 HVE (high-voltage expert)	Qualified vehicle electrician with authorisation to work in energised vehicles of the MAN Truck & Bus AG (for vehicles not intrinsically safe in terms of high voltage)	Chapter 3.2 b	Work on production vehicles not intrinsically safe in terms of high voltage ♦ Vehicles are also de-energised without certification or there is no complete contact protection in place, e.g. in the event of an accident.
Level 2 HVE (high-voltage expert)		Chapter 3.3	Working on live accumulators ♦ Work on energised parts which are not necessarily touch-protected (for fault finding, component replacement, etc.)

Observe the minimum qualification for the following work:	Minimum qualification:
De-energise high-voltage system	HVT ➔ page 11

Work on power unit

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Three-phase current drive - VX54-	X			EIP
Electric drive motor - V141-	X			EIP
Drive motor temperature sender/rotor position sender -G712- / -G713-	X			EIP



Work on power and control electronics for electric drive

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Power and control electronics for electric drive - JX1- with ♦ Electric drive control unit - J841- ♦ Intermediate circuit capacitor 1 - C25- ♦ Voltage converter - A19- ♦ DC/AC converter for drive motor - A37-	X			EIP
Air conditioner compressor fuse - S355-	X			EIP

Working on high-voltage system

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Potential equalisation lines (earth) with connection to HV components	X			HVT
Measure insulation resistance		X		HVT
Maintenance connector for high-voltage system (service connector, service disconnect)		X		HVT
PX – high-voltage cables (orange) in vehicle	X			HVT

Work on heating and air conditioning system

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Electrical air conditioner compressor - V470-		X		EIP
High-voltage heater (PTC) - Z115-		X		EIP
Work on heating and air conditioning system in vehicle interior			X	EIP
Refrigerant lines in the vehicle periphery (work which is not directly on the AC compressor and can be carried out without opening the refrigerant circuit, e.g. loosening and securing refrigerant lines)			X	EIP



When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Air conditioning performance test (for checking the pressure levels in the refrigerant circuit by means of air conditioning service equipment)			X	EIP
Refrigerant lines directly on electrical air conditioner compressor - V470-		X		EIP
Extract, evacuate or fill refrigerant			X	EIP

Work on charging connection

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
High-voltage battery charging socket 1 - UX4- with ◆ Temperature sender for charging socket 1 - G853- ◆ LED module for charging socket 1 - L263- ◆ Actuator for locking		X		EIP
Immediate charge button - E766-			X	EIP
Manual release mechanism for charging connector		X		EIP
Charge high-voltage battery in workshop area			X	EIP

Work on charging unit

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Charging unit 1 for high-voltage battery - AX4- with control unit for high-voltage battery charging unit - J1050-		X		EIP



Work on high-voltage battery

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Removing and installing underbody cladding			X	EIP
Remove and install high-voltage battery 1 - AX2-		X		HVT
Disconnect high-voltage network connection		X		HVT
Disconnect low-voltage connection of high-voltage battery			X	EIP
Battery regulation control unit - J840- (service flap)		X		HVT
Module monitor control unit for batteries - J497- (battery removed)				HVE
Switching unit for high-voltage battery - SX6- (battery removed)				HVE
Open high-voltage battery 1 - AX2- (battery removed)				HVE
Bond high-voltage battery 1 - AX2- (battery removed)				HVE
Battery module 1 - J991- and other battery modules (battery removed)				HVE

Work on accident vehicles

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Preliminary evaluation of hazardous situation				HVT
Body work (with straightening jig)	X			EIP
Paint vehicle – observe instructions in paint manual			X	EIP

Work in vicinity of high-voltage components

When working on the following components or for following work:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Body work (assembly work as well as glass and dent repairs)			X	EIP
Gearbox with electric drive motor	X			EIP
Steering rack		X		EIP
Subframe, front		X		EIP



When working on the following components or for following work:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Front brakes			X	EIP
Rear brakes			X	EIP
Brake servo			X	EIP
Brake pressure accumulator for energy recovery	X			EIP
Rear axle and running gear			X	EIP
Underbody cladding			X	EIP
When welding, cover high-voltage components and visually inspect afterwards	X			EIP
Work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables	X			EIP
Work for which engine is detached (e.g. engine mounting)		X		EIP
Control units and electric components of 12 V system (except airbag)			X	EIP
Battery - A-			X	EIP
Front left headlight - MX1-			X	EIP
Front right headlight - MX2-			X	EIP
Change bulb in headlight			X	EIP

General work

When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ➤ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Drain and fill fluids (coolant and oils)			X	EIP
Coolant circuit and coolant expansion tank			X	EIP
Change tyres			X	EIP
Miscellaneous work on 12 V system (excluding airbag)			X	EIP
Airbag		X		EIP
Work on earth points of 12 V system (without potential equalisation lines)			X	EIP
Tail lights			X	EIP
Renew/repair windows			X	EIP
Repairs in vehicle interior			X	EIP
Repairs in roof area			X	EIP
Repairs on rear lid			X	EIP
Repairs on bumpers			X	EIP



When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ➔ page 11
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	no	
Move lock carrier to service position			X	EIP
Removing and installing lock carrier			X	EIP
Repair work on lock carrier			X	EIP
Disconnect high-voltage battery 1 - AX2- from support frame		X		HVT
Removing and installing deformation elements			X	EIP



6 Brake test

⇒ [“6.1 General information”, page 17](#)

⇒ [“6.2 Checking vehicles with front-wheel drive”, page 17](#)

⇒ [“6.3 Checking vehicles with rear-wheel drive”, page 17](#)

⇒ [“6.4 Checking vehicles with all-wheel drive”, page 17](#)

⇒ [“6.5 Checking parking brake”, page 18](#)

6.1 General information

- ◆ The drive is provided by the test rig.
- ◆ For the test, ensure for vehicles with a manual gearbox that the gear lever is in neutral, and for vehicles with an automatic gearbox that the selector lever is in »N«.
- ◆ When conducting the test, observe the specifications provided by the manufacturer of the test rig.



Note

The brake regulation systems do not function when ignition is off.

6.2 Checking vehicles with front-wheel drive

The brake test must be carried out on a single-axle roller dynamometer.

The maximum test speed is 6 km/h.

The test rigs authorised by Volkswagen fulfil these requirements.

6.3 Checking vehicles with rear-wheel drive

The brake test must be carried out on a single-axle roller dynamometer.

The maximum test speed is 6 km/h.

The test rigs authorised by Volkswagen fulfil these requirements.

6.4 Checking vehicles with all-wheel drive

⇒ [“6.4.1 On a regulated roller dynamometer with counter-rotating rollers for four-wheel drive vehicles”, page 17](#)

⇒ [“6.4.2 Without roller dynamometer for four-wheel drive vehicles”, page 18](#)

6.4.1 On a regulated roller dynamometer with counter-rotating rollers for four-wheel drive vehicles



Note

The brake test must be performed on a controlled contra-rotating single-axle roller dynamometer for four-wheel drive vehicles.

“Counter-rotating” means that the dynamometer rollers of the single-axle roller dynamometer are driven forwards on one side and backwards on the other.



The rollers drive the wheels of one axle in opposite directions to avoid transmitting torque to the other axle.

“Regulated” means that the speed of the brake test stand rollers is so controlled so that there is no difference in speed during the brake test.

This prevents the transfer of a force through the stationary wheels (not on the set of rollers) via the propshaft to the rotating wheels on the rollers.

The forward rotating wheel is measured during the test, which means two brake tests per axle must be performed.

The maximum test speed is 6 km/h.

The test rigs authorised by Volkswagen fulfil these requirements.

6.4.2 Without roller dynamometer for four-wheel drive vehicles



Note

If no test rig is available for four-wheel drive vehicles, the brake test can also be carried out on a standard single-axle roller dynamometer as follows:

- Drive vehicle forwards onto the rollers.
- Switch off engine and wait for 2 seconds.
- Carry out front brake test.
- Start engine and wait for approx. 5 seconds until sufficient vacuum has been built up.
- Drive vehicle forwards until rear wheels are positioned on rollers.
- Switch off engine and wait for 2 seconds.
- Carry out rear brake test.
- Start engine and wait for approx. 5 seconds until sufficient vacuum has been built up.

6.5 Checking parking brake

- Pull handbrake lever into first detent position.
- You should not be able to turn the wheels by hand.

If wheels still can be turned by hand, proceed as follows:

- ♦ Adjusting parking brake ➔ [page 77](#) .



45 – Anti-lock brake system

1 General information

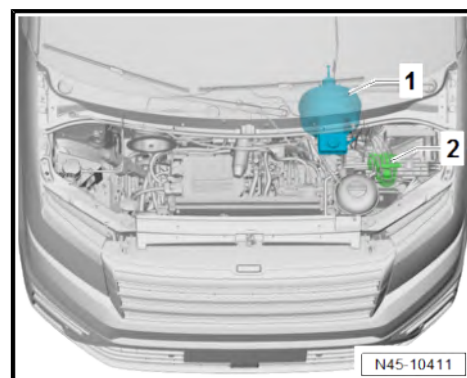
⇒ [“1.1 Repair instructions for repair work on ABS”, page 19](#)

1.1 Repair instructions for repair work on ABS

Location of ABS, e-Crafter and e-TGE

1 - Brake servo

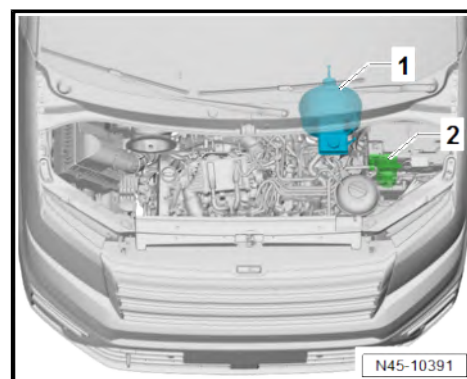
2 - ABS hydraulic unit - N55- and ABS control unit - J104-



ABS layout in LHD vehicles

1 - Brake servo

2 - ABS hydraulic unit - N55- and ABS control unit - J104-



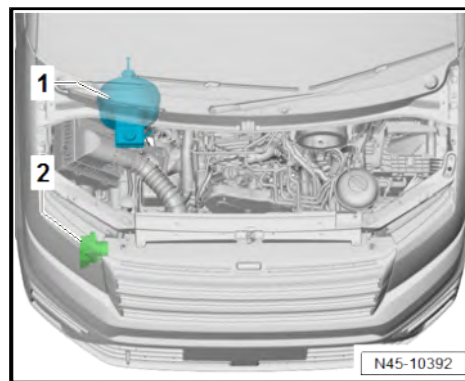


ABS layout in RHD vehicles

1 - Brake servo

2 - ABS hydraulic unit - N55- and ABS control unit - J104-

- ◆ Before carrying out repair work on the anti-lock brake system, determine the cause of the fault as well as the control unit code using ⇒ Vehicle diagnostic tester.
- ◆ New control units from the replacement parts service are not coded. They have to be coded after installation ⇒ Vehicle diagnostic tester.
- ◆ With ignition switched off, disconnect battery earth strap.
- ◆ Before carrying out welding work with an electric welding unit, see ⇒ General Information; Body Repairs, General Body Repairs .
- ◆ When working with brake fluid, observe the relevant safety precautions and notes ⇒ [page 127](#) .
- ◆ After work for which the brake system had to be opened, bleed the brake system with brake filling and bleeding unit - VAS 5234- ⇒ [page 128](#) .
- ◆ During the final road test, ensure that a controlled brake test is performed at least once (pulsations must be felt at the brake pedal).
- ◆ Absolute cleanliness is required when working on the anti-lock brake system. Auxiliary items containing mineral oil, e.g. oils, greases, etc. must never be used.
- ◆ Thoroughly clean all unions and adjacent areas before loosening. Do not use aggressive cleaning agents such as brake cleaner, petrol, thinners or similar.
- ◆ Place removed parts on a clean surface and cover them over.
- ◆ If repair work cannot be performed immediately, cover new parts which have been removed from their packing. (Use sealing plug repair set)
- ◆ Only use lint-free cloths.
- ◆ Remove packing from replacement parts immediately prior to installation and not before.
- ◆ Only use genuine packed parts.
- ◆ When the system is open, do not work with compressed air or move the vehicle.
- ◆ Valve coils in the control unit cannot be adjusted.
- ◆ Valve coils in the control unit cannot be renewed.
- ◆ Pressure sensor must not be changed or damaged.
- ◆ Pressure sensor cannot be renewed.
- ◆ Sensor housing must not be exposed to mechanical stress.
- ◆ No measurements must be taken on contact points of control unit.
- ◆ No measurements must be taken on contact points of hydraulic unit.
- ◆ Contact pins of hydraulic unit must not be bent or damaged.
- ◆ Contacts cannot be renewed.
- ◆ No contact sprays may be used on the contacts or the pressure sensor.





- ◆ Foreign bodies must not get between the control unit and the hydraulic unit.
- ◆ During painting operations, the electronic control unit can be exposed to a maximum temperature of 95°C for only a short period, and to a maximum of 85°C for longer periods (approx. 2 hours). During longer periods (2 hours), the control unit can be exposed to a maximum temperature of 85°C. Ensure that no brake fluid enters connectors.

 NOTICE

Risk of damage to hydraulic unit caused by removal of the return flow pump.

- Never detach the return flow pump from the hydraulic unit.

2 Overview of fitting locations

⇒ ["2.1 Overview of fitting locations - ABS/ESP", page 22](#)

2.1 Overview of fitting locations - ABS/ESP

1 - ABS control unit - J104- / ABS hydraulic unit - N55-

- ☐ with brake pressure sender 1 - G201-
- ☐ Removing and installing
⇒ [page 25](#)

2 - Brake light switch - F-

- ☐ Removing and installing
⇒ [page 107](#)

3 - Dash panel insert - KX2-

- ☐ Removing and installing
⇒ Electrical system;
Rep. gr. 90 ; Dash panel
insert; Removing and in-
stalling dash panel in-
sert KX2

4 - Button for TCS and elec- tronic stabilisation program - E256-

- ☐ Removing and installing
⇒ Electrical system;
Rep. gr. 96 ; Controls;
Removing and installing
special vehicle control
unit -E256-

5 - Front right speed sensor - G45-

- ☐ Removing and installing
⇒ [page 38](#)

6 - Wheel bearing unit

- ☐ Removing and installing
⇒ Running gear, axles,
steering; Rep. gr. 40 ;
Wheel bearings; Re-
moving and installing
wheel bearing unit .

7 - Wheel bearing unit

- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 42 ; Wheel bearings, trailing arm;
Removing and installing wheel bearing unit

8 - Rear right speed sensor - G44-

- ☐ Removing and installing ⇒ [page 39](#)

9 - Rear left speed sensor - G46-

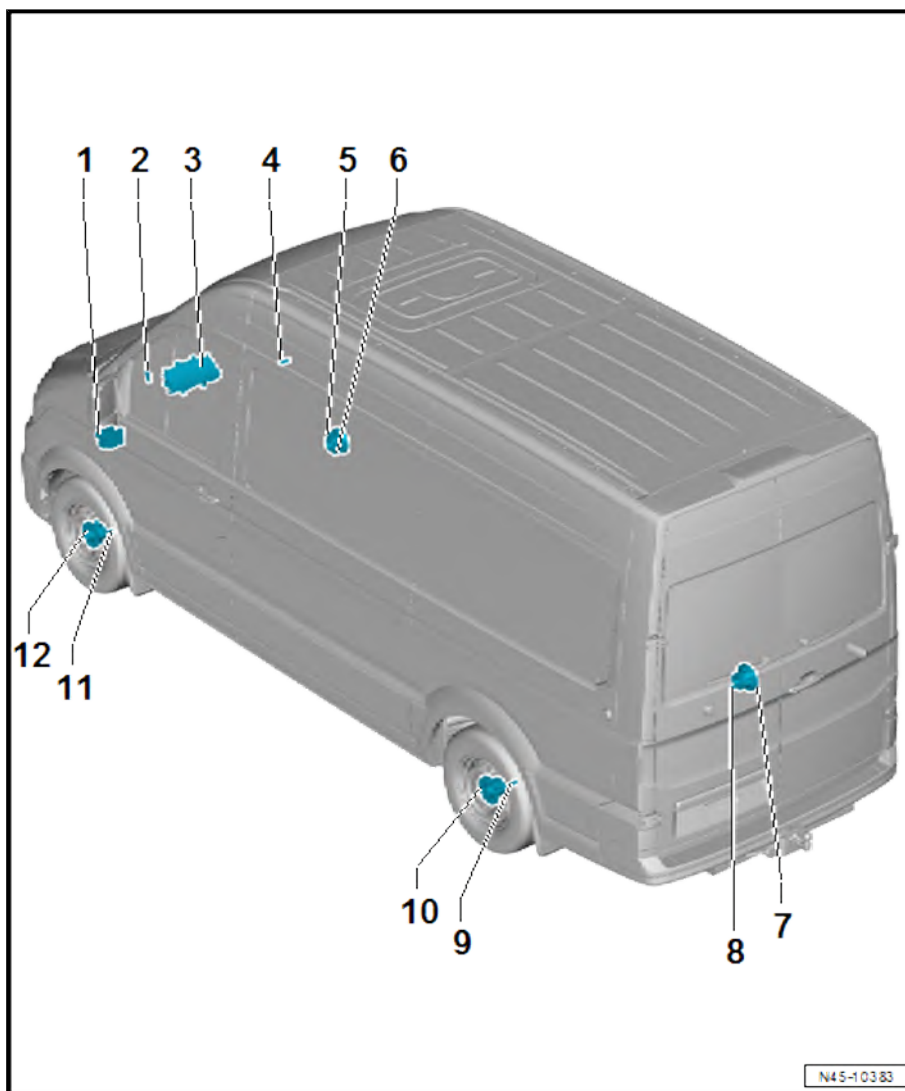
- ☐ Removing and installing ⇒ [page 39](#)

10 - Wheel bearing unit

- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 42 ; Wheel bearings, trailing arm;
Removing and installing wheel bearing unit

11 - Front left speed sensor - G47-

- ☐ Removing and installing ⇒ [page 38](#)





12 - Wheel bearing unit

- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 40 ; Wheel bearings; Removing and installing wheel bearing unit .

3 Control unit and hydraulic unit

⇒ ["3.1 Assembly overview - control unit and hydraulic unit", page 24](#)

⇒ ["3.2 Removing and installing ABS control unit J104 and ABS hydraulic unit N55", page 25](#)

⇒ ["3.3 Removing and installing ABS control unit J104 without ABS hydraulic unit N55", page 30](#)

⇒ ["3.4 Connecting brake lines to hydraulic unit", page 34](#)

3.1 Assembly overview - control unit and hydraulic unit

1 - Brake servo

- ❑ Removing and installing
⇒ [page 107](#)

2 - Brake line

- ❑ 14 Nm

3 - Brake line

- ❑ 14 Nm

4 - Brake line

- ❑ 14 Nm

5 - Brake line

- ❑ 14 Nm

6 - Brake line

- ❑ 14 Nm

7 - Brake line

- ❑ 14 Nm

8 - Bolts

- ❑ Qty. 4
- ❑ 3 Nm

9 - Nuts

- ❑ Qty. 3
- ❑ 8 Nm

10 - Brake line

- ❑ 14 Nm

11 - Bracket

12 - Nuts

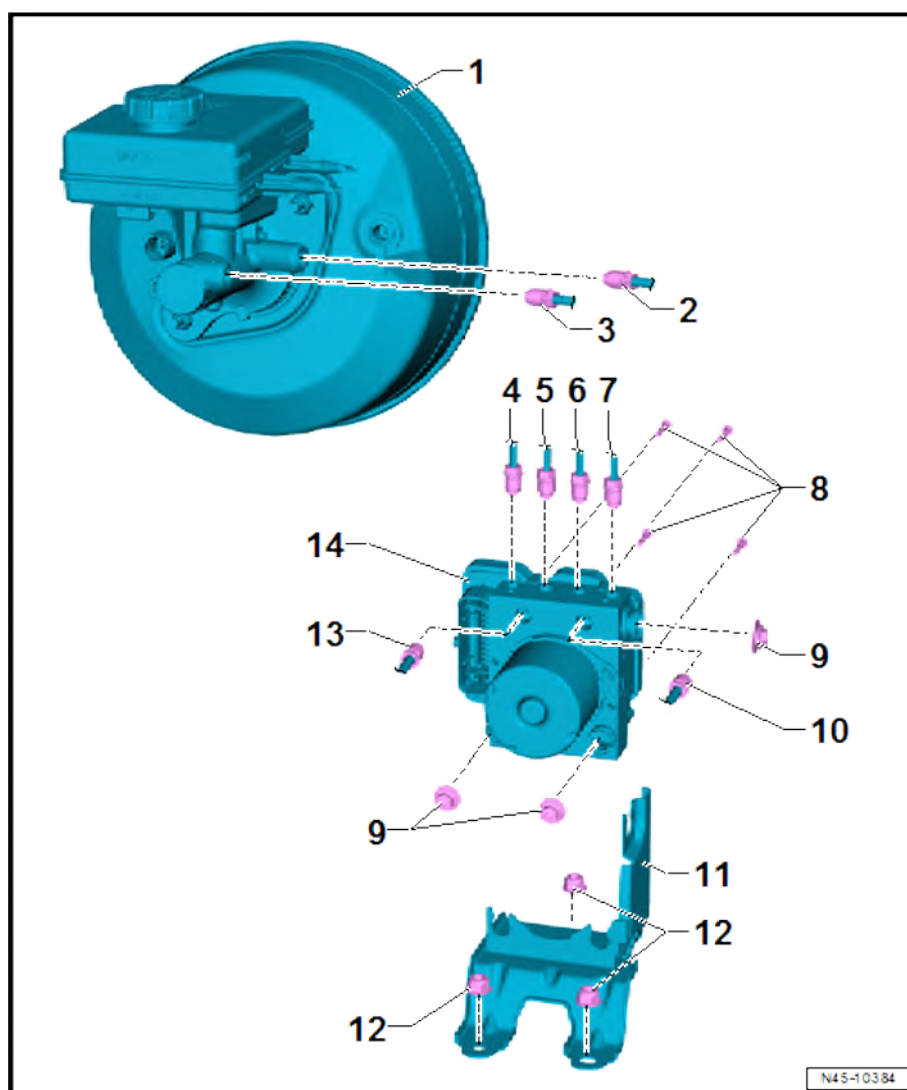
- ❑ Qty. 3
- ❑ 20 Nm

13 - Brake line

- ❑ 14 Nm

14 - ABS control unit - J104- with ABS hydraulic unit - N55-

- ❑ with brake pressure sender 1 - G201-
- ❑ Removing and installing ⇒ [page 25](#)



N45-10384



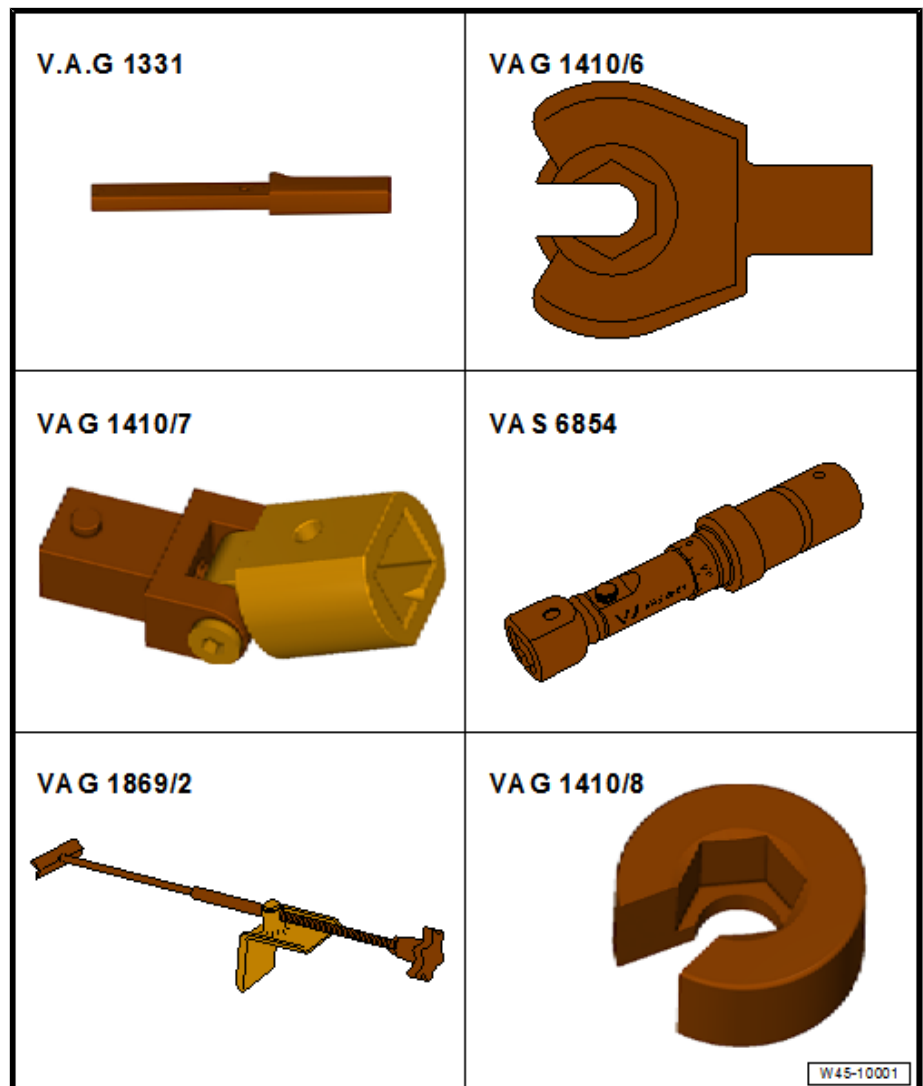
3.2 Removing and installing ABS control unit - J104- and ABS hydraulic unit - N55-

⇒ [“3.2.1 Removing and installing ABS control unit J104 / ABS hydraulic unit N55 , LHD vehicles”, page 25](#)

⇒ [“3.2.2 Removing and installing ABS control unit J104 / ABS hydraulic unit N55 , RHD vehicles”, page 27](#)

3.2.1 Removing and installing ABS control unit - J104- / ABS hydraulic unit - N55-, LHD vehicles

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-
- ◆ Ring insert tool, open, AF 11 mm - V.A.G 1410/6-
- ◆ Universal joint - V.A.G 1410/7-
- ◆ Torque wrench - VAS 6854-
- ◆ Brake pedal depressor - V.A.G 1869/2-
- ◆ Assembly tool - V.A.G 1410/8-



Sealing plug repair kit

1 - Sealing plug M10

2 - Sealing plug M12

Removing

Vehicles with second battery - A1-

- Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery tray; Removing and installing battery tray .

Continued for all vehicles

- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .
- Remove front left wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .

- Connect hose of bleeder bottle -1- to bleeder valve of front left brake caliper.
- Open bleeder valve.

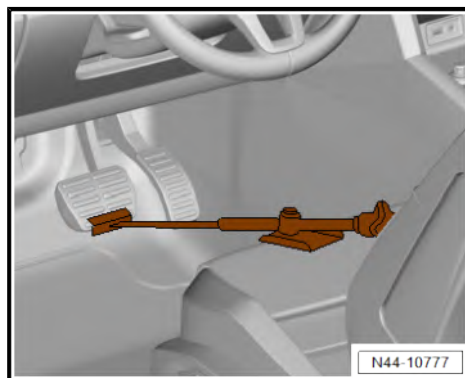
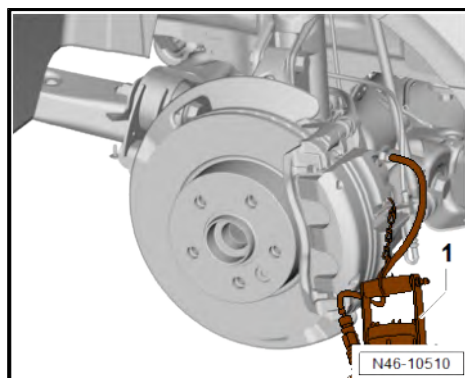
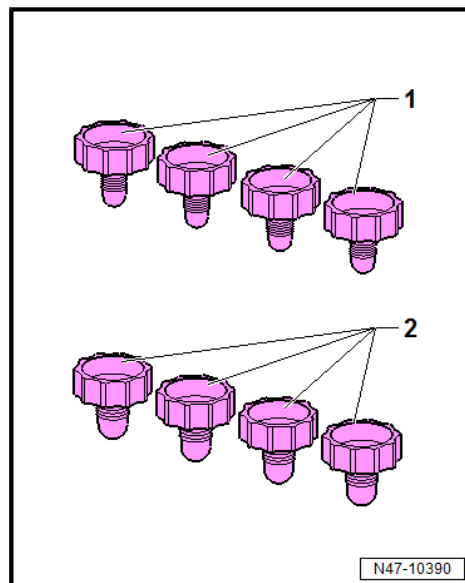
- Use brake pedal depressor - V.A.G 1869/2- to depress brake pedal.
- Close front left bleed valve.
- Do not remove brake pedal depressor - V.A.G 1869/2- .
- Place sufficient lint-free cloths under control unit and hydraulic unit.



NOTICE

Risk of damage to brake lines if bent.

- **Never excessively bend the brake lines in the area of the hydraulic unit.**



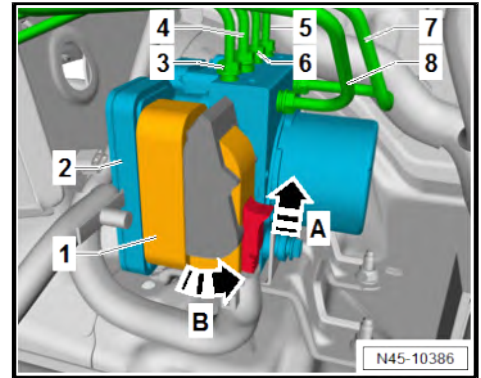


- Remove brake lines -7- and -8-, unscrewing from brake master cylinder first then from hydraulic unit.
- Unscrew brake lines -3- to -6- from hydraulic unit.
- Seal brake lines and threaded holes using sealing plugs .
- Push locking element upwards in direction of -arrow A-.
- Swivel release lever in direction of -arrow B-.
- Pull off electrical connector -1- on ABS control unit - J104- -2-.



Note

Ensure no brake fluid gets onto contacts.



- Loosen nuts -arrows-.
- Pull out hydraulic unit with control unit -1- upwards and remove from bracket -2-.

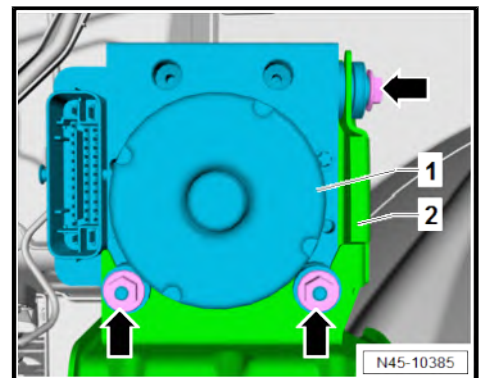
Installing

Install in the reverse order of removal, observing the following:



Note

- ◆ *Remove sealing plugs from new hydraulic unit only when the corresponding brake line is to be fitted.*
- ◆ *If sealing plugs are removed too early from the hydraulic unit, brake fluid can escape, and it can then no longer be guaranteed that the unit can be sufficiently filled and bled.*



- Remove brake pedal actuator - V.A.G 1869/2- .
- Connect brake line ➔ [page 34](#) .
- Bleed brake system ➔ [page 128](#) .
- Code ABS control unit - J104- using ➔ Vehicle diagnostic tester.

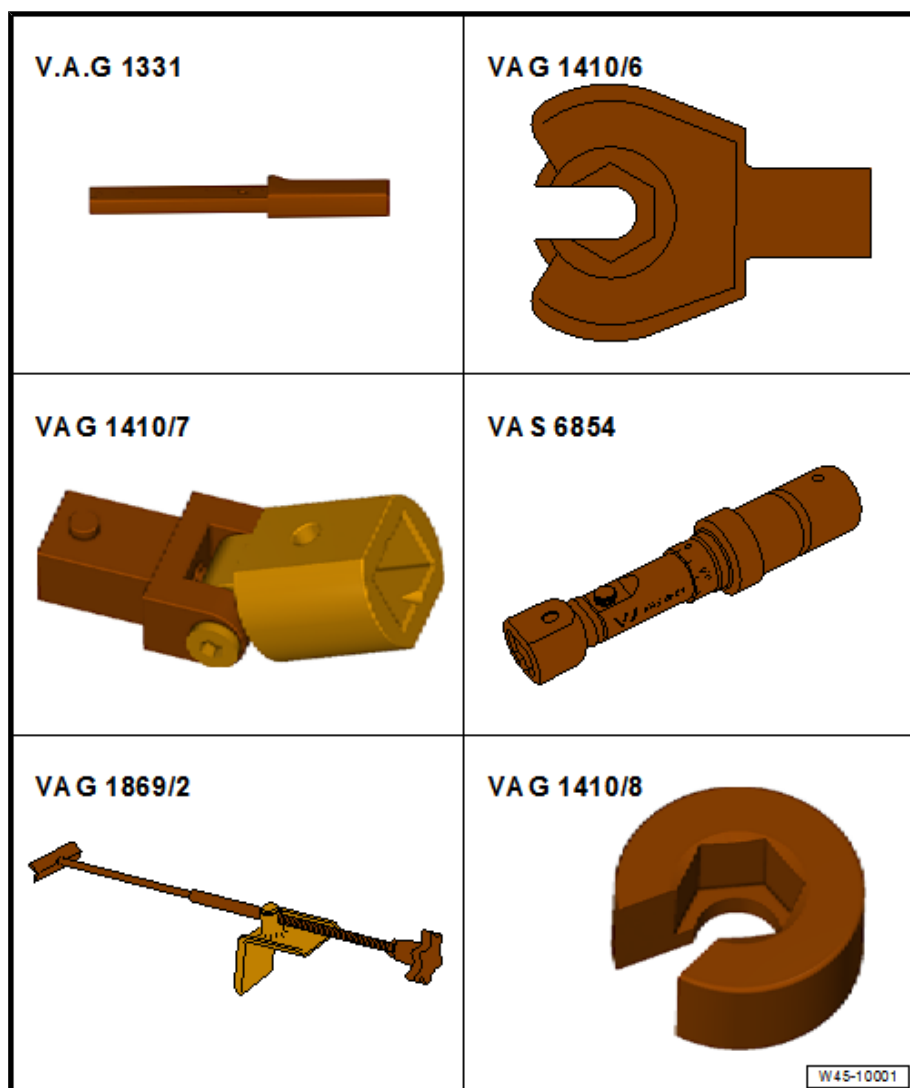
Specified torques

- ◆ ➔ ["3.1 Assembly overview - control unit and hydraulic unit", page 24](#)

3.2.2 Removing and installing ABS control unit - J104- / ABS hydraulic unit - N55-, RHD vehicles



**Special tools and workshop
equipment required**



- ◆ Torque wrench - V.A.G 1331-
- ◆ Ring insert tool, open, AF 11 mm - V.A.G 1410/6-
- ◆ Universal joint - V.A.G 1410/7-
- ◆ Torque wrench - VAS 6854-
- ◆ Brake pedal depressor - V.A.G 1869/2-
- ◆ Assembly tool - V.A.G 1410/8-



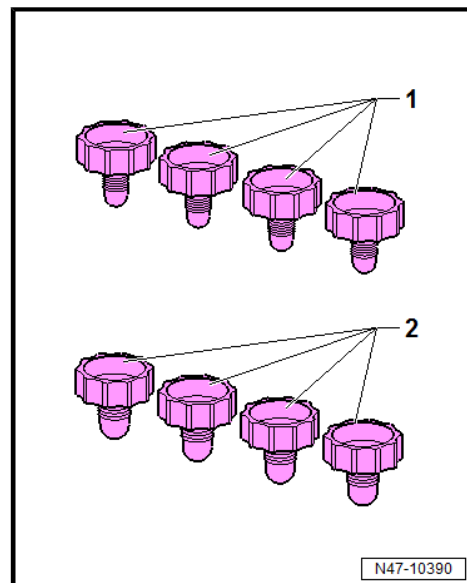
Sealing plug repair kit

1 - Sealing plug M10

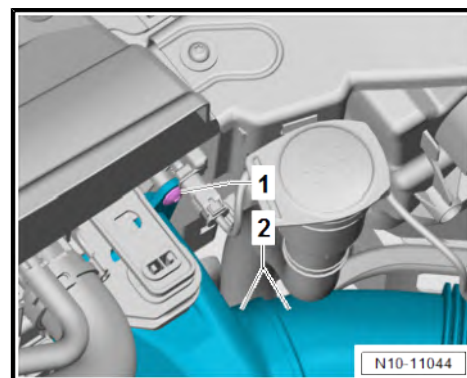
2 - Sealing plug M12

Removing

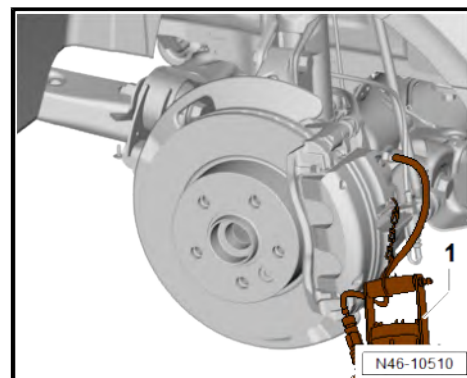
- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .



- Unscrew bolt -1-.
- Unclip intake hose -2- from air filter housing.
- Remove intake hose -2-.
- Remove front left wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .



- Connect hose of bleeder bottle -1- to bleeder valve of front left brake caliper. Open bleeder valve.



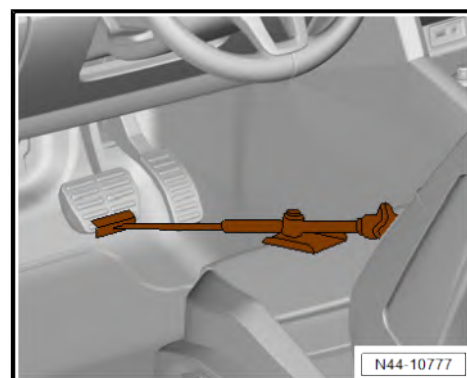
- Use brake pedal depressor - V.A.G 1869/2- to depress brake pedal.
- Close front left bleed valve.
- Do not remove brake pedal depressor - V.A.G 1869/2- .
- Place sufficient lint-free cloths under control unit and hydraulic unit.



NOTICE

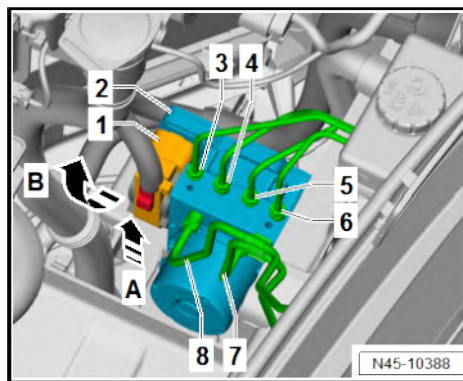
Risk of damage to brake lines if bent.

- **Never excessively bend the brake lines in the area of the hydraulic unit.**





- Remove brake lines -7- and -8-, unscrewing from brake master cylinder first then from hydraulic unit.
- Unscrew brake lines -3- to -6- from hydraulic unit.
- Seal brake lines and threaded holes using sealing plugs .
- Push locking element upwards in direction of -arrow A-.
- Swivel release lever in direction of -arrow B-.
- Pull off electrical connector -1- on ABS control unit - J104-2-.



Note

Ensure no brake fluid gets onto contacts.

- Loosen nuts -arrows-.
- Pull out hydraulic unit with control unit -1- upwards and remove from bracket -2-.

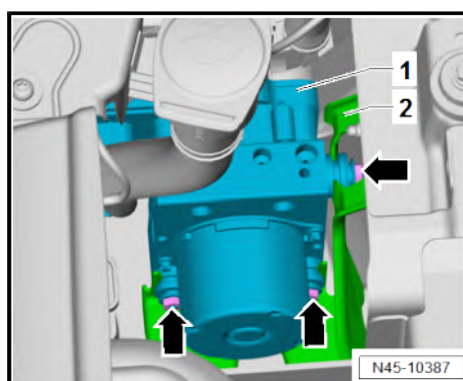
Installing

Install in the reverse order of removal, observing the following:



Note

- ♦ *Remove sealing plugs from new hydraulic unit only when the corresponding brake line is to be fitted.*
- ♦ *If sealing plugs are removed too early from the hydraulic unit, brake fluid can escape, and it can then no longer be guaranteed that the unit can be sufficiently filled and bled.*
- Remove brake pedal actuator - V.A.G 1869/2- .
- Connect brake line ⇒ [page 34](#) .
- Bleed brake system ⇒ [page 128](#) .
- Code ABS control unit - J104- using ⇒ Vehicle diagnostic tester.



Specified torques

- ♦ ⇒ [“3.1 Assembly overview - control unit and hydraulic unit”, page 24](#)

3.3 Removing and installing ABS control unit - J104- without ABS hydraulic unit - N55-

⇒ [“3.3.1 Removing and installing ABS control unit J104 without ABS hydraulic unit N55 , LHD vehicles”, page 30](#)

⇒ [“3.3.2 Removing and installing ABS control unit J104 without ABS hydraulic unit N55 , RHD vehicles”, page 32](#)

3.3.1 Removing and installing ABS control unit - J104- without ABS hydraulic unit - N55- , LHD vehicles

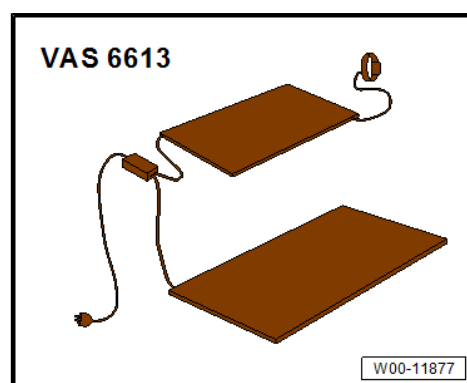
Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



- ◆ ESD workplace - VAS 6613-



Removing



Note

The hydraulic lines remain connected.

- Read out and note the existing control unit code using ⇒ Vehicle diagnostic tester.

Vehicles with second battery - A1-

- Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery tray; Removing and installing battery tray .

Continued for all vehicles

- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .



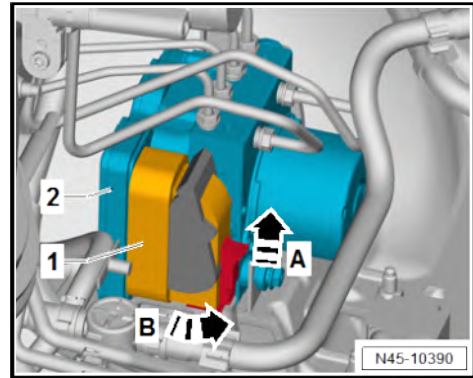
NOTICE

Risk of damage to control unit caused by static charge and dirt.

- **Discharge any static charge:** touch an ESD workplace (earth yourself).
- **Protect interior of control unit from moisture and dirt.**



- Press fastener in direction of -arrow A-.
- Swivel release lever in direction of -arrow B-.
- Pull off electrical connector -1- on ABS control unit - J104- -2-.

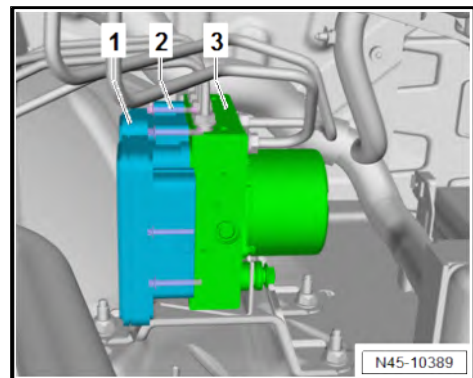


- Unscrew bolts -2-.
- Pull off ABS control unit - J104- -1- from ABS hydraulic unit - N55- -3-.

Installing

Install in the reverse order of removal, observing the following:

- Make sure that surfaces of ABS control unit - J104- and ABS hydraulic unit - N55- have been cleaned.
- Place ABS control unit - J104- on ABS hydraulic unit - N55- without canting ABS control unit.



Note

- ♦ *The threads of the ABS hydraulic unit - N55- for securing the ABS control unit - J104- must not be recut.*
- ♦ *If the threads are damaged, the ABS hydraulic unit - N55- needs to be renewed.*

Specified torques

- ♦ ⇒ ["3.1 Assembly overview - control unit and hydraulic unit", page 24](#)

3.3.2 Removing and installing ABS control unit - J104- without ABS hydraulic unit - N55- , RHD vehicles

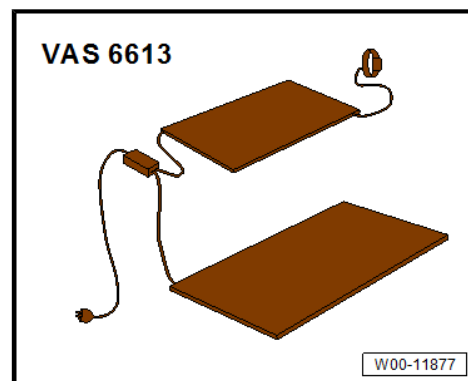
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-





◆ ESD workplace - VAS 6613-



Removing



Note

The hydraulic lines remain connected.

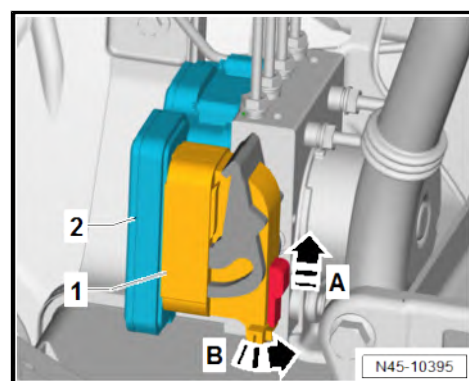
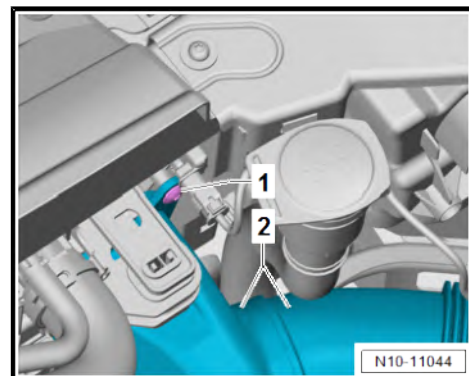
- Read out and note the existing control unit code using ⇒ Vehicle diagnostic tester.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .
- Unscrew bolt -1-.
- Unclip intake hose -2- from air filter housing.
- Remove intake hose -2-.



NOTICE

Risk of damage to control unit caused by static charge and dirt.

- **Discharge any static charge:** touch an ESD workplace (earth yourself).
- **Protect interior of control unit from moisture and dirt.**
- Press fastener in direction of -arrow A-.
- Swivel release lever in direction of -arrow B-.
- Pull off electrical connector -1- on ABS control unit - J104-2-.





- Unscrew bolts -2-.
- Pull off ABS control unit - J104- -1- from ABS hydraulic unit - N55- -3-.

Installing

Install in the reverse order of removal, observing the following:

- Make sure that surfaces of ABS control unit - J104- and ABS hydraulic unit - N55- have been cleaned.
- Place ABS control unit - J104- on ABS hydraulic unit - N55- without canting ABS control unit.



Note

- ◆ *The threads of the ABS hydraulic unit - N55- for securing the ABS control unit - J104- must not be recut.*
- ◆ *If the threads are damaged, the ABS hydraulic unit - N55- needs to be renewed.*

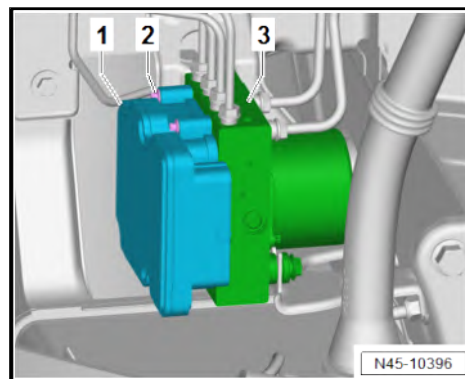
Specified torques

- ◆ ⇒ [“3.1 Assembly overview - control unit and hydraulic unit”, page 24](#)

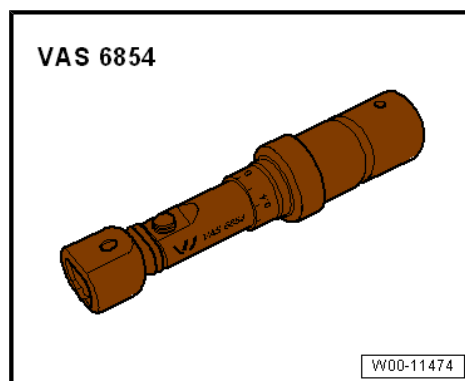
3.4 Connecting brake lines to hydraulic unit

Special tools and workshop equipment required

- ◆ Torque wrench - VAS 6854-



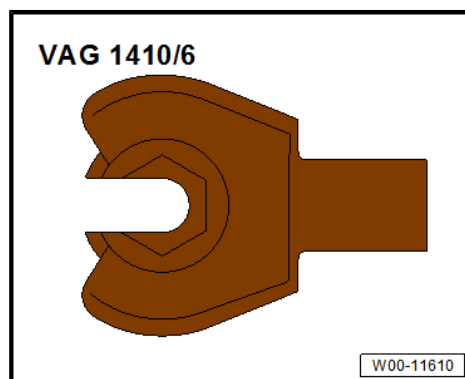
VAS 6854



W00-11474

- ◆ Ring insert tool, open, AF 11 mm - V.A.G 1410/6-

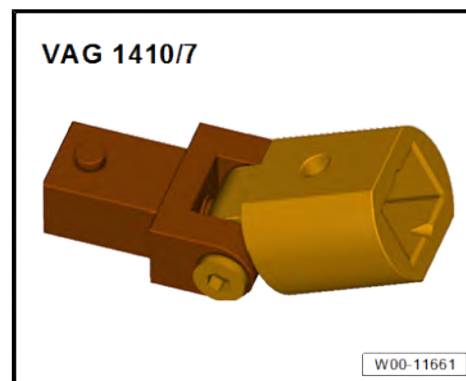
VAG 1410/6



W00-11610



- ◆ Universal joint - V.A.G 1410/7-



- ◆ Assembly tool - V.A.G 1410/8-



Procedure

! NOTICE

Risk of damage to brake lines if bent.

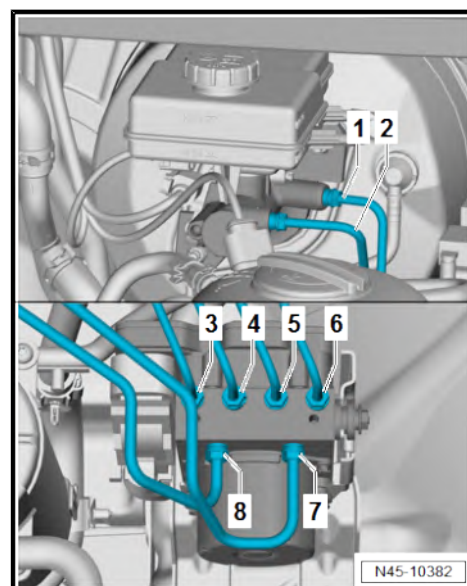
- Never excessively bend the brake lines in the area of the hydraulic unit.

On brake master cylinder

- 1 - Primary piston circuit of brake master cylinder to hydraulic unit.
- 2 - Secondary piston circuit of brake master cylinder to hydraulic unit.

On hydraulic unit

- 3 - Hydraulic unit to rear right brake caliper
 - 4 - From hydraulic unit to front left brake caliper
 - 5 - From hydraulic unit to front right brake caliper.
 - 6 - From hydraulic unit to rear left brake caliper
 - 7 - From hydraulic unit to primary piston circuit of brake master cylinder.
 - 8 - From hydraulic unit to secondary piston circuit of brake master cylinder.
- It must be possible to screw brake line nuts to end of threads into ABS hydraulic unit - N55- by hand.
 - Tighten nuts to specified torque.



Specified torques

- ◆ ⇒ [“3.1 Assembly overview - control unit and hydraulic unit”, page 24](#)

4 Sensors

⇒ ["4.1 Assembly overview - speed sensor on front axle", page 36](#)

⇒ ["4.2 Assembly overview - speed sensor on rear axle", page 37](#)

⇒ ["4.3 Removing and installing speed sensors on front axle G45 / G47", page 38](#)

⇒ ["4.4 Removing and installing speed sensors on rear axle G44 / G46", page 39](#)

4.1 Assembly overview - speed sensor on front axle

1 - Wheel bearing housing

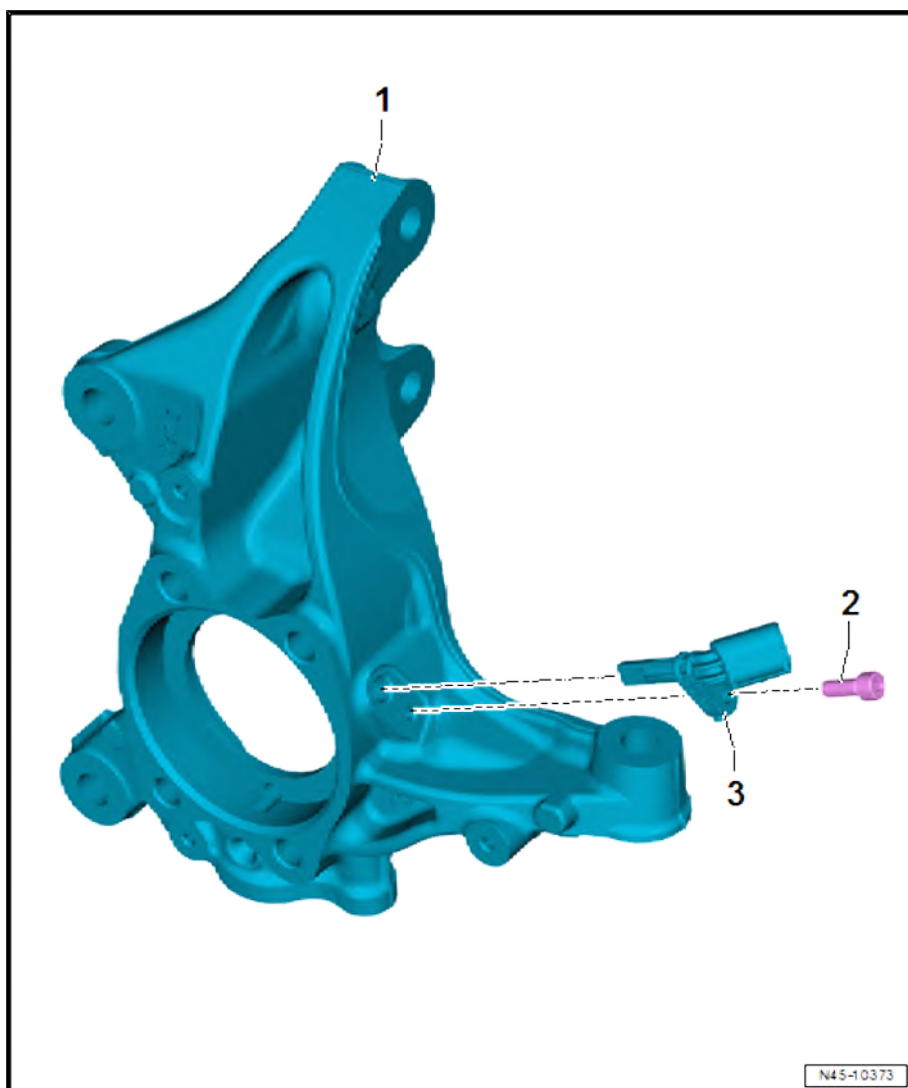
- ❑ Removing and installing
⇒ Running gear, axles, steering; Rep. gr. 40 ; Wheel bearing; Removing and installing wheel bearing housing .

2 - Bolt

- ❑ 8 Nm

3 - Speed sensors - G45- / - G47-

- ❑ Removing and installing
⇒ [page 38](#)





4.2 Assembly overview - speed sensor on rear axle

Assembly overview - speed sensor on rear axle, single tyres
⇒ [page 37](#)

Assembly overview - speed sensor on rear axle, twin tyres
⇒ [page 37](#)

Assembly overview - speed sensor on rear axle, single tyres

1 - Speed sensors - G44- / - G46-

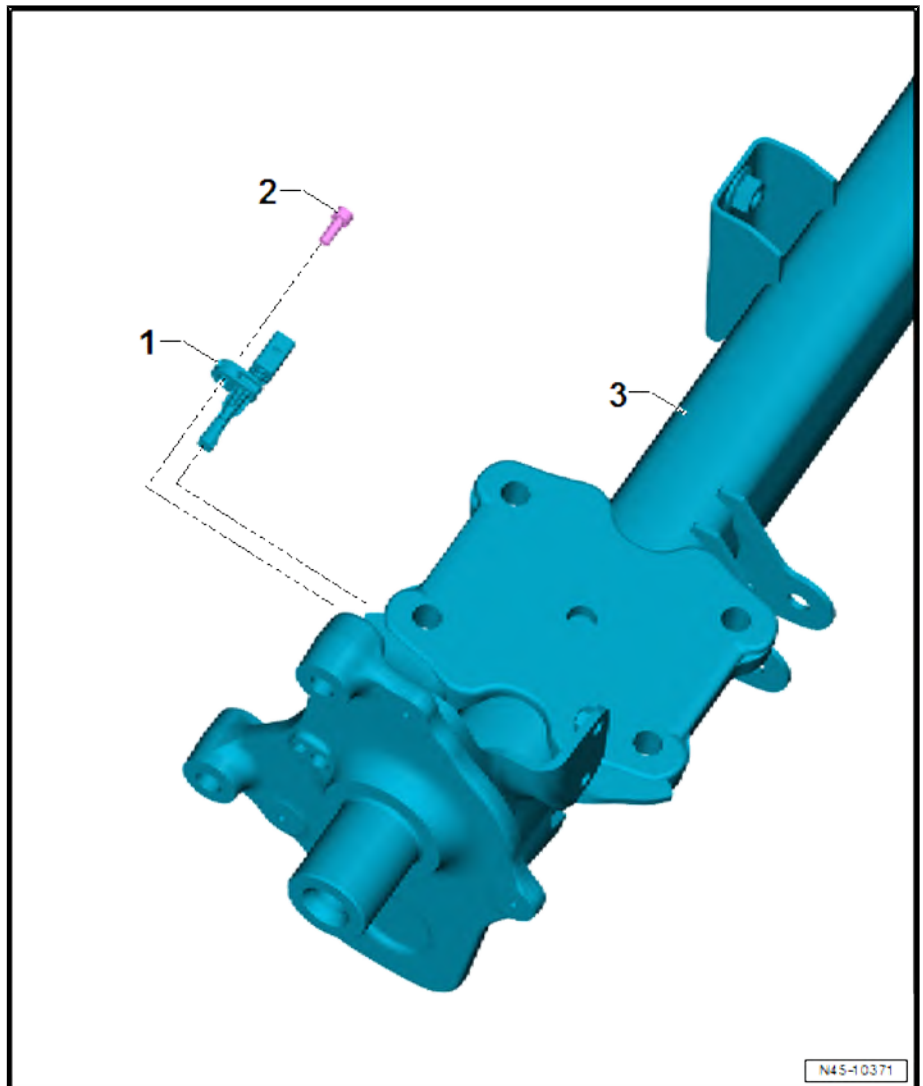
- ❑ Removing and installing
⇒ [page 39](#)

2 - Bolt

- ❑ 8 Nm

3 - Rear axle

- ❑ Removing and installing
⇒ Running gear, axles, steering; Rep. gr. 42 ;
Rear axle; Removing and installing rear axle



Assembly overview - speed sensor on rear axle, twin tyres



1 - Speed sensors - G44- / -G46-

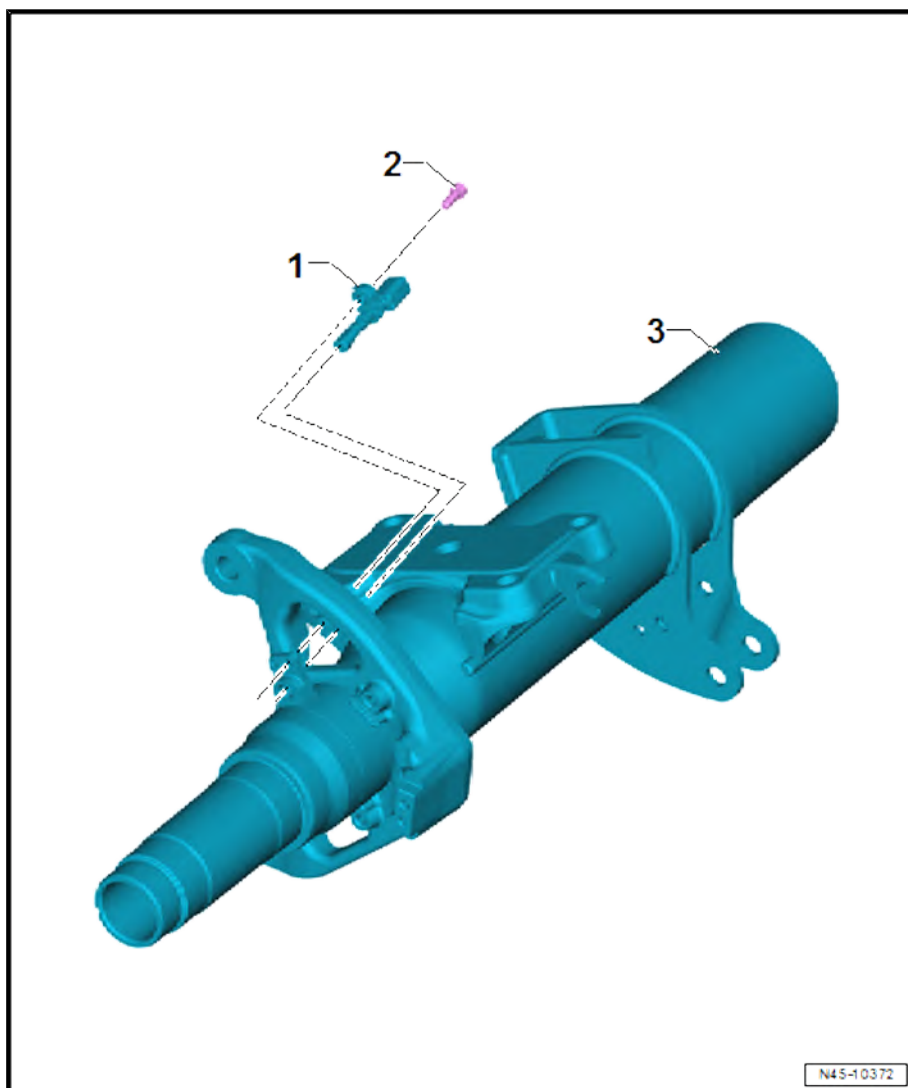
- ❑ Removing and installing
⇒ [page 40](#)

2 - Bolt

- ❑ 8 Nm

3 - Rear axle

- ❑ Removing and installing
⇒ Running gear, axles,
steering; Rep. gr. 42 ;
Rear axle; Removing
and installing rear axle



4.3 Removing and installing speed sensors on front axle -G45- / -G47-

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



- ◆ Polycarbamide grease ⇒ Electronic parts catalogue (ETKA)
or ⇒ webMANTIS for MAN TGE



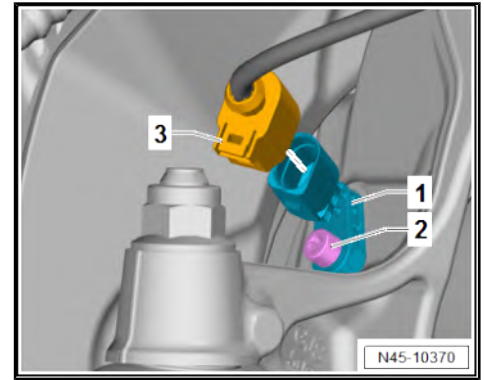
Removing

- Separate electrical connector -3-.
- Unscrew bolt -2-.
- Pull speed sensor -1- out of wheel bearing housing.

Installing

Install in the reverse order of removal, observing the following:

- Before inserting speed sensor , clean inner surface of hole, and coat speed sensor all-round with polycarbamide grease
⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Insert speed sensor into hole in wheel bearing housing and tighten bolt.



Specified torques

- ◆ ⇒ [“4.1 Assembly overview - speed sensor on front axle”, page 36](#)

4.4 Removing and installing speed sensors on rear axle -G44- / -G46-

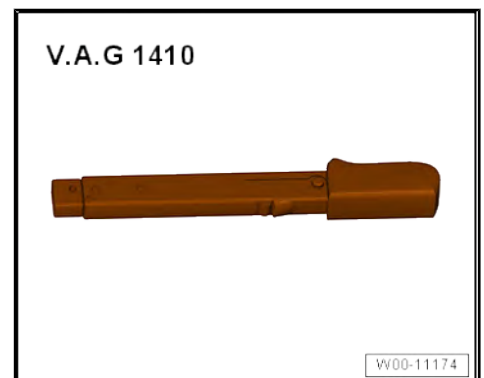
⇒ [“4.4.1 Removing and installing speed sensors on rear axle G44 / G46 , single tyres”, page 39](#)

⇒ [“4.4.2 Removing and installing speed sensors on rear axle G44 / G46 , twin tyres”, page 40](#)

4.4.1 Removing and installing speed sensors on rear axle -G44- / -G46- , single tyres

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



- ◆ Polycarbamide grease ⇒ Electronic parts catalogue (ETKA)
or ⇒ webMANTIS for MAN TGE



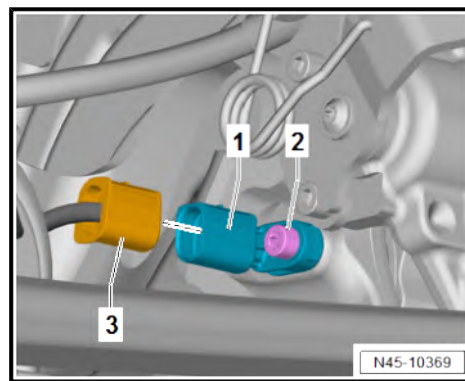
Removing

- Separate electrical connector -3-.
- Unscrew bolt -2-.
- Pull out speed sensor -1- from axle beam.

Installing

Install in the reverse order of removal, observing the following:

- Before inserting speed sensor , clean inner surface of hole, and coat speed sensor all-round with polycarbamide grease
⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Insert speed sensor into hole in wheel bearing housing and tighten bolt.



Specified torques

- ♦ ⇒ [“4.2 Assembly overview - speed sensor on rear axle”, page 37](#)

4.4.2 Removing and installing speed sensors on rear axle -G44- / -G46- , twin tyres

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-



- ♦ Polycarbamide grease ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

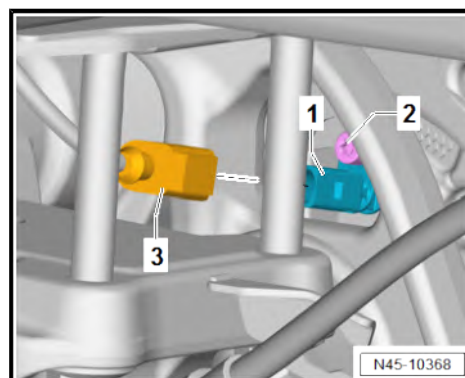
Removing

- Separate electrical connector -3-.
- Unscrew bolt -2-.
- Pull out speed sensor -1- from axle beam.

Installing

Install in the reverse order of removal, observing the following:

- Before inserting speed sensor , clean inner surface of hole, and coat speed sensor all-round with polycarbamide grease
⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Insert speed sensor into hole in wheel bearing housing and tighten bolt.



Specified torques

- ♦ ⇒ [“4.2 Assembly overview - speed sensor on rear axle”, page 37](#)



46 – Brakes - mechanism

1 Front brakes

⇒ [“1.1 Assembly overview - front brakes”, page 41](#)

⇒ [“1.2 Removing and installing brake pads/linings”, page 42](#)

⇒ [“1.3 Removing and installing brake caliper”, page 45](#)

⇒ [“1.4 Removing and installing brake carrier”, page 47](#)

⇒ [“1.5 Removing and installing brake disc”, page 48](#)

1.1 Assembly overview - front brakes

1 - Wheel bearing housing

- ☐ With wheel bearing and wheel hub
- ☐ Removing and installing
⇒ Running gear, axles, steering; Rep. gr. 40 ; Wheel bearing; Removing and installing wheel bearing housing .

2 - Spacer

- ☐ Only on vehicles with twin tyres

3 - Bolts

- ☐ Renew after removal
- ☐ Qty. 5
- ☐ 130 Nm +90°

4 - Cover plate

5 - Bolts

- ☐ Number of bolts may differ
- ☐ Qty. 3
- ☐ 9.5 Nm

6 - Brake disc

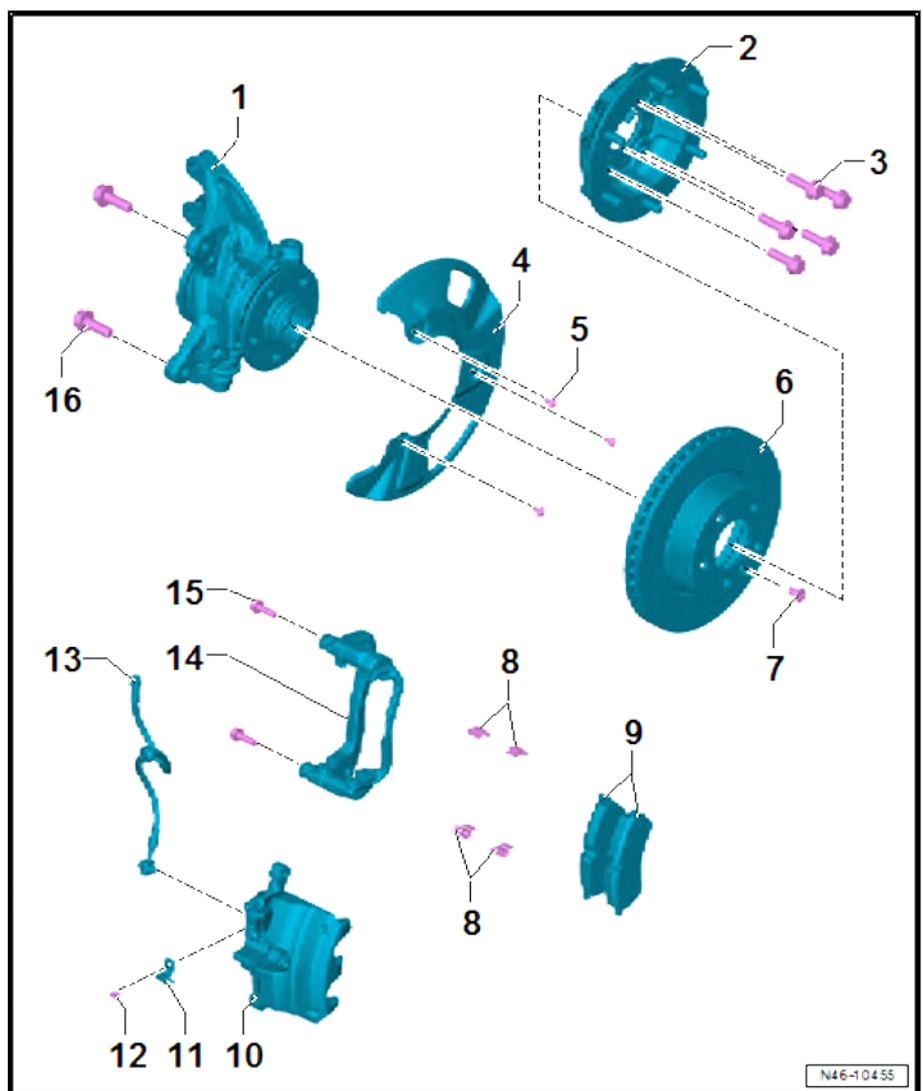
- ☐ Check thickness
⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing
⇒ [page 48](#)

7 - Bolt

- ☐ 20 Nm

8 - Pad retainers

- ☐ Always renew when changing pads.





9 - Brake pads

- ☐ Check thickness ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing ⇒ [page 42](#)

10 - Brake caliper

- ☐ Removing and installing ⇒ [page 45](#)

11 - Retaining plate

12 - Bolt

- ☐ 9.5 Nm

13 - Brake hose

- ☐ Install without twisting
- ☐ 14 Nm on brake line
- ☐ 35 Nm on brake caliper

14 - Brake carrier

- ☐ If protective sleeves or guide pins are damaged, install repair kit and use lubricant sachet supplied to lubricate guide pins
- ☐ Removing and installing ⇒ [page 47](#)

15 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 65 Nm

16 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 180 Nm +90°

1.2 Removing and installing brake pads/linings

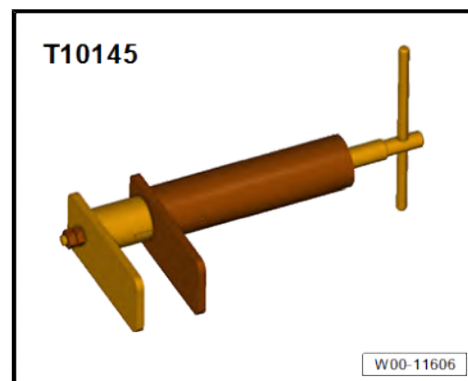
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-





- ◆ Piston resetting appliance - T10145-



- ◆ Welding wire

Removing

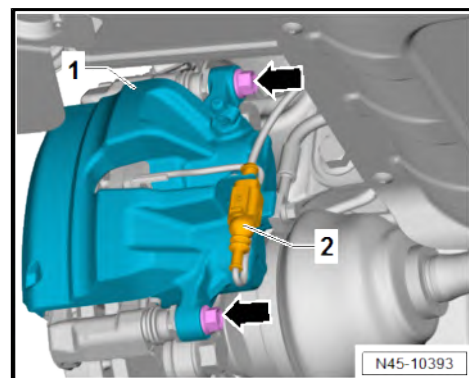
- Remove front wheel ➔ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .



Note

Mark brake pads or linings when removing if they are to be reused. Fit in same position when installing, or braking will be uneven.

- Separate electrical connector -2- for brake pad wear indicator.
- Unscrew bolts -arrows- from brake caliper -1-.





- Detach brake caliper -2-, and secure it with welding wire so that weight of brake caliper does not strain or damage brake hose.
- Guide wire for brake pad wear indicator through opening in brake caliper.
- Detach brake pads -1- in direction of -arrow-.

Cleaning

WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- **Never blow out the brake system with compressed air.**

- Thoroughly clean contact surfaces on brake carrier and brake pads, and remove any corrosion.
- Clean brake caliper.



Note

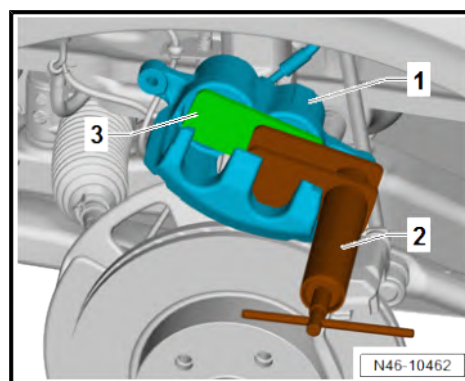
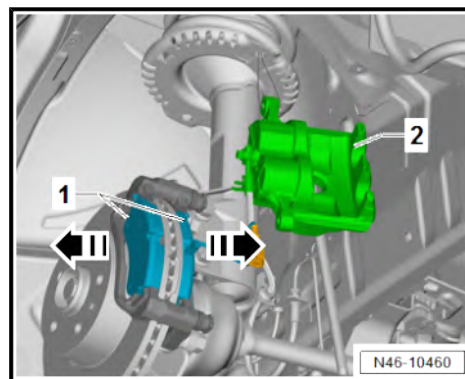
Use brake and clutch cleaner to clean the brake caliper.

Installing

Install in the reverse order of removal, observing the following:

Before inserting new brake pads, press piston back into cylinder using piston resetting tool - T10145- . Before pressing piston back, draw off brake fluid from brake fluid reservoir with bleeder bottle. Otherwise, fluid can overflow and cause damage, particularly if reservoir has been topped up.

- Place a piece of wood -3- or a brake pad in the recess so that both pistons are pressed back at the same time.
- Push back both pistons from brake caliper -1- using piston resetting appliance - T10145- -2-.





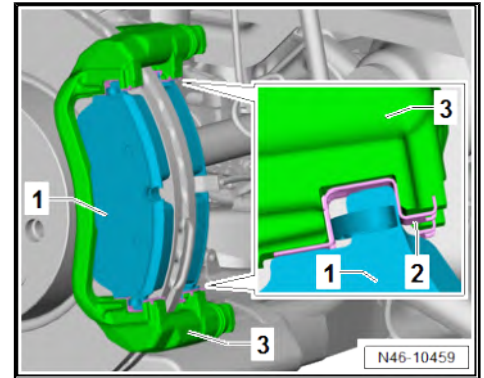
- Insert pad retainers -2- into brake carrier -3-.
- Insert brake pads -1- into pad retainers -2-.

How to proceed after brake pad renewal

- After replacing brake pads, depress brake pedal firmly several times with vehicle stationary so that the pads are properly seated in their normal operating position.
- After changing brake pads/linings, check brake fluid level.

Specified torques

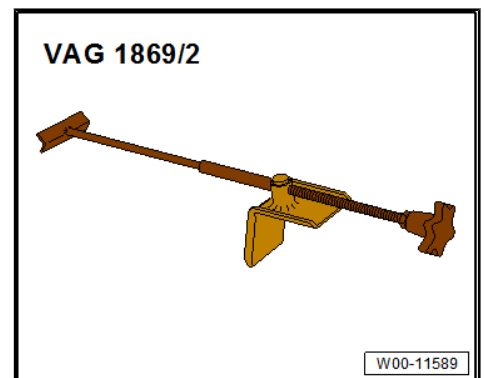
- ◆ ⇒ ["1.1 Assembly overview - front brakes", page 41](#)



1.3 Removing and installing brake caliper

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-
- ◆ Brake pedal depressor - V.A.G 1869/2-



Removing

- Remove front wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .

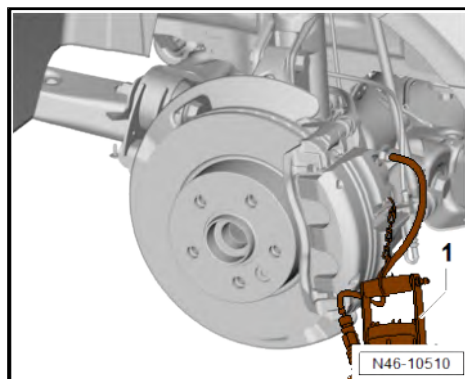


Note

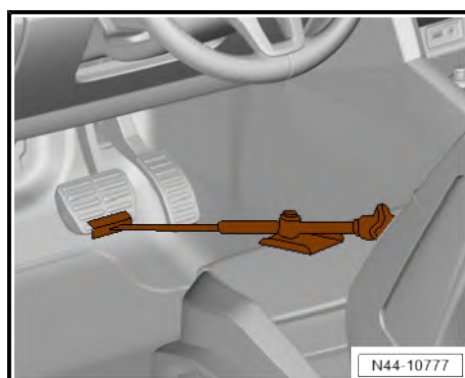
This procedure is only relevant when replacing or repairing the brake caliper.



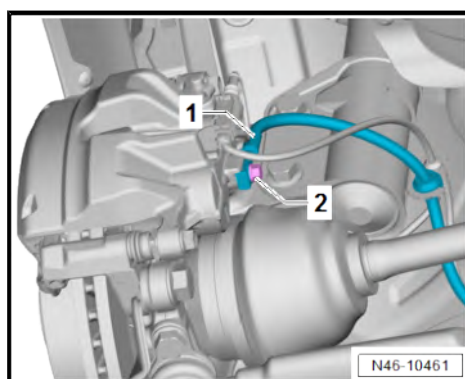
- Connect bleeder bottle hose -1- to brake caliper bleed valve and open bleed valve.



- Then apply brake pedal actuator - V.A.G 1869/2- .
- Following this, close bleeder valve and remove bleeder bottle.



- Unscrew bolt -2- from brake hose -1-.



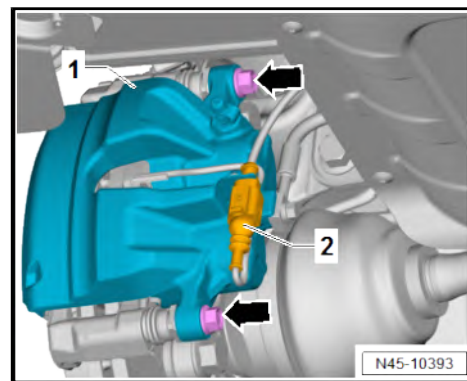


- Separate electrical connector -2- for brake pad wear indicator.
- Unscrew bolts -arrows- from brake caliper -1-.
- Remove brake caliper from brake carrier.

Installing

Install in the reverse order of removal, observing the following:

- The pistons are pressed back.
- The brake pads are seated on the brake carrier.
- Screw brake hose onto brake caliper.
- Remove brake pedal depressor - V.A.G 1869/2- .
- Bleed brake system ➔ [page 128](#) .



Note

- ◆ *Firmly depress brake pedal several times with vehicle stationary so that the brake pads/linings are properly seated in their normal operating position.*
- ◆ *Check brake fluid level.*

Specified torques

- ◆ ➔ [“1.1 Assembly overview - front brakes”, page 41](#)

1.4 Removing and installing brake carrier

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

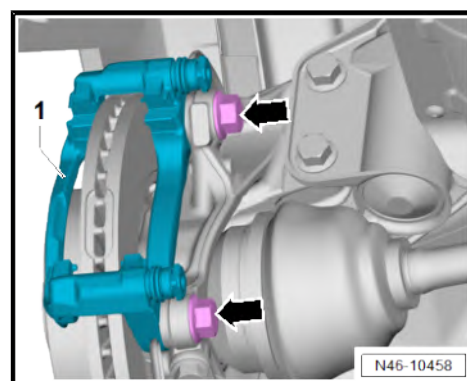
- Remove brake pads ➔ [page 42](#) .
- Remove bolts -arrows-
- Remove backplate -1-.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“1.1 Assembly overview - front brakes”, page 41](#)





1.5 Removing and installing brake disc

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



- ◆ Torque wrench - V.A.G 1332-

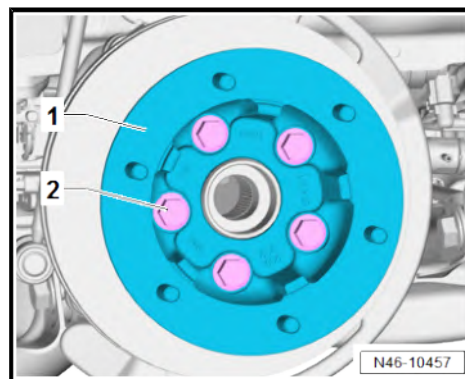


Removing

- Remove backplate ➤ [page 47](#) .

Vehicles with twin wheels

- Unscrew bolts -2-.
- Remove spacer -1-.





Continued for all vehicles

- Unscrew bolt -2-.



Note

- ◆ *Risk of damage to brake discs.*
- ◆ *Do not use force to separate the brake disc from the wheel hub. If necessary, use penetration spray.*
- Remove brake disc -1-.

Cleaning



WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- Never blow out the brake system with compressed air.

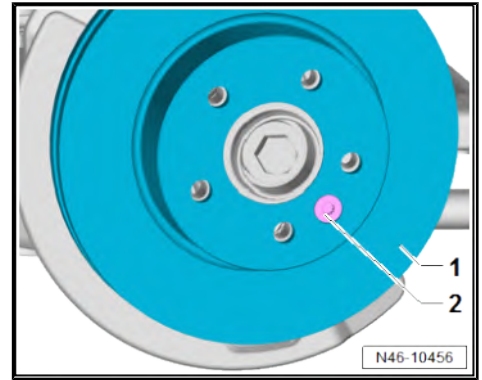
- Thoroughly clean contact surfaces on wheel hub, remove any corrosion.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - front brakes”, page 41](#)



2 Rear brakes

⇒ "2.1 Assembly overview - rear brakes", page 50

⇒ “2.2 Removing and installing brake pads/linings”, page 53

⇒ **“2.3 Removing and installing brake caliper”, page 58**

⇒ **“2.4 Removing and installing brake carrier”, page 64**

⇒ “2.5 Removing and installing brake disc”, page 66

⇒ “2.6 Removing and installing brake shoes”, page 68

⇒ “2.7 Adjusting basic setting/clearance of drum brake”, page 71

2.1 Assembly overview - rear brakes

⇒ “2.1.1 Assembly overview – rear brake, single tyres”, page 50

⇒ “2.1.2 Assembly overview – rear brake, twin tyres”, page 52

2.1.1 Assembly overview – rear brake, single tyres

1 - Rear axle

- ❑ Removing and installing
⇒ Running gear, axles,
steering; Rep. gr. 42 ;
Rear axle; Removing
and installing rear axle

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 150 Nm +90°

3 - Brake carrier

- ❑ Removing and installing
⇒ [page 64](#)

4 - Brake hose

- ☐ 35 Nm on brake caliper
- ☐ 14 Nm on brake line

5 - Brake cable

- ❑ Removing and installing
⇒ page 83

6 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 30 Nm

7 - Brake caliper

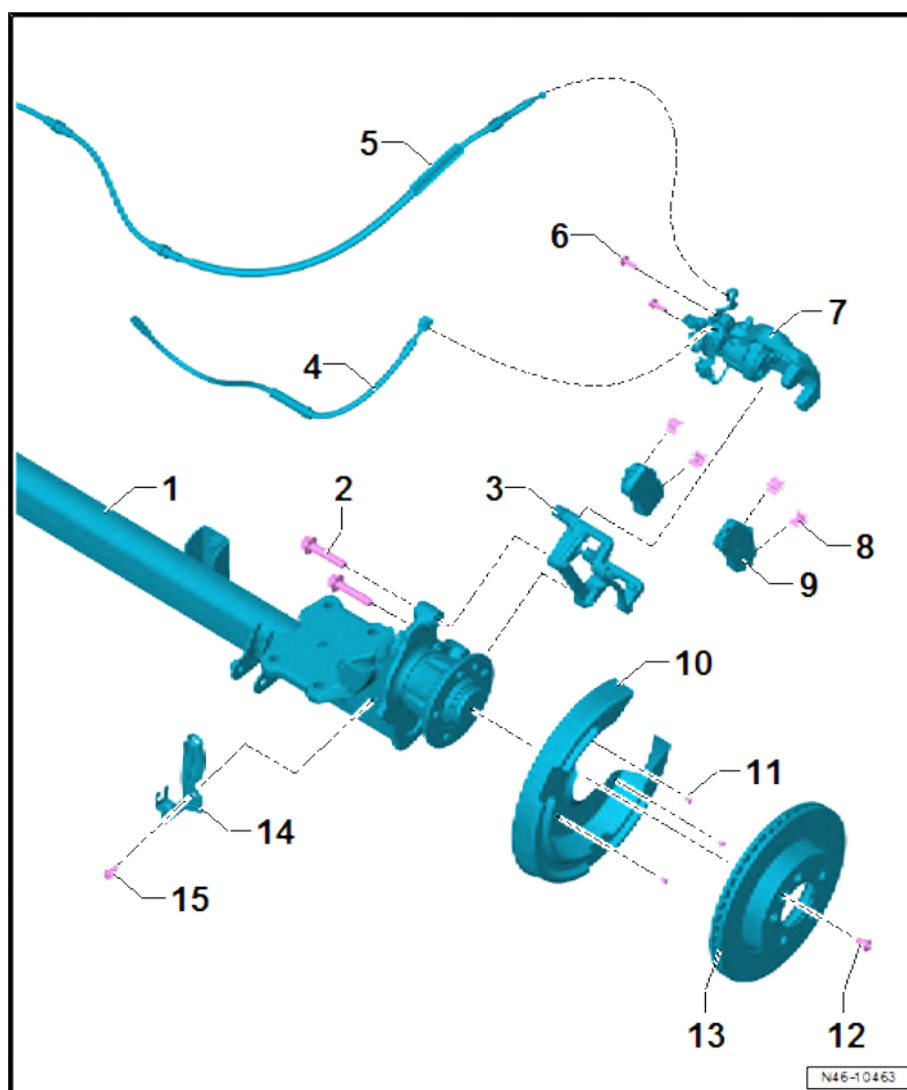
- ❑ Removing and installing
⇒ page 58

8 - Pad retainers

- Always renew when changing pads.

9 - Brake pads

- ❑ Check thickness ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage





- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing ⇒ [page 53](#)

10 - Cover plate

11 - Bolts

- ☐ Qty. 3
- ☐ 9.5 Nm

12 - Bolt

- ☐ 20 Nm

13 - Brake disc

- ☐ Check thickness ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing ⇒ [page 66](#)

14 - Bracket

- ☐ To axle beam

15 - Bolt

- ☐ 20 Nm

2.1.2 Assembly overview – rear brake, twin tyres

1 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 180 Nm +90°

2 - Brake hose

- ☐ 35 Nm on brake caliper
- ☐ 14 Nm on brake line

3 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 65 Nm

4 - Brake caliper

- ☐ Removing and installing
⇒ [page 61](#)

5 - Brake carrier

- ☐ Removing and installing
⇒ [page 65](#)

6 - Guide plates

- ☐ Always renew when changing pads.
- ☐ Qty. 4

7 - Brake pads

- ☐ Check thickness
⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing
⇒ [page 56](#)

8 - Brake disc

- ☐ Check thickness ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing ⇒ [page 67](#)

9 - Wheel flange

10 - Bolts

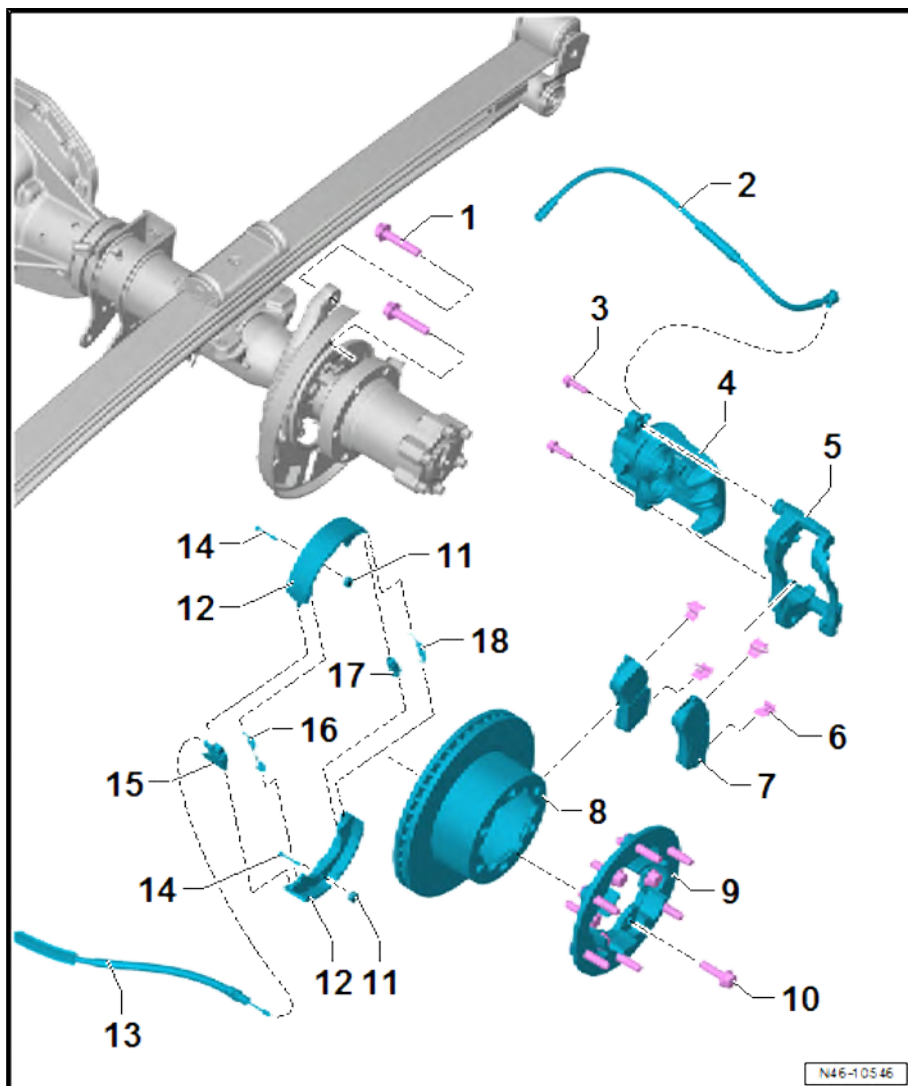
- ☐ Renew after removal
- ☐ Qty. 6
- ☐ 130 Nm +90°

11 - Retaining springs

- ☐ Renew after removal
- ☐ Qty. 2

12 - Brake pads with linings

- ☐ Qty. 2





- ☐ Check thickness ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Brake system: Visual check for leaks and damage
- ☐ When worn, renew on both sides of axle.
- ☐ Removing and installing ⇒ [page 68](#)

13 - Brake cable

- ☐ Removing and installing ⇒ [page 84](#)

14 - Setting tools

- ☐ Renew after removal
- ☐ Qty. 2

15 - Expander

- ☐ Renew after removal

16 - Extension spring

- ☐ Renew after removal

17 - Adjuster

- ☐ Renew after removal

18 - Compression spring

- ☐ Renew after removal

2.2 Removing and installing brake pads/linings

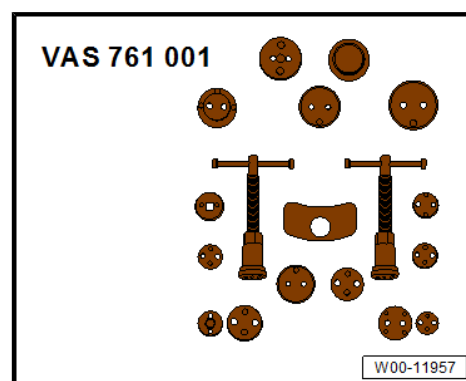
⇒ [“2.2.1 Removing and installing brake pads/linings, single tyres”, page 53](#)

⇒ [“2.2.2 Removing and installing brake pads/linings, twin tyres”, page 56](#)

2.2.1 Removing and installing brake pads/linings, single tyres

Special tools and workshop equipment required

- ◆ Brake piston tool set - VAS 761 001-





- ◆ Torque wrench - V.A.G 1331-



- ◆ Welding wire

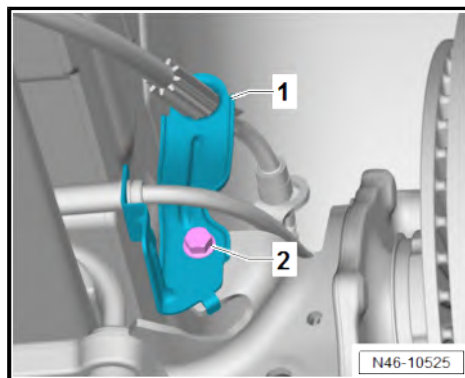
Removing

- Remove rear wheel ➔ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .
- Unscrew bolt -2-.
- Remove bracket -1-.

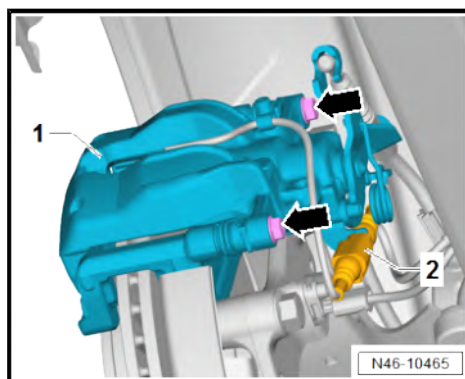


Note

Mark brake pads or linings when removing if they are to be reused. Fit in same position when installing, or braking will be uneven.



- Separate electrical connector -2- for brake pad wear indicator.
- Remove bolts -arrows-
- Detach brake caliper -1-, and secure it with welding wire so that weight of brake caliper -1- does not strain or damage brake hose.





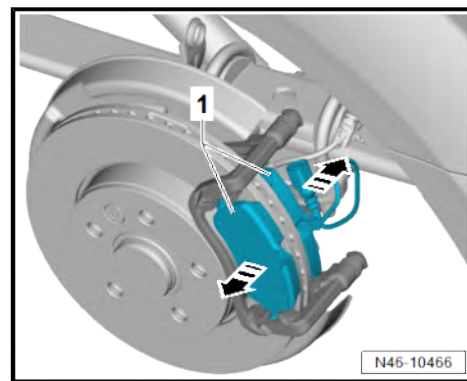
- Detach brake pads -1- in direction of -arrow-.

Cleaning

WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- Never blow out the brake system with compressed air.



Note

Use only brake and clutch cleaner to clean the brake caliper.

- Thoroughly clean contact surfaces on brake carrier and brake pads, and remove any corrosion.
- Clean brake caliper.

Installing

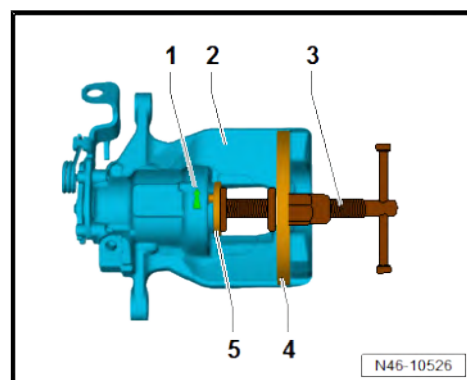
Install in the reverse order of removal, observing the following:

- Extract brake fluid from brake fluid reservoir using bleeder bottle.

Note

Observe the arrow -1- cast on the brake caliper -2- for the screw in direction.

- Fit suitable adapter -5- from brake piston tool set - VAS 761 001- -3-.
- Mount retaining plate -4- from brake piston tool set - VAS 761 001- -3-.
- Insert brake piston tool set - VAS 761 001- -3- in brake caliper -2- as shown in illustration.



Note

When screwing in, do not damage the boot that protects the piston.

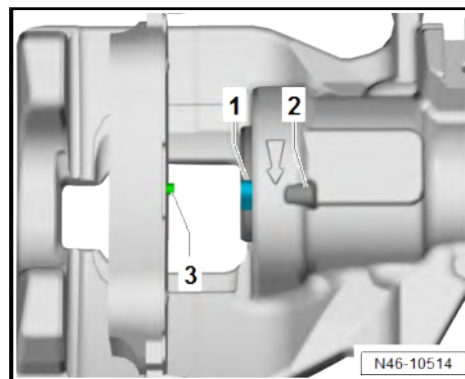
- Screw in brake piston by turning handle of brake piston tool set - VAS 761 001- -3- in direction of arrow -1- cast on brake caliper -2-.



- Turn brake piston until notch -1- aligns with protrusion -2-.
- Pin -3- on brake pad engages in recess -1-

With new brake pads, use the pad retainers included in the repair kit.

- Make sure that protective caps are not damaged and are seated correctly.
- Make sure that guide pins are free to move.



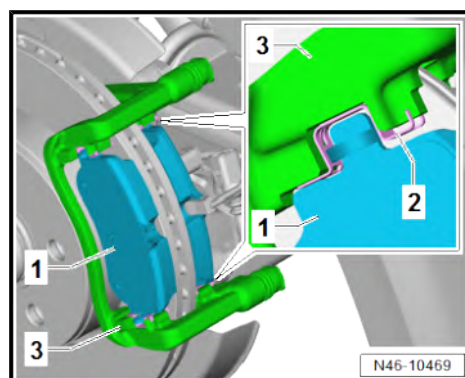
- Insert pad retainers -2- into brake carrier -3-.
- Insert brake pads -1- into pad retainers -2-.

How to proceed after brake pad renewal

- After replacing brake pads, depress brake pedal firmly several times with vehicle stationary so that the pads are properly seated in their normal operating position.
- After changing brake pads/linings, check brake fluid level.

Specified torques

- ♦ ➔ [“2.1 Assembly overview - rear brakes”, page 50](#)



2.2.2 Removing and installing brake pads/linings, twin tyres

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1332-



- ♦ Piston resetting appliance - T10145-



- ♦ Welding wire



Removing

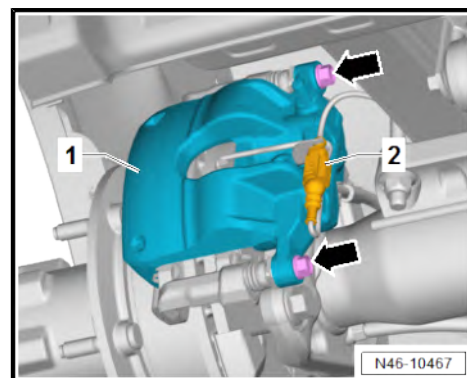
- Remove rear wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .



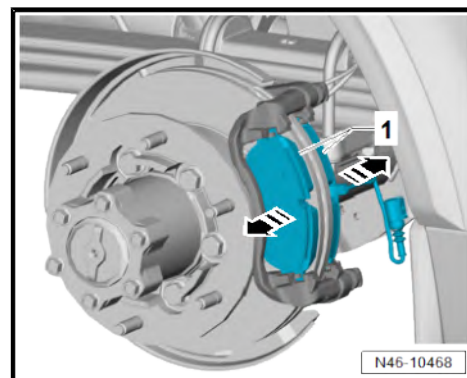
Note

Mark brake pads or linings when removing if they are to be reused. Fit in same position when installing, or braking will be uneven.

- Separate electrical connector -2- for brake pad wear indicator.
- Unscrew bolts -arrows- from brake caliper -1-.



- Detach brake caliper, and secure it with welding wire so that weight of the brake caliper does not stress or damage brake hose.
- Remove brake pads -1- from brake carrier in direction of -arrow-.



Cleaning



WARNING

**Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.**

- Never blow out the brake system with compressed air.



Note

Use brake and clutch cleaner to clean the brake caliper.

- Thoroughly clean contact surfaces on brake carrier and brake pads, and remove any corrosion.
- Clean brake caliper.

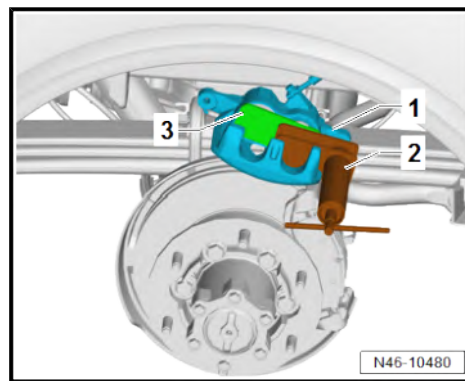
Installing

Install in the reverse order of removal, observing the following:

Before inserting new brake pads, press piston back into cylinder using piston resetting tool - T10145- . Before pressing piston back, draw off brake fluid from brake fluid reservoir with bleeder bottle. Otherwise, fluid can overflow and cause damage, particularly if reservoir has been topped up.



- Place a piece of wood -3- or a brake pad in the recess so that both pistons are pressed back at the same time.
- Push back both pistons from brake caliper -1- using piston re-setting appliance - T10145- -2-.



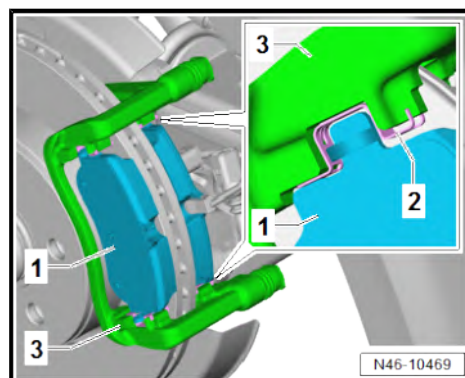
- Insert pad retainers -2- into brake carrier -3-.
- Insert brake pads -1- into pad retainers -2-.

How to proceed after brake pad renewal

- After replacing brake pads, depress brake pedal firmly several times with vehicle stationary so that the pads are properly seated in their normal operating position.
- After changing brake pads/linings, check brake fluid level.

Specified torques

- ♦ ⇒ [“2.1 Assembly overview - rear brakes”, page 50](#)



2.3 Removing and installing brake caliper

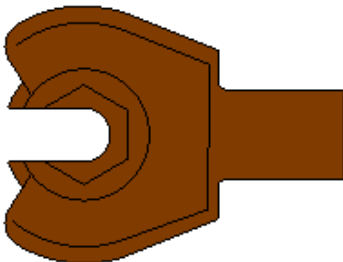
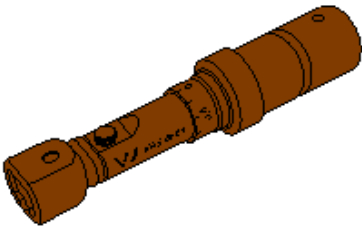
⇒ [“2.3.1 Removing and installing brake caliper, single tyres”, page 58](#)

⇒ [“2.3.2 Removing and installing brake caliper, twin tyres”, page 61](#)

2.3.1 Removing and installing brake caliper, single tyres



Special tools and workshop equipment required

V.A.G 1331 	VAG 1410/6 
VAG 1410/7 	VAS 6854 
VAG 1869/2 	VAG 1410/8  W45-10001

- ◆ Torque wrench - V.A.G 1331-
- ◆ Ring insert tool, open, AF 11 mm - V.A.G 1410/6-
- ◆ Universal joint - V.A.G 1410/7-
- ◆ Torque wrench - VAS 6854-
- ◆ Brake pedal depressor - V.A.G 1869/2-
- ◆ Assembly tool - V.A.G 1410/8-

Removing

- Remove rear wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .

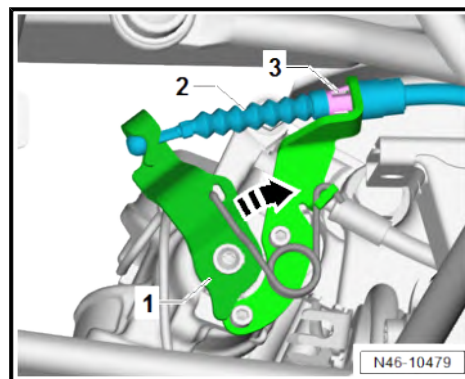


Note

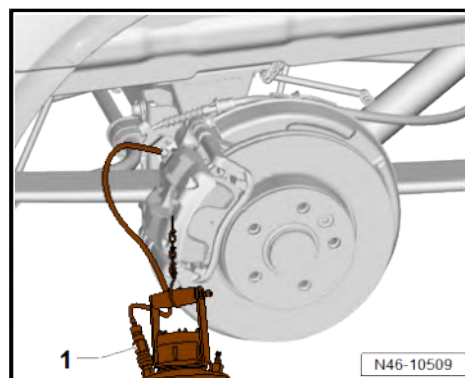
Mark brake pads or linings when removing if they are to be reused. Fit in same position when installing, or braking will be uneven.



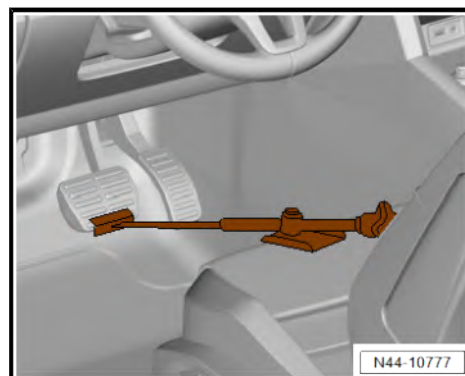
- Push actuating lever -1- in direction of -arrow-.
- Disengage handbrake cable -2- from actuating lever -1-.
- Press together both locking lugs -3- on cable support bracket using pliers.
- Remove handbrake cable -2-.



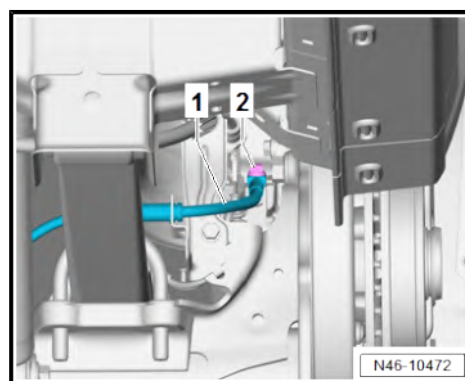
- Connect bleeder hose of bleeder bottle -1- to bleeder valve of brake caliper and open bleeder valve.



- Apply brake pedal depressor - V.A.G 1869/2- .
- Close bleeder screw and remove bleeder bottle.



- Unscrew banjo bolt -2- from brake caliper.
- Remove brake hose -1-.



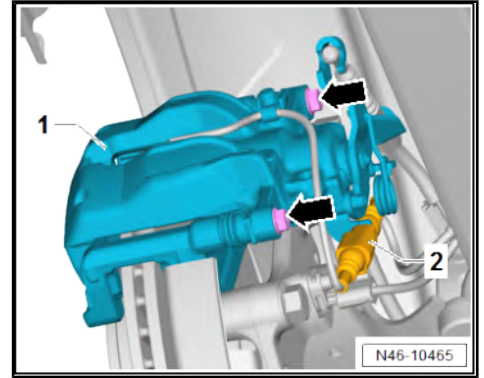


- Separate electrical connector -2- for brake pad wear indicator.
- Unscrew bolts -arrows- from brake caliper -1-.
- Pull off brake caliper -1- from brake carrier.

Installing

Install in the reverse order of removal, observing the following:

- The pistons are pressed back.
- The brake pads are seated on the brake carrier.
- Fit brake caliper on brake carrier.
- Screw brake hose onto brake caliper.
- Remove brake pedal depressor - V.A.G 1869/2- .
- Bleed brake system ➔ [page 128](#) .
- Adjusting parking brake ➔ [page 77](#) .



Note

- ◆ *Firmly depress brake pedal several times with vehicle stationary so that the brake pads/linings are properly seated in their normal operating position.*
- ◆ *Check brake fluid level.*

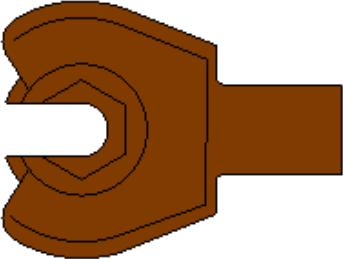
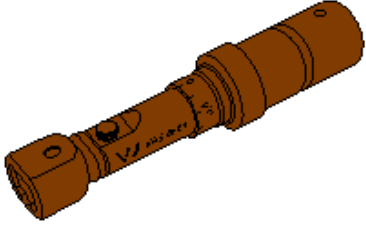
Specified torques

- ◆ ➔ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.3.2 Removing and installing brake caliper, twin tyres



Special tools and workshop equipment required

V.A.G 1331 	VAG 1410/6 
VAG 1410/7 	VA S 6854 
VAG 1869/2 	VAG 1410/8  W45-10001

- ◆ Torque wrench - V.A.G 1331-
- ◆ Ring insert tool, open, AF 11 mm - V.A.G 1410/6-
- ◆ Universal joint - V.A.G 1410/7-
- ◆ Torque wrench - VAS 6854-
- ◆ Brake pedal depressor - V.A.G 1869/2-
- ◆ Assembly tool - V.A.G 1410/8-

Removing

- Remove rear wheel ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .

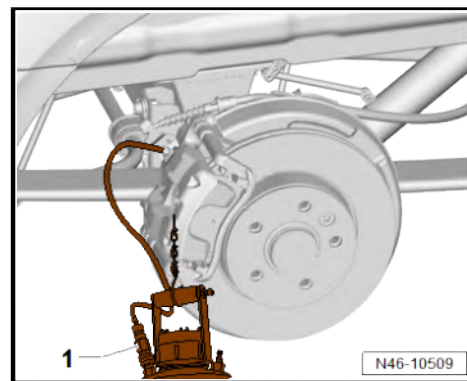


Note

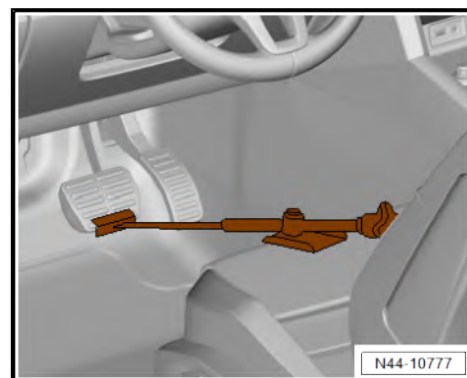
Mark brake pads or linings when removing if they are to be reused. Fit in same position when installing, or braking will be uneven.



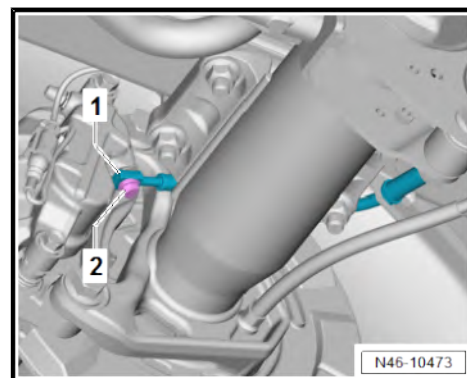
- Connect bleeder hose of bleeder bottle -1- to bleeder valve of brake caliper and open bleeder valve.



- Apply brake pedal depressor - V.A.G 1869/2- .
- Close bleeder screw and remove bleeder bottle.



- Unscrew banjo bolt -2- from brake caliper.
- Remove brake hose -1-.



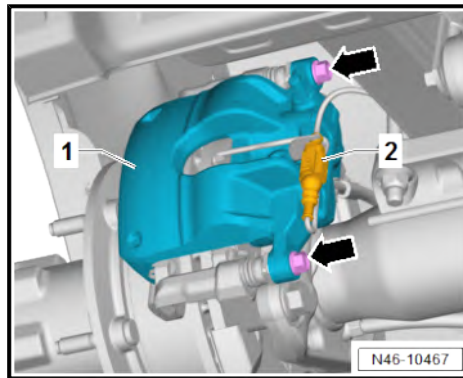


- Separate electrical connector -2- for brake pad wear indicator.
- Unscrew bolts -arrows- from brake caliper -1-.
- Pull brake caliper from brake carrier.

Installing

Install in the reverse order of removal, observing the following:

- The pistons are pressed back.
- The brake pads are seated on the brake carrier.
- Fit brake caliper on brake carrier.
- Screw brake hose onto brake caliper.
- Remove brake pedal depressor - V.A.G 1869/2- .
- Bleed brake system ➔ [page 128](#) .



Note

- ◆ *Firmly depress brake pedal several times with vehicle stationary so that the brake pads/linings are properly seated in their normal operating position.*
- ◆ *Check brake fluid level.*

Specified torques

- ◆ ➔ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.4 Removing and installing brake carrier

➔ [“2.4.1 Removing and installing brake carrier, single tyres”, page 64](#)

➔ [“2.4.2 Removing and installing brake carrier, twin tyres”, page 65](#)

2.4.1 Removing and installing brake carrier, single tyres

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

- Remove brake pads ➔ [page 53](#) .



- Unscrew bolts -arrows- of brake carrier -1-.
- Remove backplate -1-.

Installing

Install in the reverse order of removal, observing the following:

- Thoroughly clean contact surfaces for brake pads on brake carrier and remove any corrosion.



Note

- ◆ *Every time after changing pads, firmly depress brake pedal several times with vehicle stationary. This process ensures that brake pads are properly seated in their normal operating position.*
- ◆ *Check the brake fluid level in the brake fluid reservoir every time after the brake pads have been changed.*

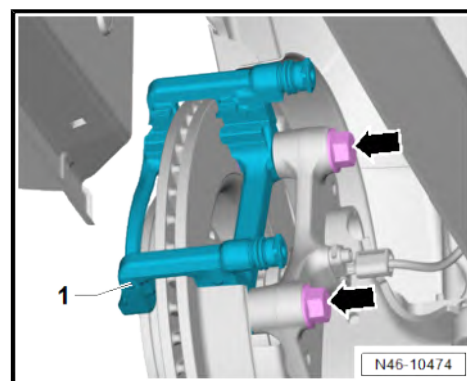
Specified torques

- ◆ ⇒ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.4.2 Removing and installing brake carrier, twin tyres

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

- Remove brake pads ⇒ [page 56](#) .



- Unscrew bolts -arrows- of brake carrier -1-.
- Remove backplate -1-.

Installing

Install in the reverse order of removal, observing the following:

- Thoroughly clean contact surfaces for brake pads on brake carrier and remove any corrosion.



Note

- ◆ *Every time after changing pads, firmly depress brake pedal several times with vehicle stationary. This process ensures that brake pads are properly seated in their normal operating position.*
- ◆ *Check the brake fluid level in the brake fluid reservoir every time after the brake pads have been changed.*

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.5 Removing and installing brake disc

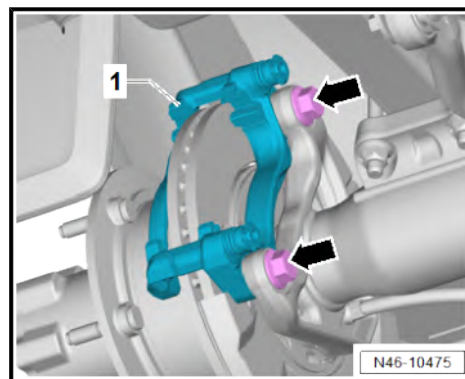
⇒ [“2.5.1 Removing and installing brake disc, single tyres”, page 66](#)

⇒ [“2.5.2 Removing and installing brake disc, twin tyres”, page 67](#)

2.5.1 Removing and installing brake disc, single tyres

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove backplate ➔ [page 64](#) .
- Unscrew bolt -2- from brake disc -1-.
- Remove brake disc -1-.

Cleaning

⚠ WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- Never blow out the brake system with compressed air.

- Thoroughly clean contact surfaces on wheel hub, remove any corrosion.

Installing

Install in the reverse order of removal, observing the following:

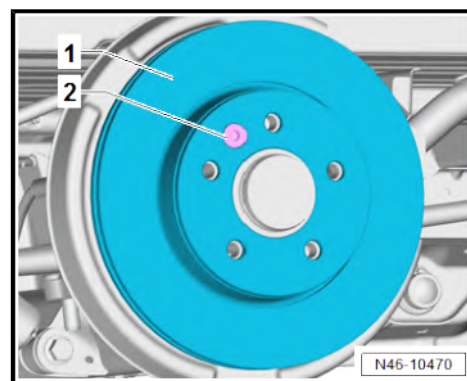
Specified torques

- ◆ ➔ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.5.2 Removing and installing brake disc, twin tyres

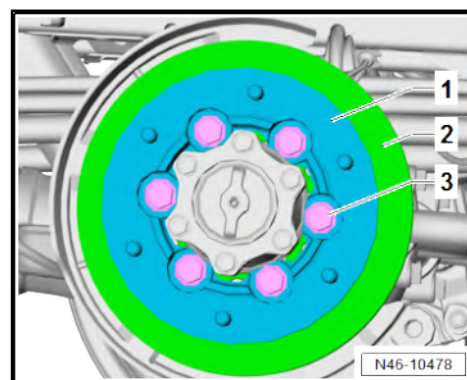
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

- Remove backplate ➔ [page 65](#) .
- Unscrew bolts -3-.
- Remove wheel flange -1- from brake disc -2-.
- Reset drum brake ➔ [page 71](#) .





- Remove brake disc -1-.

Cleaning

WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- Never blow out the brake system with compressed air.

- Thoroughly clean contact surfaces on wheel hub, remove any corrosion.
- Clean sender wheel of speed sensor.

Installing

Install in the reverse order of removal, observing the following:

- Adjust basic setting/clearance of drum brake ⇒ [page 71](#) .

Procedure for renewing brake discs

- During a road test, proceed as follows to run in the brake shoes:
 - ◆ at a speed of 30 km/h, hold handbrake lever for approx. 10 seconds in second detent position;
 - ◆ allow brake to cool for about 2 minutes;
 - ◆ at a speed of 30 km/h, hold handbrake lever for approx. 20 seconds in second detent position;

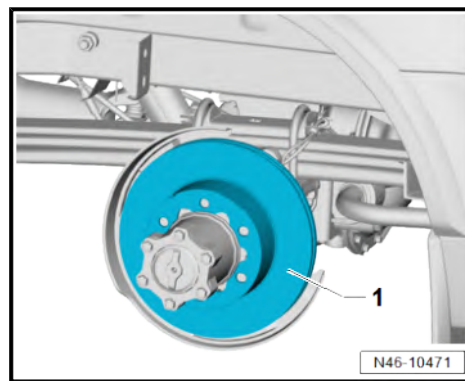
Specified torques

- ◆ ⇒ [“2.1 Assembly overview - rear brakes”, page 50](#)

2.6 Removing and installing brake shoes

Special tools and workshop equipment required

- ◆ Hook - 3438-

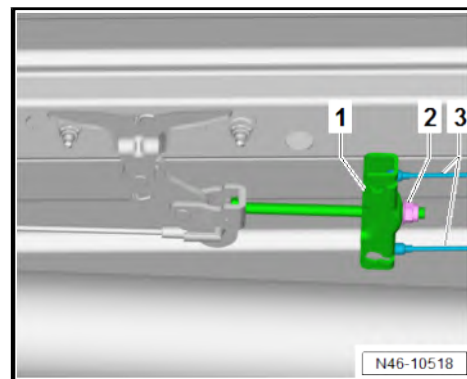


- ◆ Lubricating paste ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

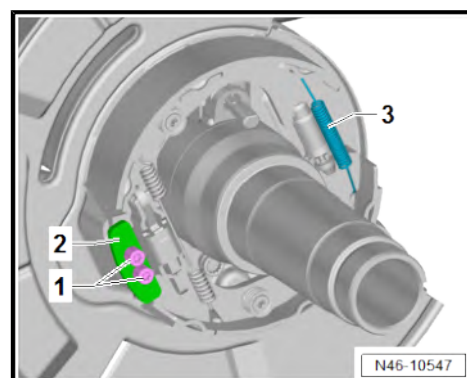


Removing

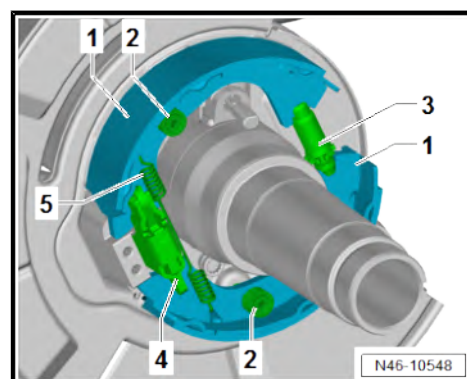
- Remove wheel bearing housing ⇒ Running gear, axles, steering; Rep. gr. 42 ; Wheel bearing, trailing arm; Removing and installing wheel bearing housing .
- Loosen nut -2-.
- Detach brake cable -3- from compensation element -1-.
- Mark brake shoes if they are to be reused.



- Unscrew bolts -1-. At the same time, detach locking plate -2-.
- Release compression spring -3- with hook - 3438- .



- Release retaining springs -2-. At the same time, detach retaining springs -2-.
- Remove adjuster -3-.
- Remove brake shoes -1- together with expander -4- and extension spring -5-.
- Detach brake cable from expander -4-.



Cleaning

WARNING

Health hazard due to poisonous dust from brake system.
Risk of irreversibly deposited dust particles in the lungs. Risk of respiratory health problems.

- Never blow out the brake system with compressed air.

- Clean contact surfaces on brake carrier and brake shoes using brake and clutch cleaner, remove corrosion.
- Clean sender wheel of speed sensor with brake cleaner .

Installing

Install in the reverse order of removal, observing the following:



Note

Ensure that no grease comes into contact with friction surfaces of brake shoes and brake disc.



- Grease contact surfaces -arrows- on brake carrier -1- with lubricating paste ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



Note

If the brake shoes shall be removed, observe correct installation position of the extension and compression springs.

- Insert extension spring -5- into brake shoes -1-.
- Spring coil of extension spring -5- must be on the inside.
- Tilt brake shoes -1-. At the same time, insert expander -4-.
- Insert adjuster -3-.
- Installation position towards service hole

Adjuster -3- on left side of vehicle

Adjuster -3- on left side of vehicle has a left-hand thread.

Adjuster -3- on right side of vehicle

Adjuster -3- on right side of vehicle has a right-hand thread.

Continued for all vehicles

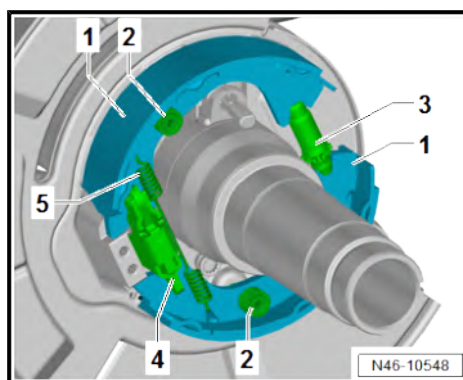
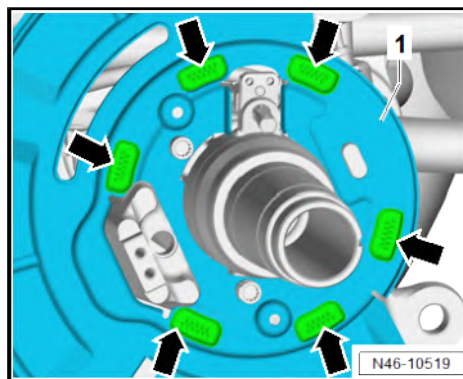
- Fit compression springs in brake shoes -1-.
- Fit pre-assembled set of brake shoes together with retaining springs -2- to carrier plate.
- Attach brake cable through opening in expander -4-.
- Detach rear wheels ⇒ Running gear, axles, steering; Rep. gr. 44 ; Wheels, tyres; Wheel change .
- Adjust basic setting/clearance of drum brake ⇒ [page 71](#) .
- Adjusting parking brake ⇒ [page 78](#) .

Procedure for renewing brake shoes

- During a road test, proceed as follows to run in the brake shoes:
 - ♦ at a speed of 30 km/h, hold handbrake lever for approx. 10 seconds in second detent position;
 - ♦ allow brake to cool for about 2 minutes;
 - ♦ at a speed of 30 km/h, hold handbrake lever for approx. 20 seconds in second detent position;

Specified torques

- ♦ Wheel bearing, trailing arm; Assembly overview - wheel bearing ⇒ Running gear, axles, steering; Rep. gr. 42 ; Wheel bearing, trailing arm; Assembly overview - wheel bearing



Component	Specified torque
Securing plate bolts	8 Nm



2.7 Adjusting basic setting/clearance of drum brake

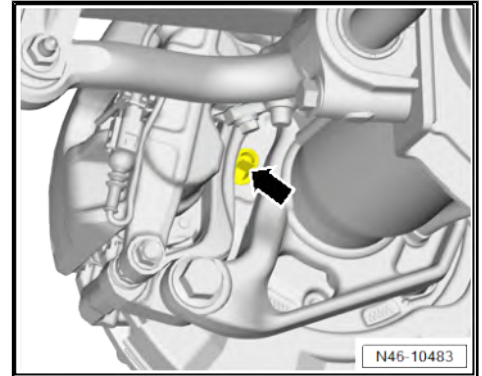


Note

The basic setting may only be performed with the rear brake cables released.

Procedure

- Remove cap -arrow-.



- Insert screwdriver -1- through maintenance hole.

To increase clearance

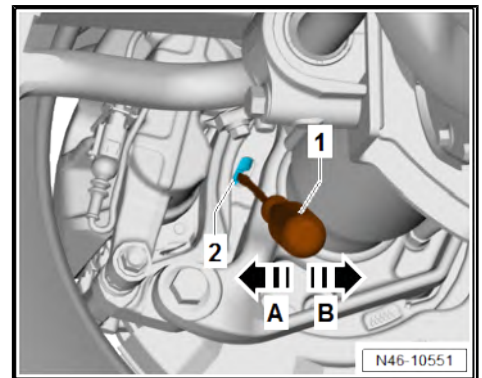
- Turn screwdriver -1- in direction of -arrow A-.

To reduce clearance

- Turn screwdriver -1- in direction of -arrow B-.

Adjusting

- Turn screwdriver -1- in direction of -arrow B- until wheel on adjuster -2- can no longer be turned.
- Turn by 8 teeth in direction of -arrow A- using screwdriver -1-.



3 Parking brake

⇒ "3.1 Assembly overview - parking brake", page 72

⇒ “3.2 Removing and installing handbrake lever”, page 76

⇒ **“3.3 Adjusting parking brake”, page 77**

⇒ **“3.4 Removing and installing front brake cable”, page 80**

⇒ **“3.5 Removing and installing front brake cable, e-Crafter and e-TGE”, page 81**

⇒ “3.6 Removing and installing rear brake cable”, page 83

3.1 Assembly overview - parking brake

⇒ “3.1.1 Assembly overview - parking brake, single tyres”, page 72

⇒ “3.1.2 Assembly overview - parking brake; super single wheels, twin tyres”, page 74

⇒ “3.1.3 Assembly overview - parking brake, e-Crafter and e-TGE”, page 75

3.1.1 Assembly overview - parking brake, single tyres

1 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

2 - Handbrake lever boot

- ❑ Removing and installing
⇒ General body repairs,
interior; Rep. gr. 68 ;
Compartments/covers;
Removing and installing
handbrake lever trim

3 - Nut

- ☐
- Renew after removal

4 - Parking brake lever

- ❑ Removing and installing
⇒ page 76

5 - Handbrake warning switch
- F9-

6 - Bracket

-  on driver seat

7 - Bracket

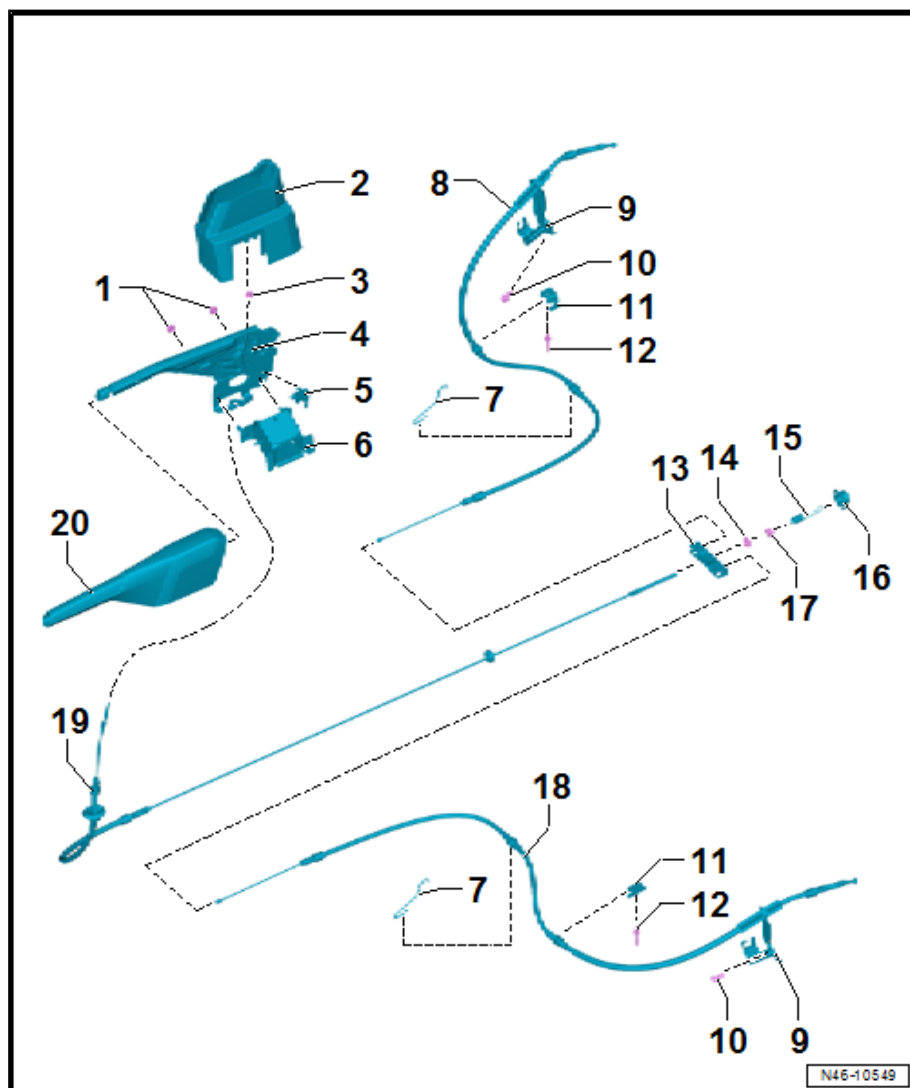
- ☐ For brake cable
- ☐ Qty. 2

8 - Brake cable

- ☐ Rear right
- ☐ Renew only in pairs
- ☐ Removing and installing
⇒ page 83

9 - Bracket

- ☐ For brake cable
- ☐ Qty. 2





10 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

11 - Support brackets

- ☐ Qty. 2

12 - Pop rivets

- ☐ Qty. 2

13 - Compensation element

14 - Filler

- ☐ Only on vehicles up to week 47/2017

15 - Extension spring

16 - Retaining clip

17 - Nut

- ☐ Qty. 2 depending on equipment level
- ☐ Renew after removal

18 - Brake cable

- ☐ Rear left
- ☐ Renew only in pairs
- ☐ Removing and installing ⇒ [page 83](#)

19 - Pull cable

- ☐ Front
- ☐ Removing and installing ⇒ [page 80](#)

20 - Handbrake lever boot

- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim

3.1.2 Assembly overview - parking brake; super single wheels, twin tyres

1 - Handbrake lever boot

- ☐ Removing and installing
⇒ General body repairs,
interior; Rep. gr. 68 ;
Compartments/covers;
Removing and installing
handbrake lever trim

2 - Nut

- ☐ Renew after removal

3 - Parking brake lever

- ☐ Removing and installing
⇒ [page 76](#)

4 - Handbrake warning switch - F9-

5 - Bracket

- ☐ on driver seat

6 - Bracket

- ☐ Centre
- ☐ For brake cable

7 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

8 - Brake cable

- ☐ Rear right
- ☐ Renew only in pairs
- ☐ Removing and installing
⇒ [page 83](#)

9 - Bracket

- ☐ For right brake cable
- ☐ Qty. 2

10 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

11 - Bracket

- ☐ For left brake cable
- ☐ Qty. 2

12 - Brake cable

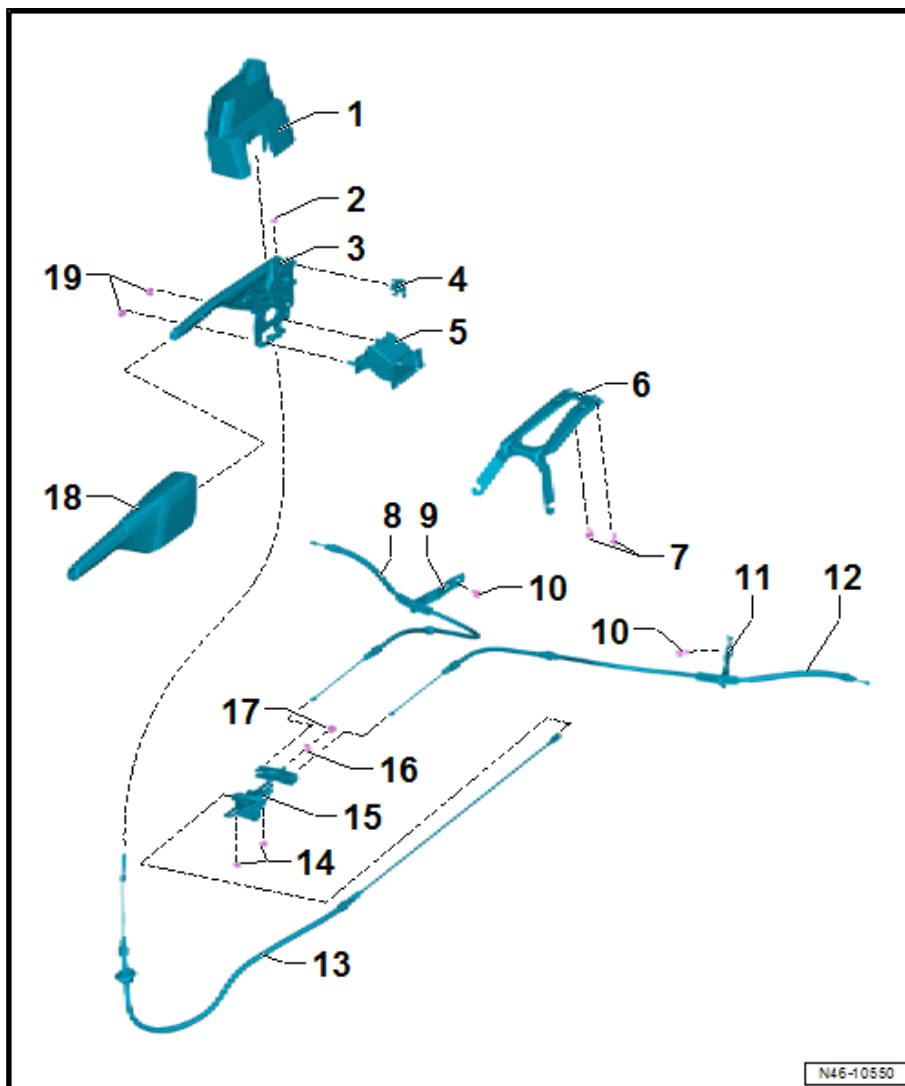
- ☐ Rear left
- ☐ Renew only in pairs
- ☐ Removing and installing ⇒ [page 83](#)

13 - Pull cable

- ☐ Front
- ☐ Removing and installing ⇒ [page 80](#)

14 - Nuts

- ☐ Renew after removal
- ☐ 25 Nm





15 - Housing

- ☐ Pull cable/brake cable converter

16 - Filler

- ☐ Only on vehicles up to week 47/2017

17 - Nut

- ☐ Qty. 2 depending on equipment level
- ☐ Renew after removal

18 - Handbrake lever boot

- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim

19 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

3.1.3 Assembly overview - parking brake, e-Crafter and e-TGE

1 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

2 - Shield

- ☐ For parking brake lever
- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim

3 - Shield

- ☐ For parking brake lever
- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim

4 - Nut

- ☐ Renew after removal

5 - Parking brake lever

- ☐ Removing and installing ⇒ [page 76](#)

6 - Handbrake warning switch - F9-

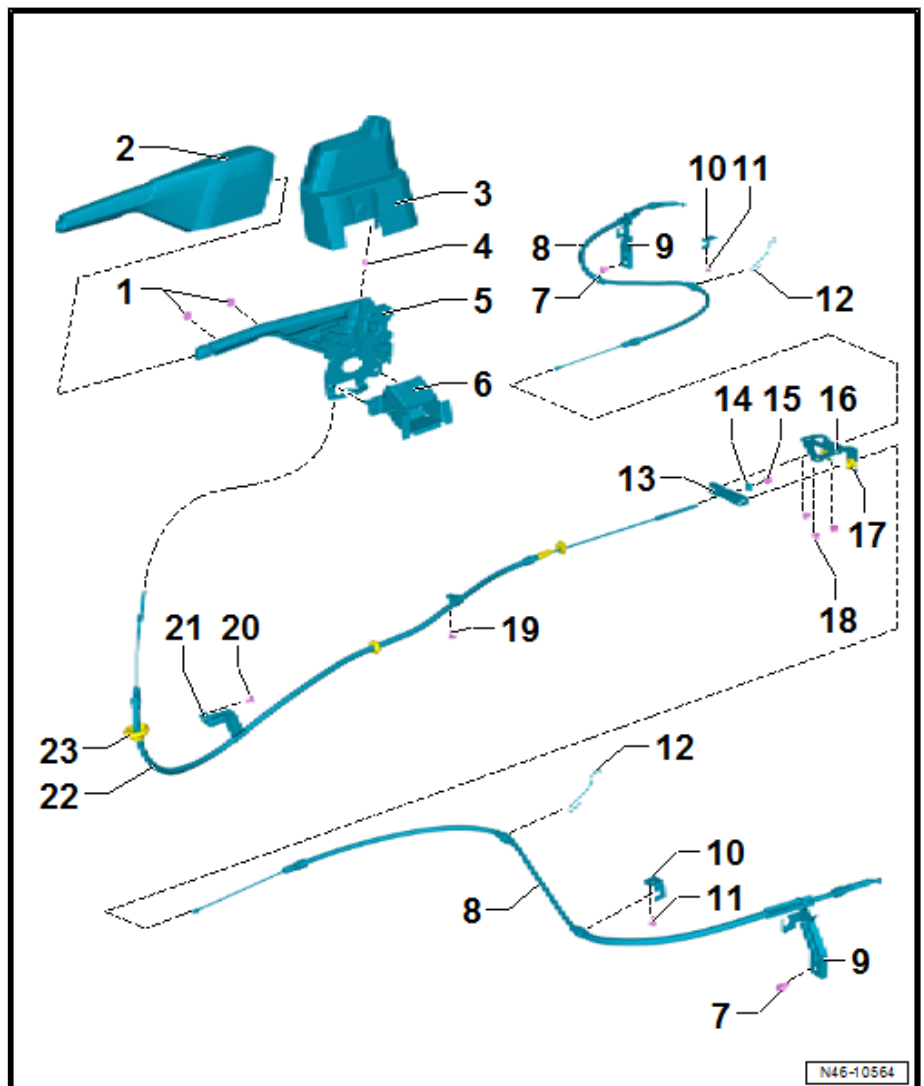
- ☐ With holder

7 - Nut

- ☐ 20 Nm

8 - Brake cables

- ☐ Rear
- ☐ Qty. 2
- ☐ Renew only in pairs
- ☐ Removing and installing ⇒ [page 86](#)





9 - Bracket

- ☐ For brake cable
- ☐ Qty. 2

10 - Support brackets

- ☐ Qty. 2

11 - Pop rivets

- ☐ Qty. 2

12 - Bracket

- ☐ For brake cable
- ☐ Qty. 2

13 - Compensation element

14 - Nut

- ☐ Qty. 2 depending on equipment level
- ☐ Renew after removal

15 - Extension spring

16 - Trestle

17 - Bushes

- ☐ Qty. 2

18 - Nuts

- ☐ Qty. 3
- ☐ 20 Nm

19 - Nut

- ☐ Renew after removal
- ☐ 8 Nm

20 - Pop rivet

21 - Bracket

22 - Brake cable

- ☐ Front
- ☐ Removing and installing ⇒ [page 81](#)

23 - Bush

3.2 Removing and installing handbrake lever

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove handbrake lever trim ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim .
- Unscrew nut -2-.
- Press together locking lugs -arrow- using pliers.
- Remove brake cable -1-.
- Unscrew nuts -4-.
- Separate electrical connector -5-.
- Remove parking brake lever -3-.

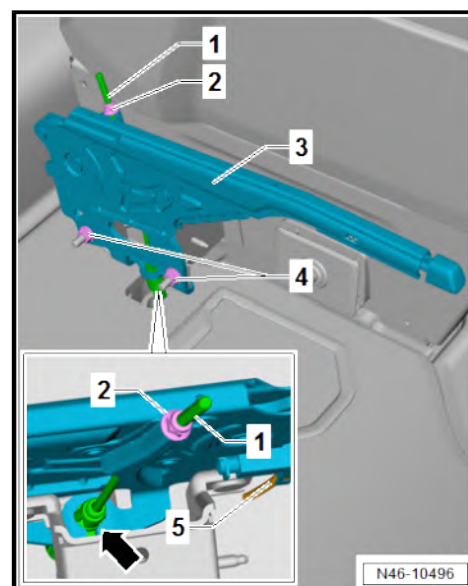
Installing

Install in the reverse order of removal, observing the following:

- Adjusting parking brake ⇒ [page 77](#) .

Specified torques

- ♦ ⇒ [“3.1 Assembly overview - parking brake”, page 72](#)



3.3 Adjusting parking brake

⇒ [“3.3.1 Adjusting parking brake, single tyres”, page 77](#)

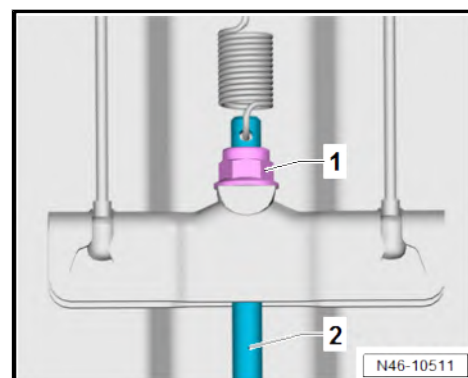
⇒ [“3.3.2 Adjusting parking brake; super single wheels, twin tyres”, page 78](#)

⇒ [“3.3.3 Adjusting parking brake, e-Crafter and e-TGE”, page 79](#)

3.3.1 Adjusting parking brake, single tyres

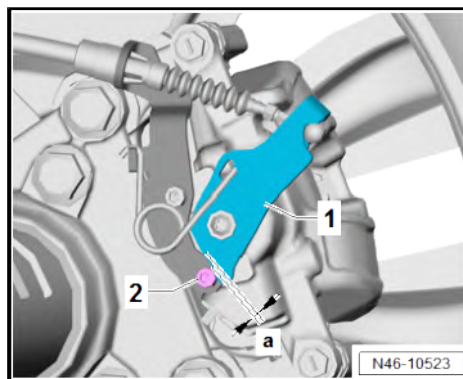
Procedure

- Raise vehicle ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Raising vehicle .
- Foot brake must be functional and bled.
- The handbrake cables on the brake caliper must be disengaged for the actuating levers on the caliper to be in the zero position.
- Release handbrake.
- Depress brake pedal 3 times.
- Attach handbrake cables to brake caliper. If necessary, adjust handbrake cable -2- with one or several nuts -1- so that it does not sag.
- Operate handbrake lever 5 times (to stretch cables).





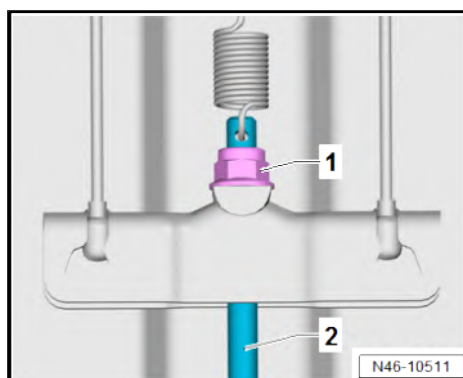
- Adjust handbrake via adjuster so that:
- Wheels can no longer be turned by hand in the first detent.
- With handbrake lever released, distance -a- between operating lever -1- and bolt -2- of cable support bracket on left and right brake calipers is max. 1 mm on each side.
- Release handbrake.



- Check that both wheels turn freely. If required, unscrew nut or nuts -1- from brake cable -2-.

If distance -a- is considerably larger than 1 mm, and wheels turn freely:

- ◆ Detach brake cables.
- ◆ Adjusting piston hydraulically by depressing brake pedal
- Attach brake cables.
- Repeat adjustment procedure.



3.3.2 Adjusting parking brake; super single wheels, twin tyres

Procedure

- Raise vehicle ⇒ Maintenance ; Booklet 10.3 ; Descriptions of work; Raising vehicle .



Note

- ◆ *Handbrake cables must not sag (hanging cable).*
- ◆ *Make sure to adhere to the sequence of the work steps given below.*
- Adjust basic setting/clearance of drum brake ⇒ [page 71](#) .
- Operate handbrake lever 5 times (to stretch cables).



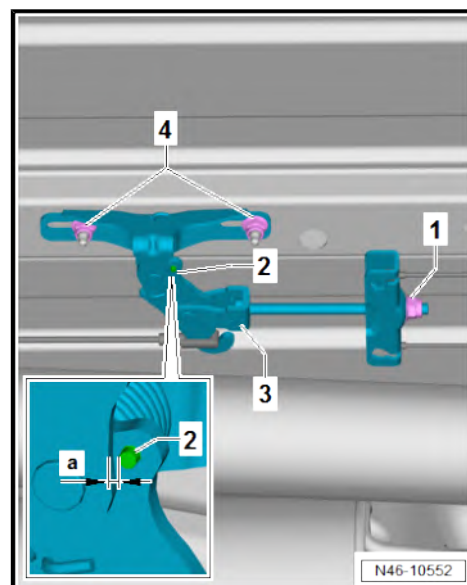
- Adjust handbrake via one or several nuts -1- so that:
 - ◆ with handbrake lever released, wheels can be turned by hand;
 - ◆ wheels start to grind as brake lever is being moved from released position to first detent position;
 - ◆ wheels can no longer be turned by hand in first detent position;
 - ◆ with handbrake lever released, distance -a- between stop -2- and housing of intermediate converter -3- is approx. 2 mm.
- Check if the wheels can be turned easily by hand when the handbrake lever is released. If required, unscrew nut or nuts -1-.

Distance -a- larger or smaller than 2 mm

- Loosen nuts -4-.
- Shift housing of intermediate converter -3- in corresponding direction until distance -a- of approx. 2 mm is attained.

Specified torques

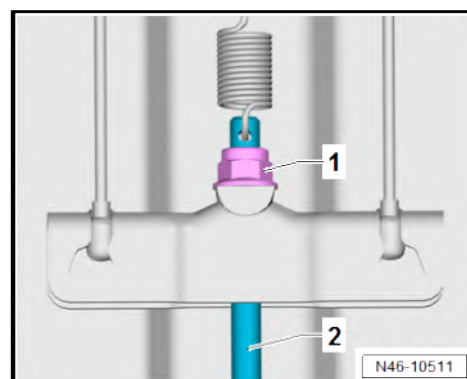
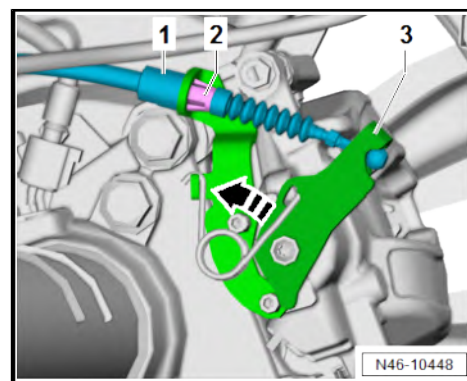
- ◆ ⇒ [“3.1 Assembly overview - parking brake”, page 72](#)



3.3.3 Adjusting parking brake, e-Crafter and e-TGE

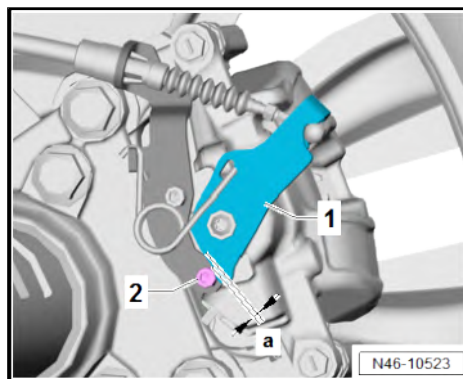
Procedure

- Push actuating lever -3- in direction of -arrow-.
- Remove brake cable -1- from actuating lever -3-.
- Foot brake must be functional and bled.
- The handbrake cables on the brake caliper must be disengaged for the actuating levers on the caliper to be in the zero position.
- Removing and installing underbody cladding on rear left and rear right ⇒ Rep. gr. 66 ; Exterior equipment; Underbody cladding; Removing and installing underbody cladding .
- Release handbrake.
- Depress brake pedal 3 times.
- Attach handbrake cables to brake caliper. If necessary, adjust handbrake cable -2- with one or several nuts -1- so that it does not sag.
- Operate handbrake lever 5 times (to stretch cables).





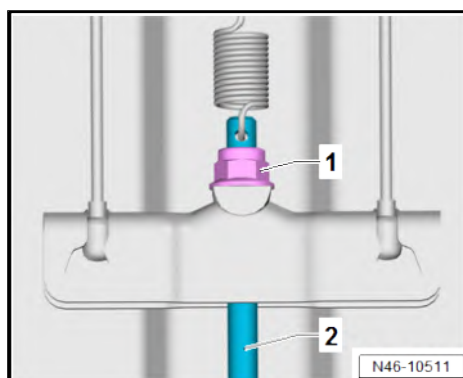
- Adjust handbrake via adjuster so that:
- Wheels can no longer be turned by hand in the first detent.
- With handbrake lever released, distance -a- between operating lever -1- and bolt -2- of cable support bracket on left and right brake calipers is max. 1 mm on each side.
- Release handbrake.



- Check that both wheels turn freely. If required, unscrew nut or nuts -1- from brake cable -2-.

If distance -a- is considerably larger than 1 mm, and wheels turn freely:

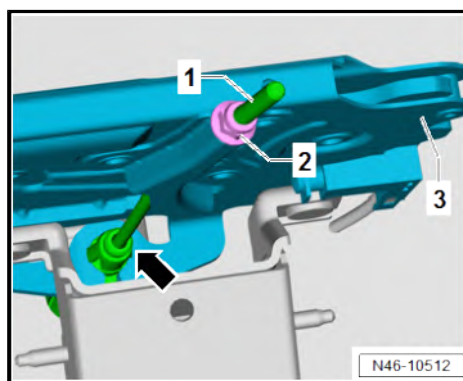
- ◆ Detach brake cables.
- ◆ Adjusting piston hydraulically by depressing brake pedal
- Attach brake cables.
- Repeat adjustment procedure.



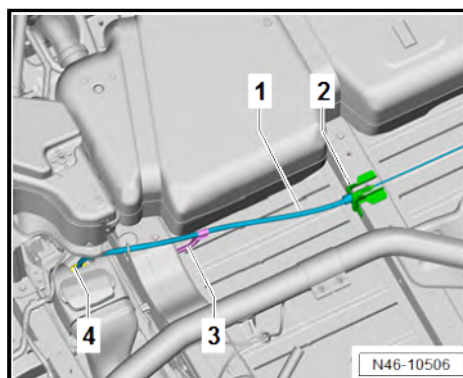
3.4 Removing and installing front brake cable

Removing

- Remove handbrake lever trim ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim .
- Unscrew nut -2-.
- Press together locking lugs -arrow- using pliers.
- Remove brake cable -1- from handbrake lever -3-.



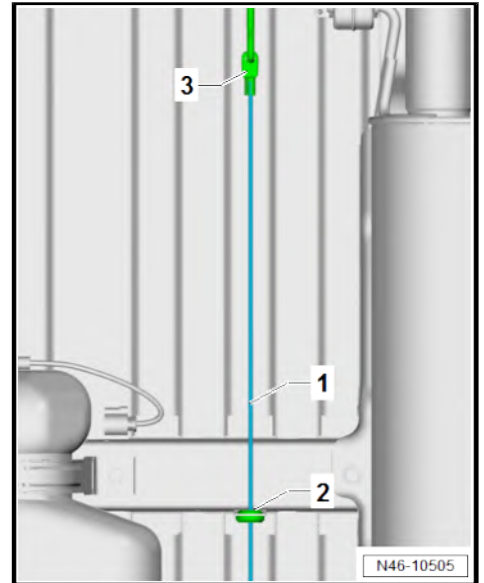
- Guide brake cable -1- through opening -4- in underbody.
- Remove brake cable -1- from bracket -3-.
- Press together locking lugs -2- using pliers.





Vehicles with long wheelbase

- Remove brake cable -1- from support bracket -2-.
- Remove brake cable -1- from cable eye -3-.



Continued for all vehicles

- Remove brake cable -1- from support bracket -2-.
- Remove return spring -5- from brake cable -1-.
- Remove brake cables -4- from compensation element -3-.

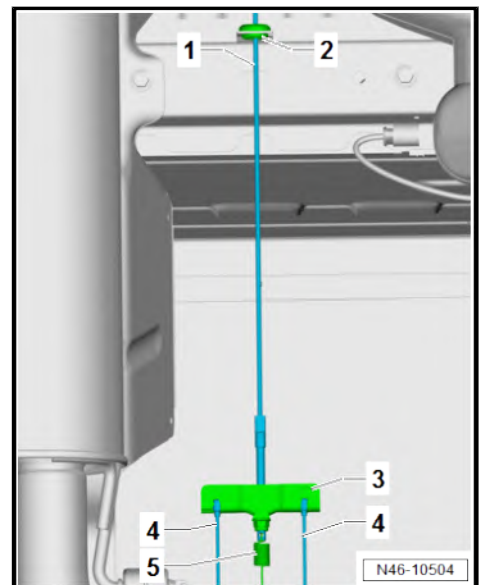
Installing

Install in the reverse order of removal, observing the following:

- Adjusting parking brake ➔ [page 77](#) .

Specified torques

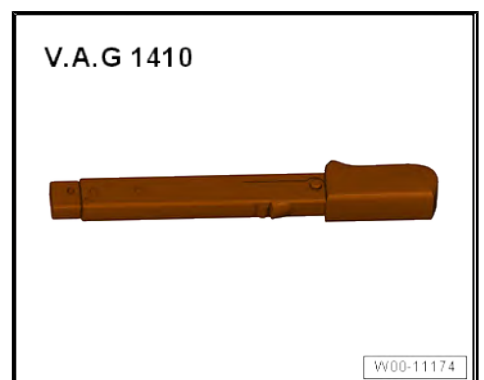
- ◆ ➔ [“3.1 Assembly overview - parking brake”, page 72](#)



3.5 Removing and installing front brake cable, e-Crafter and e-TGE

Special tools and workshop equipment required

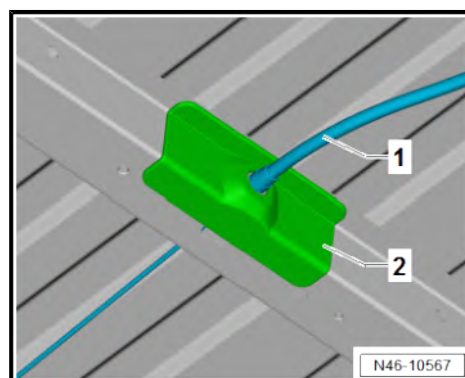
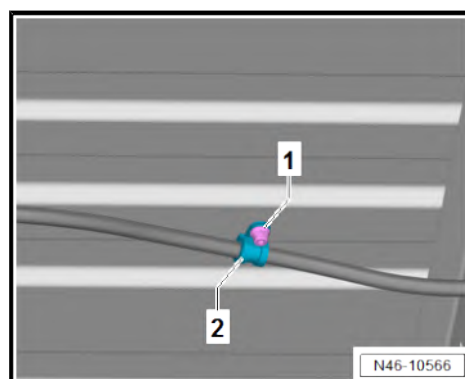
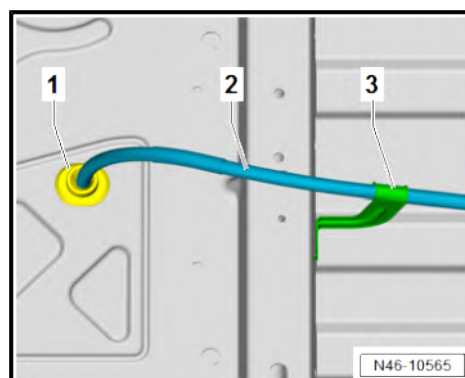
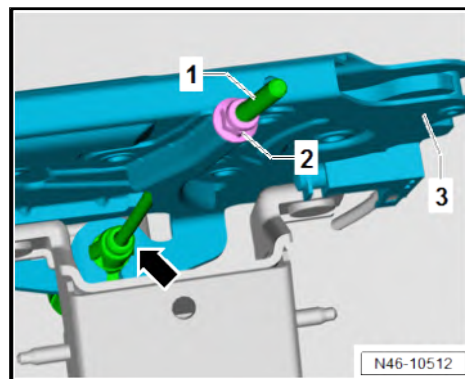
- ◆ Torque wrench - V.A.G 1410-





Removing

- Remove high-voltage battery ⇒ Rep. gr. 93 ; Electric drive; High-voltage battery unit; Removing and installing high-voltage battery .
- Remove handbrake lever trim ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing handbrake lever trim .
- Unscrew nut -2-.
- Press together locking lugs -arrow- using pliers. Remove brake cable -1- from handbrake lever -3- when doing this.
- Guide brake cable -2- through opening -1- in underbody.
- Detach brake cable -2- from bracket -3-.
- Unscrew nut -1-.
- Detach clamp -2- from underbody.
- Detach brake cable -1- from cable support bracket -2-.





- Remove return spring -5- from brake cable -1-.
- Remove brake cables -4- from compensation element -2-.
- Unscrew nut or nuts -3-.
- Remove compensation element -2- from brake cable -1-.

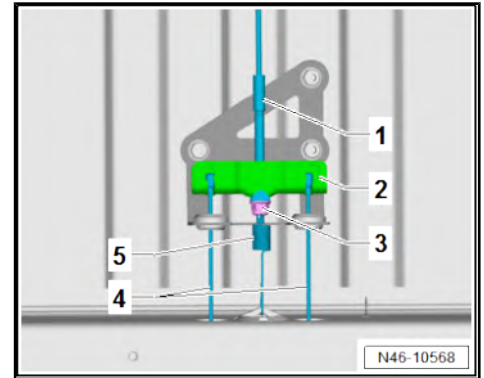
Installing

Install in the reverse order of removal, observing the following:

- Adjusting parking brake
⇒ [“3.3.3 Adjusting parking brake, e-Crafter and e-TGE”, page 79](#).

Torque settings

- ◆ ⇒ [“3.1.3 Assembly overview - parking brake, e-Crafter and e-TGE”, page 75](#)



3.6 Removing and installing rear brake cable

⇒ [“3.6.1 Removing and installing rear brake cable, single tyres”, page 83](#)

⇒ [“3.6.2 Removing and installing rear brake cable, super single wheel, twin tyres”, page 84](#)

⇒ [“3.6.3 Removing and installing rear brake cable, e-Crafter and e-TGE”, page 86](#)

3.6.1 Removing and installing rear brake cable, single tyres

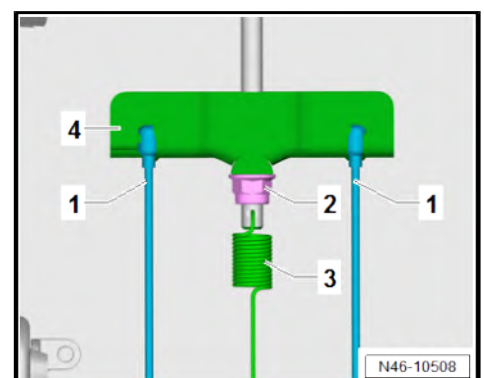
Special tools and workshop equipment required

- ◆ Tube - 3433-



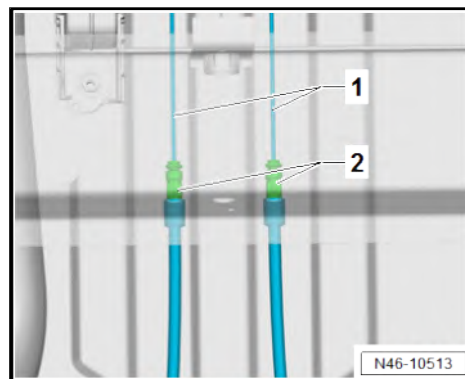
Removing

- Unhook return spring -3-.
- Loosen nut or nuts -2-.
- Remove brake cables -1- from compensation element -4-.

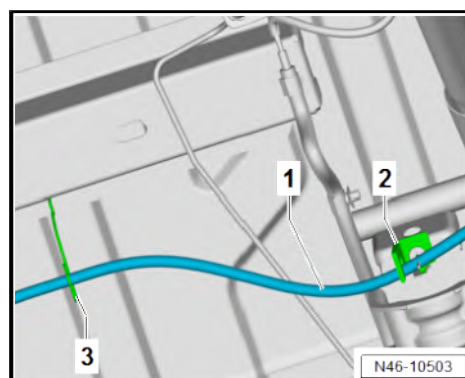




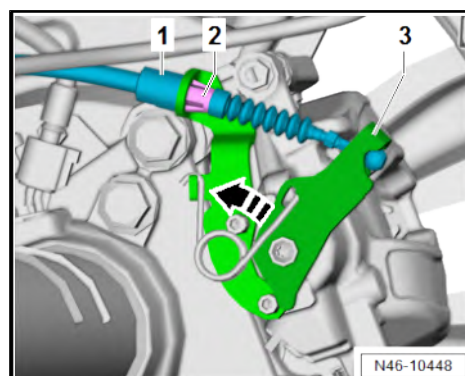
- Release retaining lugs -2- by pushing on tube - 3433- in direction of rear of vehicle. And simultaneously pull brake cable -1-.



- Unhook brake cable -1- from bracket -2- and -3- in underbody.



- Push actuating lever -3- in direction of -arrow-.
- Remove brake cable -1- from actuating lever -3-.
- Press together locking lugs -2- on cable support bracket using pliers.
- Remove brake cable -1- from cable support bracket.



- Remove brake cable -1- from guide -2-.

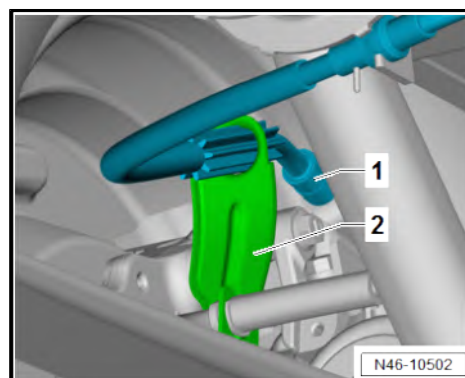
Installing

Install in the reverse order of removal, observing the following:

- Check secure engagement with pulling action.
- Adjusting parking brake ⇒ [page 77](#) .

Specified torques

- ♦ ⇒ [“3.1 Assembly overview - parking brake”, page 72](#)



3.6.2 Removing and installing rear brake cable, super single wheel, twin tyres

Special tools and workshop equipment required

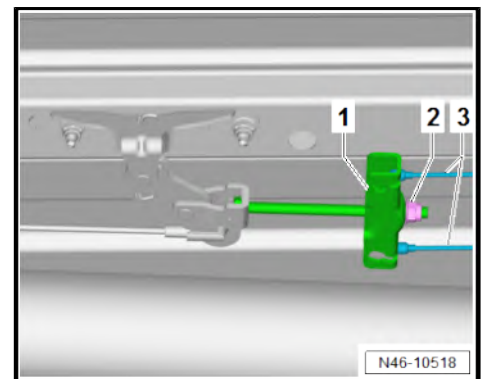


◆ Tube - 3433-

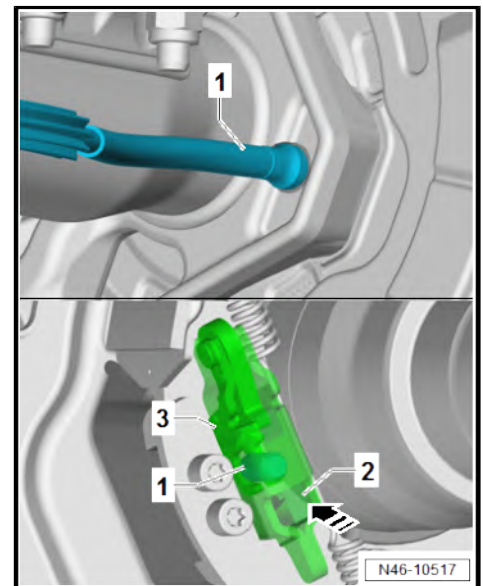


Removing

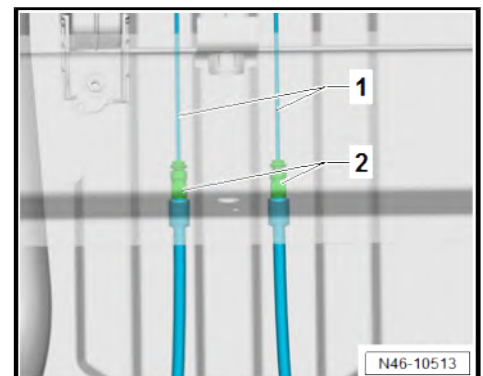
- Remove rear brake disc ➔ [page 66](#) .
- Loosen nut or nuts -2-.
- Detach brake cable -3- from compensation element -1-.



- Pull brake cable sleeve -1- out of hole in axle.
- Push brake cable -1- towards vehicle exterior. While doing this, use a screwdriver to lever nipple on brake cable -1- downwards against spring plate.
- Remove brake cable in direction of -arrow- from hole in axle.



- Release retaining lugs -2- by pushing on tube - 3433- in direction of rear of vehicle. And simultaneously pull brake cable -1-.





- Unclip brake cable -1- from bracket -2-.
- Pull out brake cable -1- from bracket -3-.

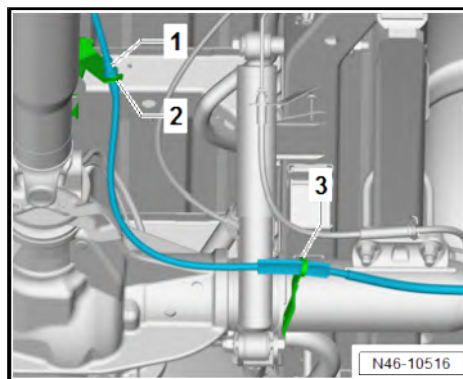
Installing

Install in the reverse order of removal, observing the following:

- Check secure engagement with pulling action.
- Adjusting parking brake ➔ [page 77](#) .

Specified torques

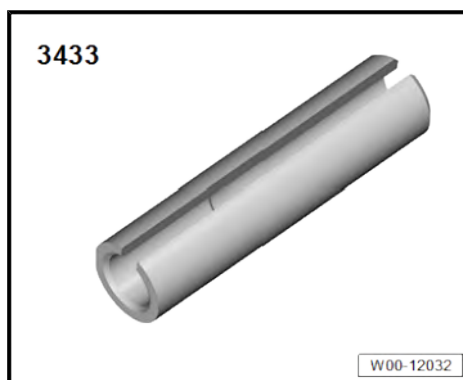
- ♦ ➔ [“2.1 Assembly overview - rear brakes”, page 50](#)



3.6.3 Removing and installing rear brake cable, e-Crafter and e-TGE

Special tools and workshop equipment required

- ♦ Tube - 3433-



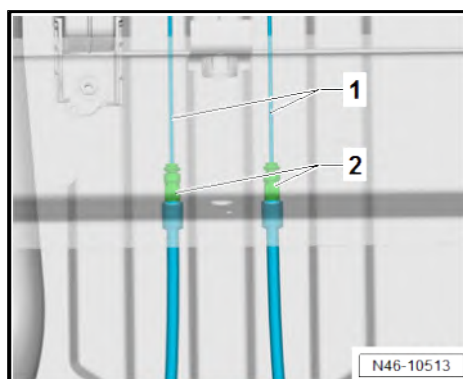
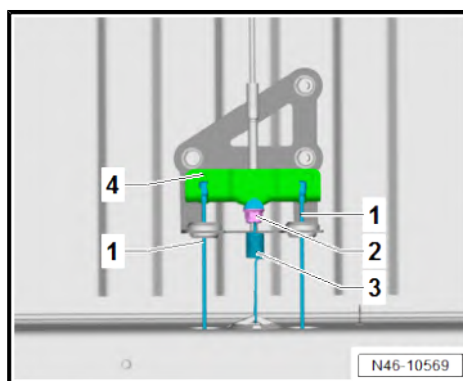
Removing

- Removing underbody cladding on rear left and right ➔ Rep. gr. 66 ; Exterior equipment; Underbody cladding; Removing and installing underbody cladding
- Unhook return spring -3-.
- Loosen nut or nuts -2-.
- Remove brake cables -1- from compensation element -4-.



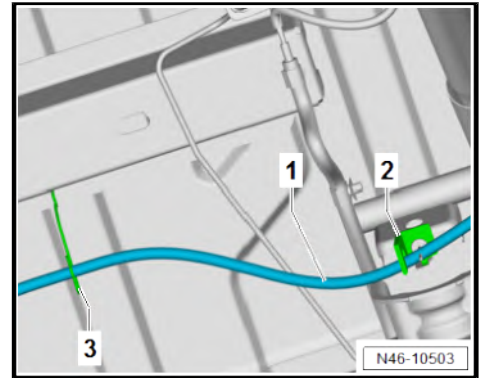
Note

- ♦ *The work steps are described for one brake cable only.*
- ♦ *Repeat the work steps for the other brake cable.*
- Release retaining lug -2- by pushing on tube - 3433- in direction of rear of vehicle. And simultaneously pull brake cable -1-.

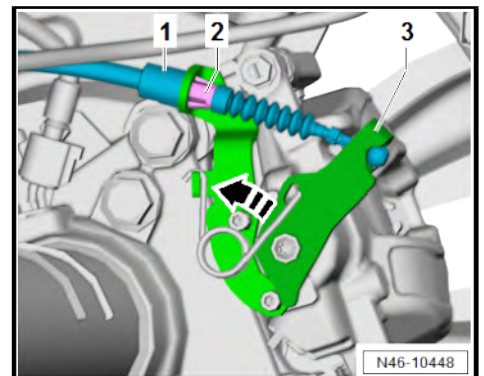




- Unhook brake cable -1- from bracket -2- and -3- in underbody.



- Push actuating lever -3- in direction of -arrow-.
- Remove brake cable -1- from actuating lever -3-.
- Press together locking lugs -2- on cable support bracket using pliers.
- Remove brake cable -1- from cable support bracket.



- Remove brake cable -1- from guide -2-.

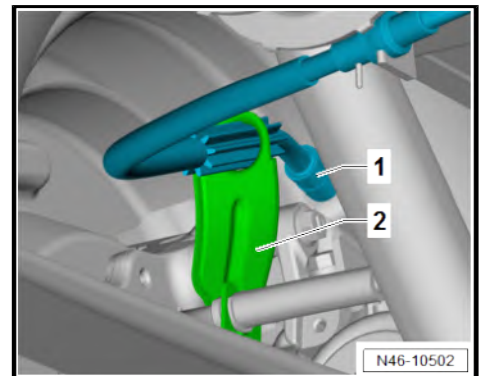
Installing

Install in the reverse order of removal, observing the following:

- Check secure engagement with pulling action.
- Adjusting parking brake
⇒ [“3.3.3 Adjusting parking brake, e-Crafter and e-TGE”, page 79](#).

Torque settings

- ◆ ⇒ [“3.1.3 Assembly overview - parking brake, e-Crafter and e-TGE”, page 75](#)





4 Brake pedal

⇒ [“4.1 Assembly overview - brake pedal”, page 88](#)

⇒ [“4.2 Removing and installing mounting bracket”, page 89](#)

⇒ [“4.3 Separating brake pedal from brake servo”, page 92](#)

⇒ [“4.4 Connecting brake pedal to brake servo”, page 93](#)

⇒ [“4.5 Removing and installing brake pedal”, page 93](#)

4.1 Assembly overview - brake pedal

1 - Bulkhead

2 - Plastic clip

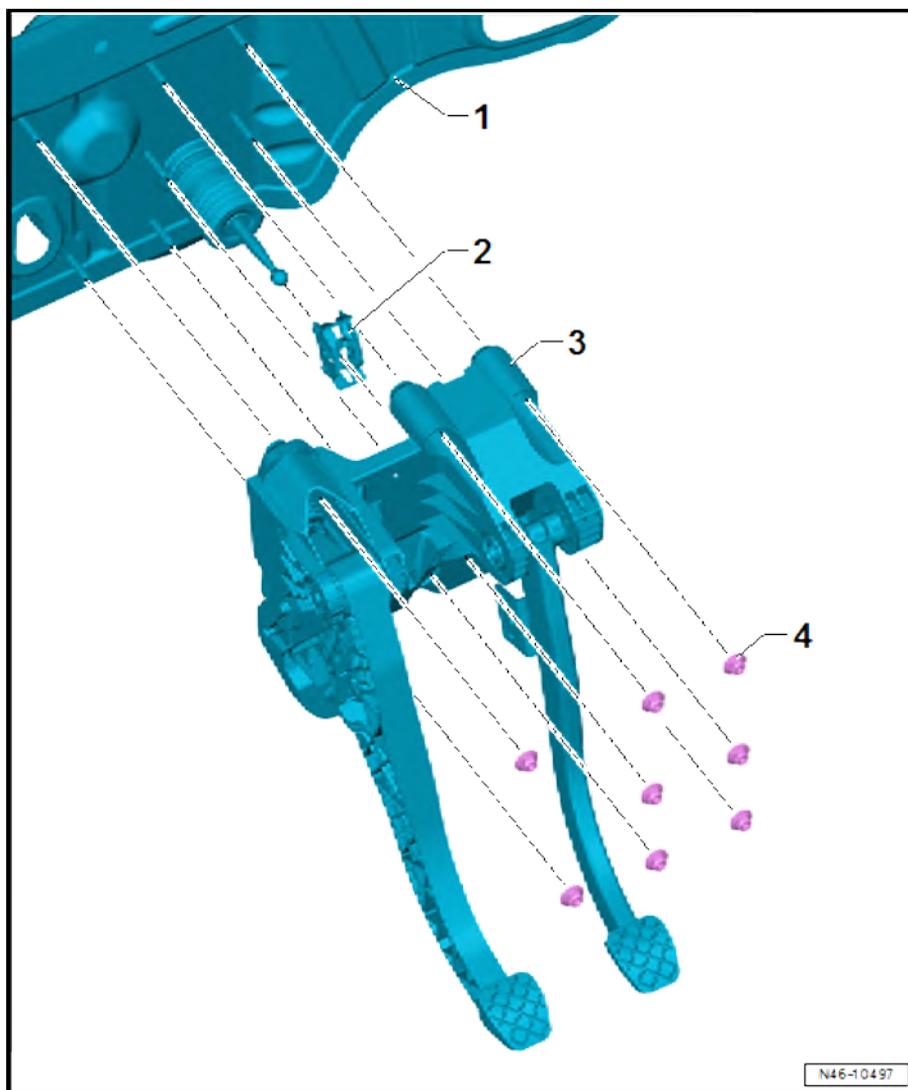
- ☐ Renew after removal

3 - Mounting bracket

- ☐ Removing and installing
⇒ [page 89](#)

4 - Nuts

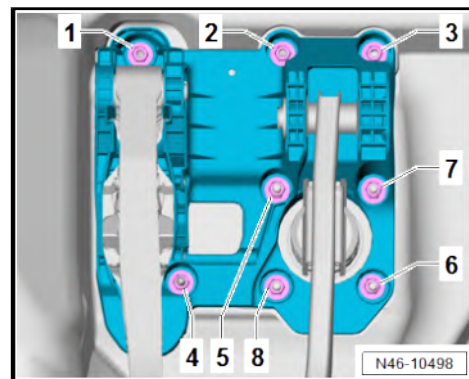
- ☐ Qty. 8
- ☐ Observe specified torque and tightening sequence ⇒ [page 89](#)



N46-10497



Specified torque and tightening sequence for mounting bracket



Component	Specified torque
Nuts -1 ... 8-	25 Nm

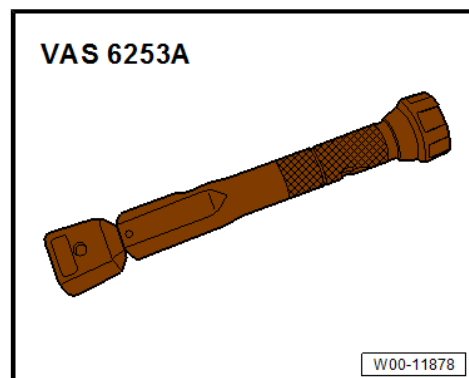
4.2 Removing and installing mounting bracket

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

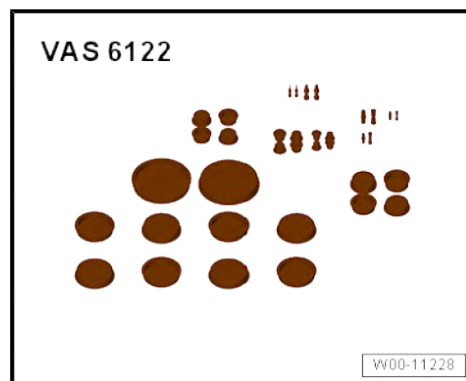


- ◆ Torque wrench - VAS 6253A-





◆ Engine bung set - VAS 6122-



◆ Brake filling and bleeding equipment - VAS 5234-

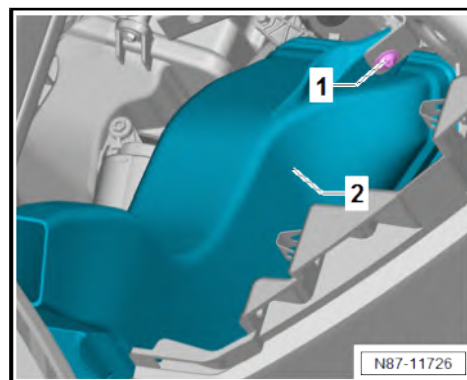


Removing

- Remove steering column ⇒ Running gear, axles, steering; Rep. gr. 48 ; Steering column; Removing and installing steering column .
- Unscrew bolt -1-.
- Swivel down driver-side footwell vent -2- off air distribution housing.
- Separate brake pedal from brake servo ⇒ [page 92](#) .

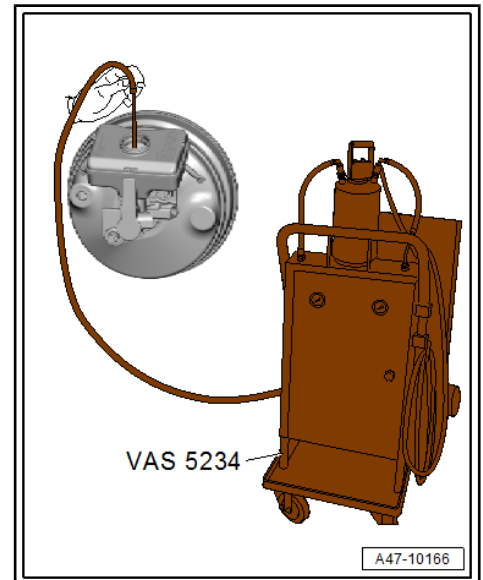
Vehicles with manual gearbox

- Lay a sufficient number of lint-free cloths under brake master cylinder.

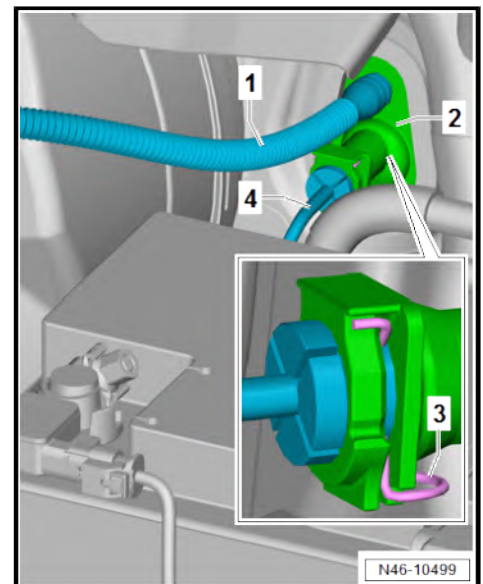




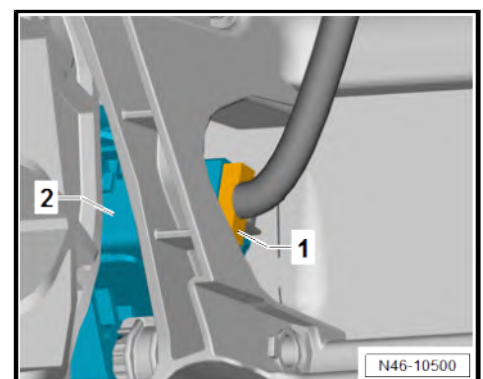
- Empty brake fluid reservoir with brake filling and bleeding equipment - VAS 5234- .



- Detach feed hose -1- from clutch master cylinder -2-.
- Seal supply hose -1- with sealing plug set for engine - VAS 6122- and raise and tie.
- Release locking clip -3-.
- Pull line -4- out of clutch master cylinder -2-.
- Seal line -4- with sealing plug set for engine - VAS 6122- .



- Separate electrical connector -1- from clutch position sender - G476- -2-.





Continued for all vehicles

- Unscrew nuts -1-.
- Remove mounting bracket -2-.

Installing

Install in the reverse order of removal, observing the following:

Vehicles with manual gearbox



Note

Make sure seals are correctly seated in supply hose.

- Bleed clutch mechanism ⇒ Rep. gr. 30 ; Clutch mechanism;
Bleeding clutch mechanism .

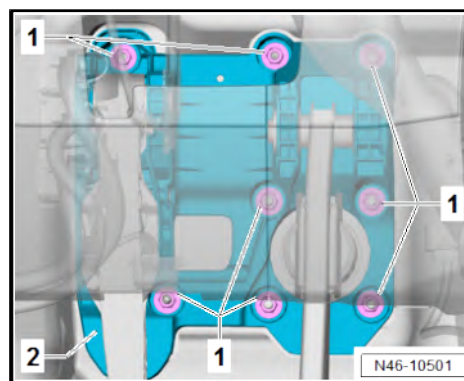
Continued for all vehicles

- Connect brake pedal to brake servo ⇒ [page 93](#) .

Specified torques

- ♦ ⇒ [“4.1 Assembly overview - brake pedal”, page 88](#)

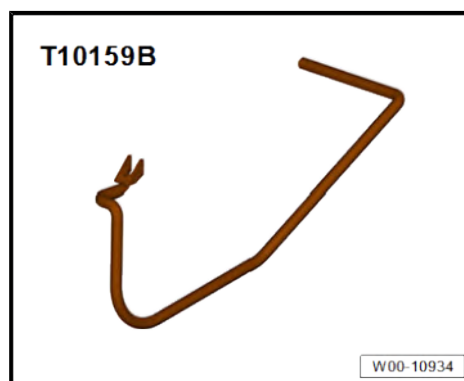
Component	Specified torque
Footwell vent bolt	1.5 Nm



4.3 Separating brake pedal from brake servo

Special tools and workshop equipment required

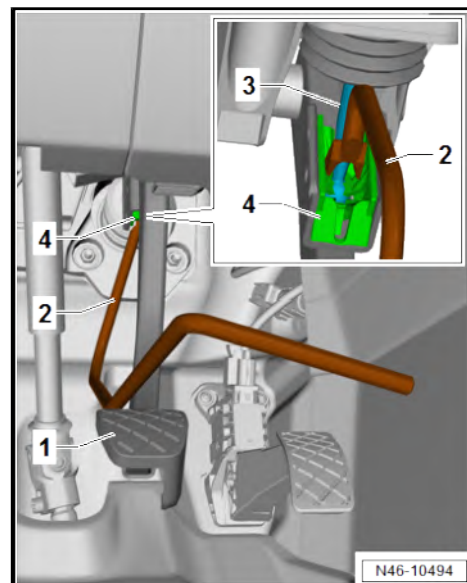
- ♦ Release tool - T10159B-





Procedure

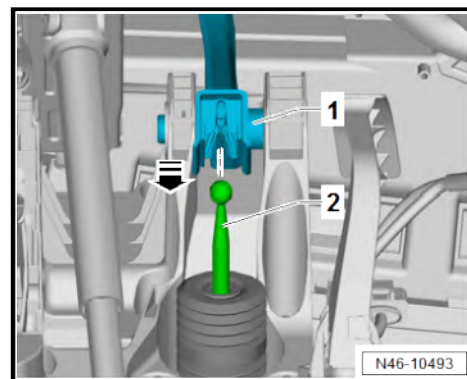
- Press brake pedal -1- in direction of brake servo and hold.
- Insert release tool - T10159B- -2- and pull in direction of driver seat while counter-holding on brake pedal -1-. (At this moment, brake pedal must not be allowed to move towards front of vehicle). This action will press retaining lugs -4- of mounting off ball head of plunger rod -3-.
- Pull release tool - T10159B- -2- and brake pedal -1- together towards driver seat. (This action will pull the brake pedal off the ball head of the plunger rod).



4.4 Connecting brake pedal to brake servo

Procedure

- Hold ball head of push rod -2- in front of mounting.
- Push brake pedal -1- in direction of -arrow- until ball head can be heard to engage.
- Check engagement by pulling on brake pedal.



4.5 Removing and installing brake pedal

The brake pedal can only be renewed in combination with the mounting bracket ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

In the event of complaints, the entire mounting bracket has to be replaced ⇒ [page 89](#) .

47 – Brakes - hydraulics

1 Front brake caliper

⇒ [“1.1 Assembly overview - front brake caliper”, page 94](#)

⇒ [“1.2 Removing and installing brake caliper piston”, page 94](#)

1.1 Assembly overview - front brake caliper

1 - Brake caliper

- ☐ Removing and installing
⇒ [page 45](#)

2 - Brake hose

3 - Bleeder screw

- ☐ Before screwing in, apply thin coat of assembly paste to threads ⇒ Electronic parts catalogue (ETKA) or ⇒ webMAN-TIS for MAN TGE
- ☐ 10 Nm

4 - Dust cap

5 - Oil seals

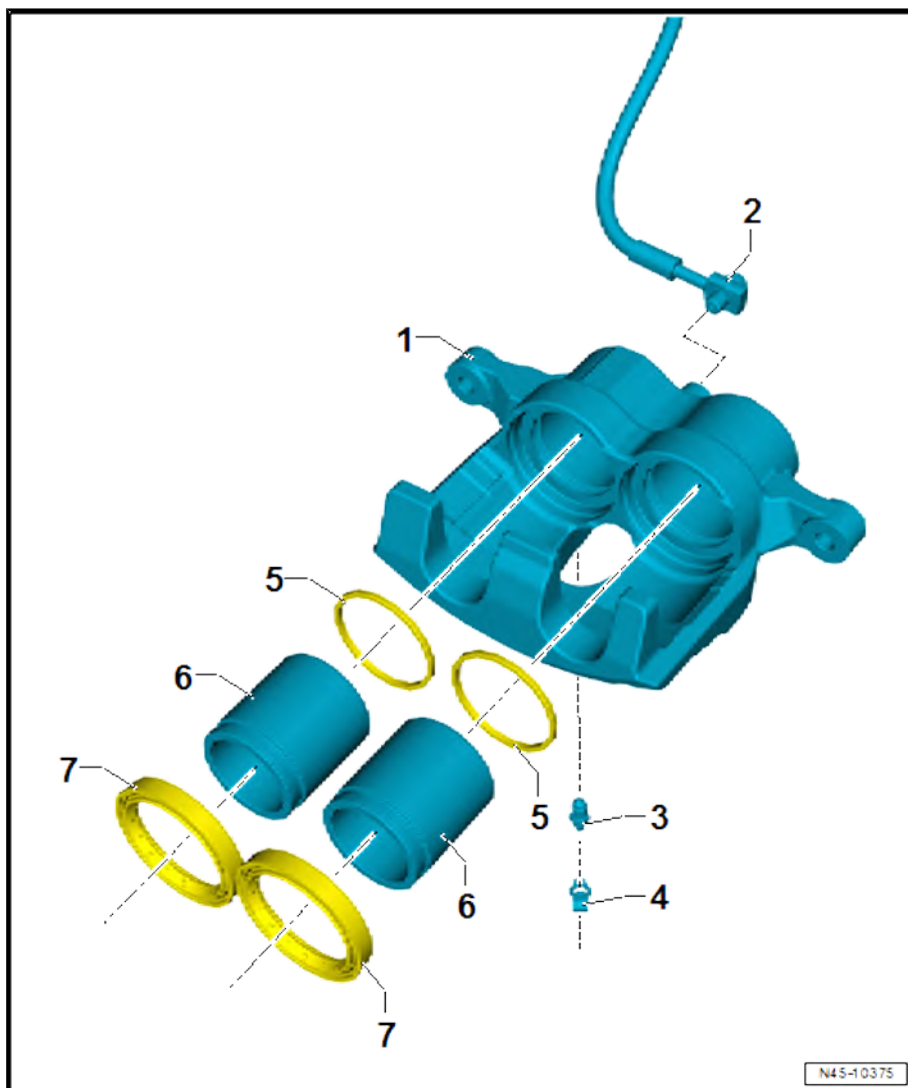
- ☐ Removing and installing
⇒ [page 94](#)

6 - Brake caliper piston

- ☐ Removing and installing
⇒ [page 94](#)

7 - Protective sleeves

- ☐ Removing and installing
⇒ [page 94](#)



1.2 Removing and installing brake caliper piston

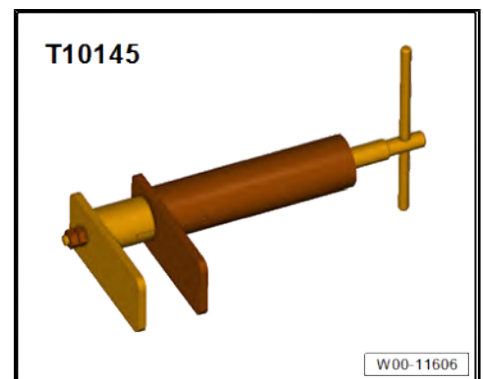
Special tools and workshop equipment required



◆ Removal wedge - 3409-



◆ Piston resetting appliance - T10145-

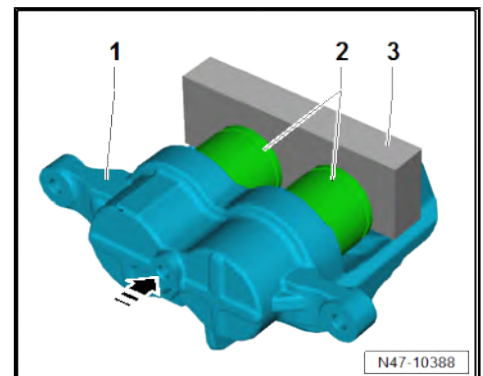


◆ Compressed air gun

◆ Wooden block

Removing

- Brake caliper removed ⇒ [page 45](#)
- Position wooden block -3- in recess to avoid piston -2- damage.
- Carefully press piston -2- out of brake caliper housing -1- using compressed air -arrow-.





- Remove seals -2- from brake caliper -1- with removal wedge -3409- -3-.

When removing ensure that the surface of the cylinder is not damaged.



Note

Use brake and clutch cleaner to clean the brake caliper.

Installing

Install in the reverse order of removal, observing the following:

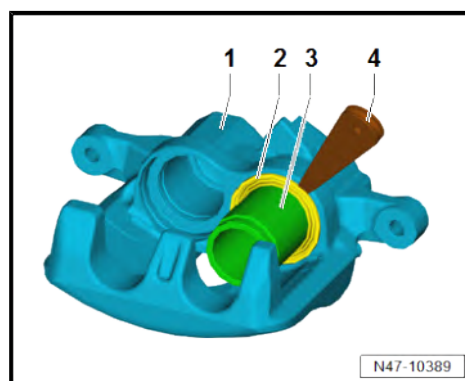
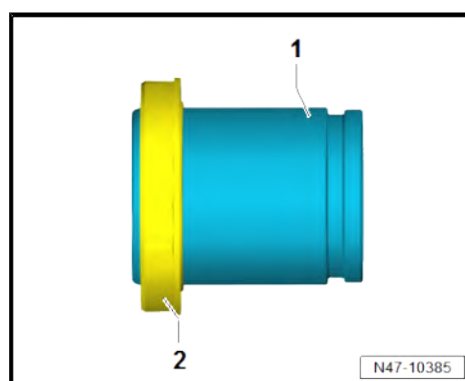
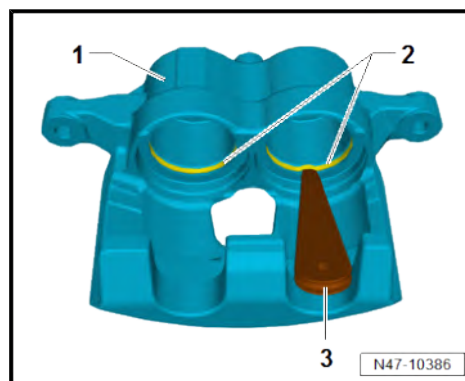


Note

- ◆ When conducting repairs on the brake caliper, always install the complete repair kit.
- ◆ New brake calipers are filled with brake fluid and are pre-bled.
- ◆ Apply thin coat of assembly paste to brake cylinder, piston and seal ⇒ *Electronic parts catalogue (ETKA)* or ⇒ *webMANTIS* for MAN TGE.
- ◆ When repairs are performed, it is essential to appropriately pre-bleed the brake calipers (without brake pads) before installing them in the vehicle ⇒ [page 97](#).

- Insert oil seals in brake caliper housing.
- Place protective sleeve -2- with outer sealing lip on piston -1-.

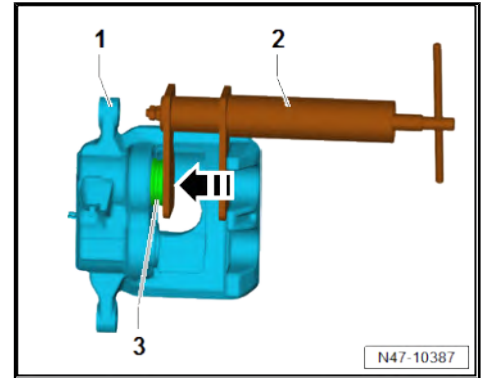
- Using removal wedge -3409- -4-, insert inner sealing lip -2- into cylinder groove.
- When doing this, hold piston -3- in front of brake caliper housing -1-.





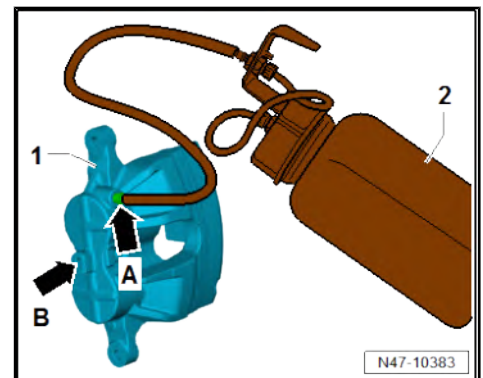
- Press piston -3- into brake caliper housing -1- using piston re-setting tool - T10145- -2-.

The outer sealing lip on the protective sleeve will then slip into the piston groove.



Pre-bleeding brake caliper

- Position brake caliper -1- as illustrated to pre-bleed.
- Place suitable container under brake caliper.
- Open bleeder valve -arrow A- and, using a commercially available bleeder bottle -2-, fill with brake fluid until bubble-free brake fluid flows out of threaded hole of brake hose connection -arrow B-. Then close bleeder screw.



2 Rear brake caliper

⇒ [“2.1 Assembly overview - rear brake caliper”, page 98](#)

⇒ [“2.2 Removing and installing brake caliper piston”, page 99](#)

2.1 Assembly overview - rear brake caliper

⇒ [“2.1.1 Assembly overview – rear brake caliper, single tyres”, page 98](#)

⇒ [“2.1.2 Assembly overview – rear brake caliper, twin tyres”, page 99](#)

2.1.1 Assembly overview – rear brake caliper, single tyres

1 - Protective sleeve

- ☐ Removing and installing
⇒ [page 99](#)

2 - Piston

- ☐ Removing and installing
⇒ [page 99](#)

3 - Seal

- ☐ Removing and installing
⇒ [page 99](#)

4 - Brake caliper

- ☐ Removing and installing
⇒ [page 58](#)

5 - Dust cap

6 - Bleeder screw

- ☐ Before screwing in, apply thin coat of assembly paste to threads ⇒ Electronic parts catalogue (ETKA) or ⇒ webMAN-TIS for MAN TGE
- ☐ 10 Nm

7 - Bolts

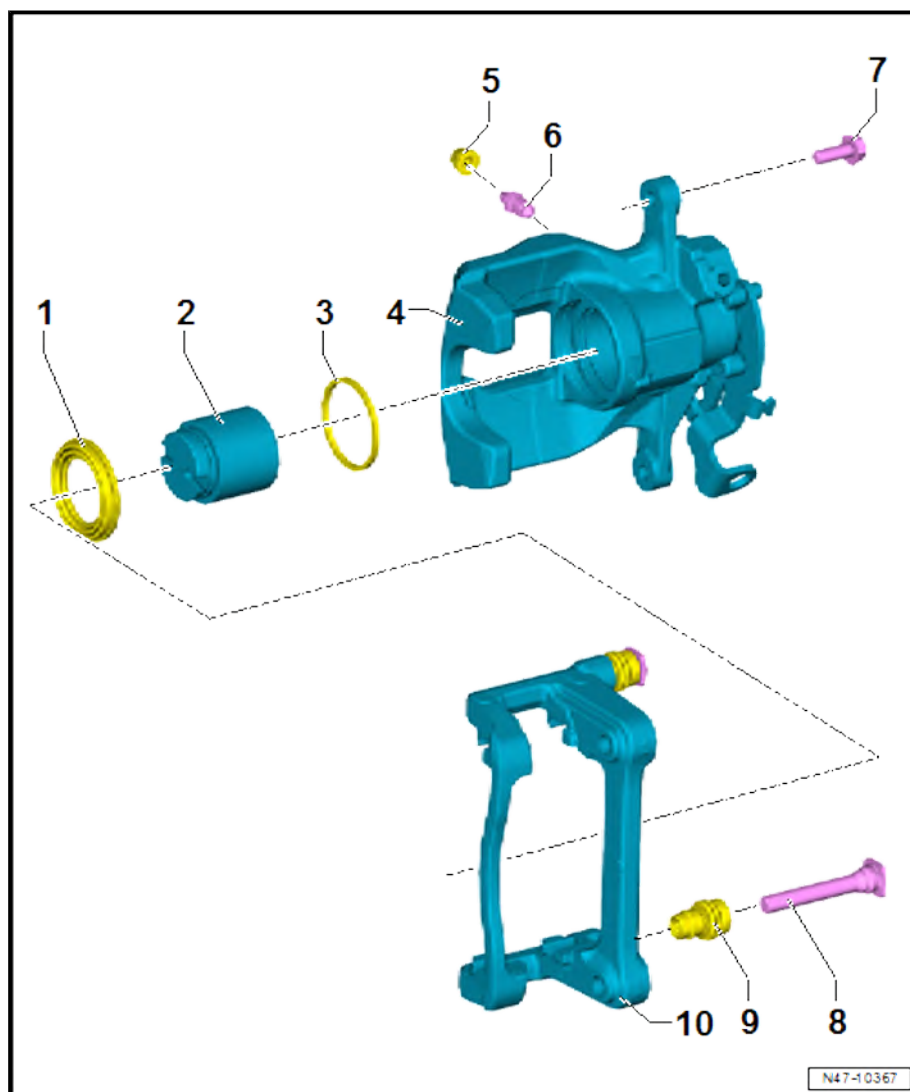
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Specified torque
⇒ [Item 6 \(page 50\)](#)

8 - Guide pin

9 - Protective boot

10 - Brake carrier

- ☐ Removing and installing
⇒ [page 64](#)





2.1.2 Assembly overview – rear brake caliper, twin tyres

1 - Brake caliper

- ❑ Removing and installing
⇒ [page 61](#)

2 - Brake hose

3 - Bleeder screw

- ❑ Before screwing in, apply thin coat of assembly paste to threads ⇒ Electronic parts catalogue (ETKA) or ⇒ webMAN-TIS for MAN TGE
- ❑ 10 Nm

4 - Dust cap

5 - Oil seals

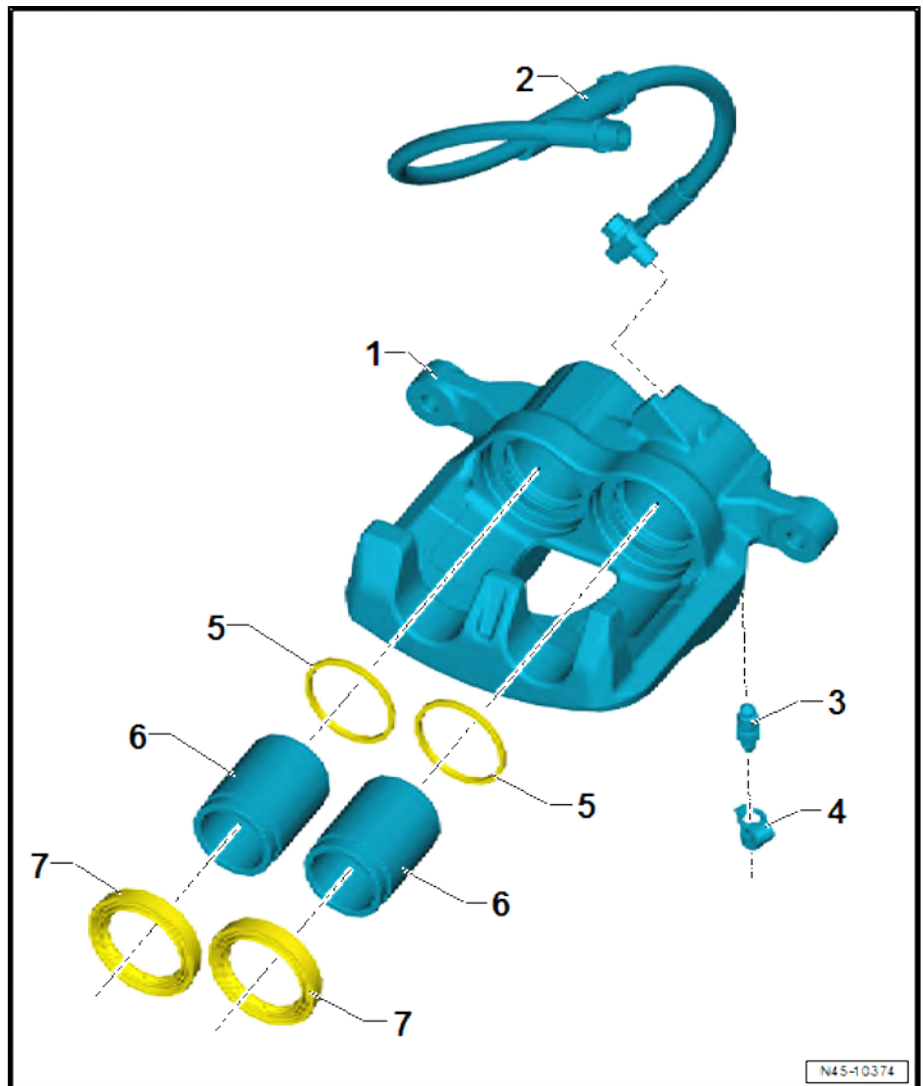
- ❑ Removing and installing
⇒ [page 102](#)

6 - Brake caliper piston

- ❑ Removing and installing
⇒ [page 102](#)

7 - Protective sleeves

- ❑ Removing and installing
⇒ [page 102](#)



2.2 Removing and installing brake caliper piston

⇒ [“2.2.1 Removing and installing brake caliper piston, single tyres”, page 99](#)

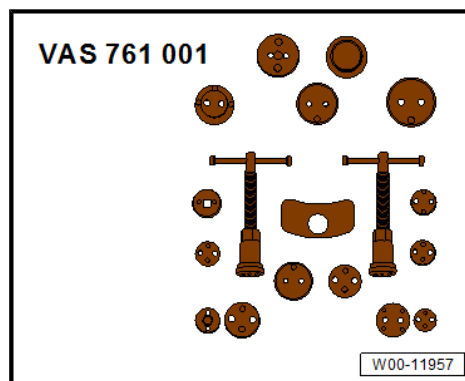
⇒ [“2.2.2 Removing and installing brake caliper piston, twin tyres”, page 102](#)

2.2.1 Removing and installing brake caliper piston, single tyres

Special tools and workshop equipment required



◆ Brake piston tool set - VAS 761 001-

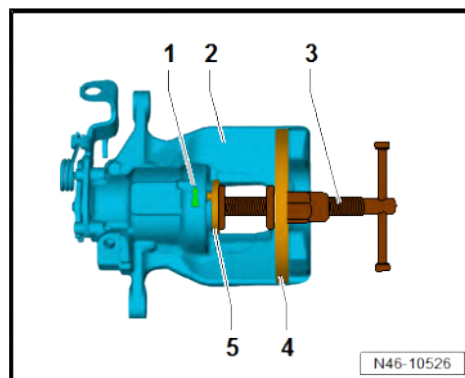


◆ Removal wedge - 3409-



Removing

- Rear brake caliper removed ⇒ [page 58](#)
- Fit suitable adapter -5- from brake piston tool set - VAS 761 001- .
- Fit retaining plate -4- from brake piston tool set - VAS 761 001- .
- Insert brake piston tool set - VAS 761 001- -3- in brake caliper -2- as shown in illustration.
- Unscrew piston in direction opposite to that of arrow -1- cast on brake caliper -2-.





- Lever out seal using removal wedge - 3409- .

Installing

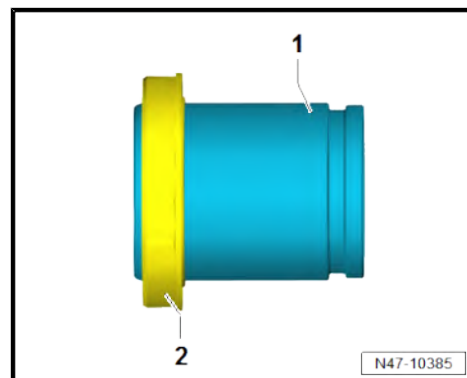
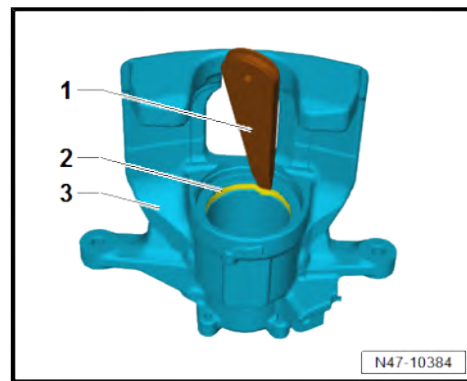
Install in the reverse order of removal, observing the following:



Note

- ◆ *When conducting repairs on the brake caliper, always install the complete repair kit.*
- ◆ *New brake calipers are filled with brake fluid and are pre-bled.*
- ◆ *Apply thin coat of assembly paste to brake cylinder, piston and seal ➔ Electronic parts catalogue (ETKA) or ➔ webMANTIS for MAN TGE.*
- ◆ *When repairs are performed, it is essential to appropriately pre-bleed the brake calipers (without brake pads) before installing them in the vehicle ➔ [page 97](#) .*
- ◆ *Use brake and clutch cleaner to clean the brake caliper.*

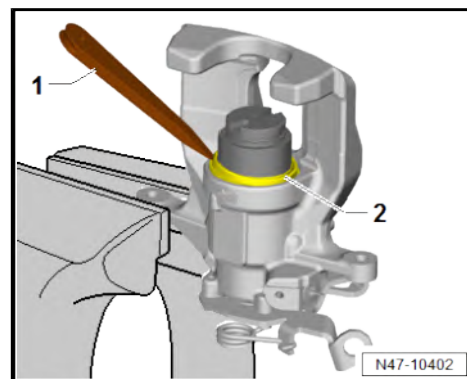
- Place protective sleeve -2- with outer sealing lip on piston -1-.



- Using removal wedge - 3409- -1-, insert inner sealing lip -2- into cylinder groove.

When doing this, hold piston in front of brake caliper.

Apply brake piston tool set - VAS 761 001- so that collar of tool rests on brake caliper.

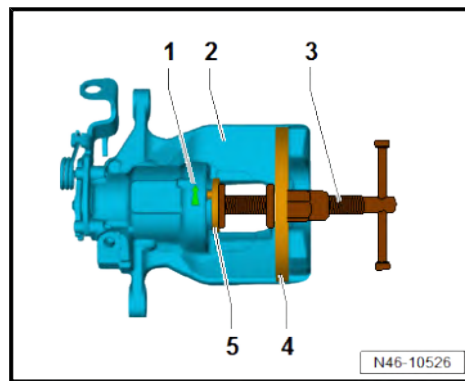




Note

Observe the arrow -1- cast on the brake caliper -2- for the screw in direction.

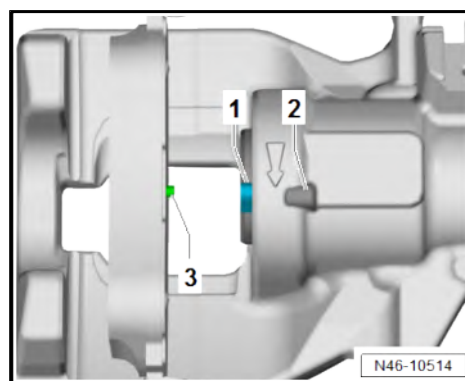
- Fit suitable adapter -5- from brake piston tool set - VAS 761 001- -3-.
- Mount retaining plate -4- from brake piston tool set - VAS 761 001- -3-.
- Insert brake piston tool set - VAS 761 001- -3- in brake caliper -2- as shown in illustration.



Note

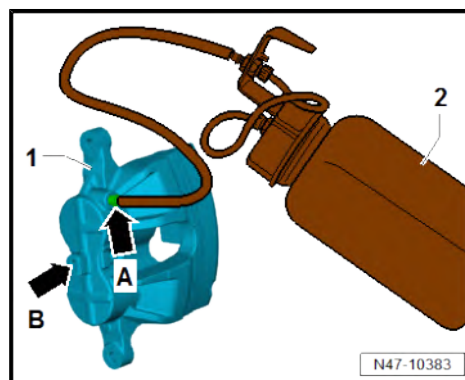
When screwing in, do not damage the boot that protects the piston.

- Screw in brake piston by turning handle of brake piston tool set - VAS 761 001- -3- in direction of arrow -1- cast on brake caliper -2-.
- Turn brake piston until notch -1- aligns with protrusion -2-.
- Pin -3- on brake pad engages in recess -1-



Pre-bleeding brake caliper

- Position brake caliper -1- as illustrated to pre-bleed.
- Open bleeder valve -arrow A- and, using a commercially available bleeder bottle -2-, fill with brake fluid until bubble-free brake fluid flows out of threaded hole of brake hose connection -arrow B-. Then close bleeder screw.



2.2.2 Removing and installing brake caliper piston, twin tyres

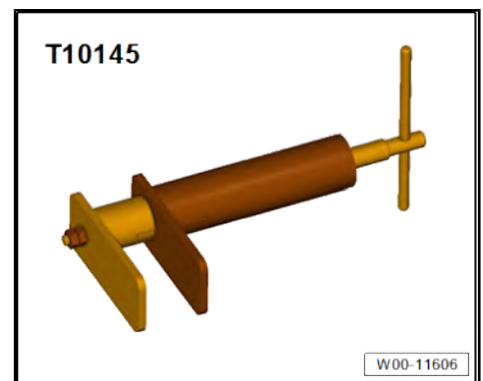
Special tools and workshop equipment required



◆ Removal wedge - 3409-



◆ Piston resetting appliance - T10145-

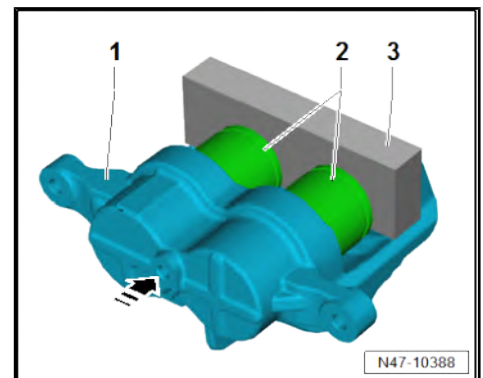


◆ Compressed air gun

◆ Wooden block

Removing

- Rear brake caliper removed ⇒ [page 61](#)
- Position wooden block -3- in recess to avoid piston -2- damage.
- Carefully press piston -2- out of brake caliper housing -1- using compressed air -arrow-.





- Remove seals -2- from brake caliper -1- with removal wedge -3409- -3-.

When removing ensure that the surface of the cylinder is not damaged.



Note

Use brake and clutch cleaner to clean the brake caliper.

Installing

Install in the reverse order of removal, observing the following:

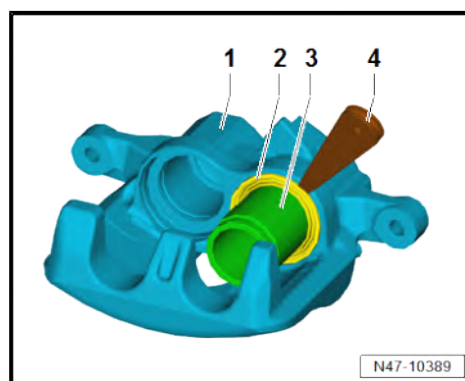
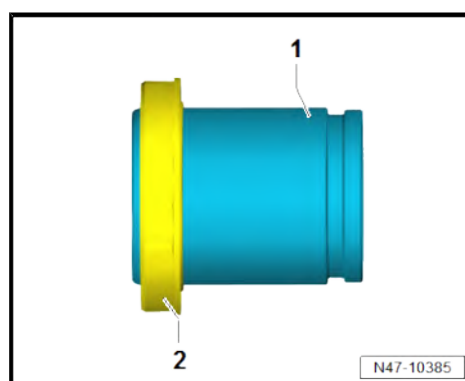
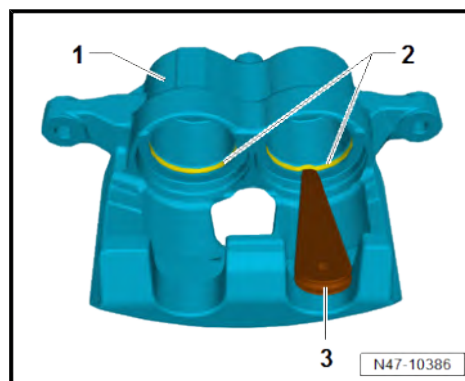


Note

- ◆ When conducting repairs on the brake caliper, always install the complete repair kit.
- ◆ New brake calipers are filled with brake fluid and are pre-bled.
- ◆ Apply thin coat of assembly paste to brake cylinder, piston and seal ⇒ *Electronic parts catalogue (ETKA)* or ⇒ *webMANTIS* for MAN TGE.
- ◆ When repairs are performed, it is essential to appropriately pre-bleed the brake calipers (without brake pads) before installing them in the vehicle ⇒ [page 97](#).

- Insert oil seals in brake caliper housing.
- Place protective sleeve -2- with outer sealing lip on piston -1-.

- Using removal wedge -3409- -4-, insert inner sealing lip -2- into cylinder groove.
- When doing this, hold piston -3- in front of brake caliper housing -1-.



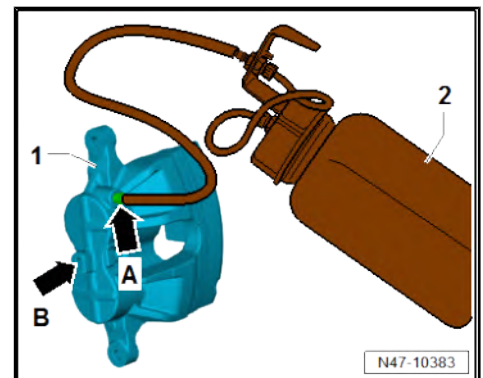
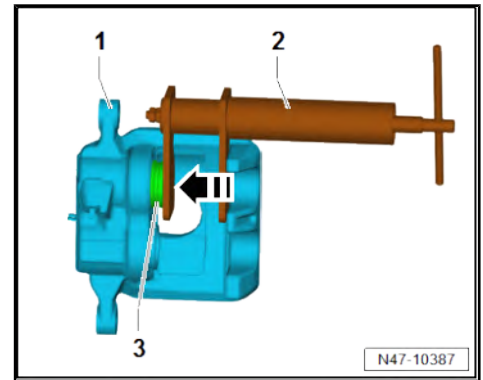


- Press piston -3- into brake caliper housing -1- using piston re-setting tool - T10145- -2-.

The outer sealing lip on the protective sleeve will then slip into the piston groove.

Pre-bleeding brake caliper

- Position brake caliper -1- as illustrated to pre-bleed.
- Place suitable container under brake caliper.
- Open bleeder valve -arrow A- and, using a commercially available bleeder bottle -2-, fill with brake fluid until bubble-free brake fluid flows out of threaded hole of brake hose connection -arrow B-. Then close bleeder screw.



3 Brake servo and brake master cylinder

⇒ [“3.1 Assembly overview - brake servo/brake master cylinder”, page 106](#)

⇒ [“3.2 Removing and installing brake light switch F”, page 107](#)

⇒ [“3.3 Removing and installing brake servo”, page 107](#)

⇒ [“3.4 Removing and installing brake master cylinder”, page 108](#)

⇒ [“3.5 Removing and installing brake fluid reservoir”, page 111](#)

3.1 Assembly overview - brake servo/brake master cylinder

1 - Cover

2 - Brake fluid reservoir

- ❑ Removing and installing
⇒ [page 111](#)

3 - Bolt

- ❑ 8 Nm

4 - Seal

5 - Seal

6 - Brake servo

- ❑ Removing and installing
⇒ [page 107](#)

7 - Seal

- ❑ Renew after removal

8 - Vacuum line

9 - Nuts

- ❑ Qty. 4
- ❑ Specified torque
⇒ [Item 4 \(page 88\)](#)

10 - Mounting bracket

- ❑ Removing and installing
⇒ [page 89](#)

11 - Bulkhead

12 - Brake line

- ❑ Specified torque
⇒ [Item 2 \(page 24\)](#)

13 - Brake line

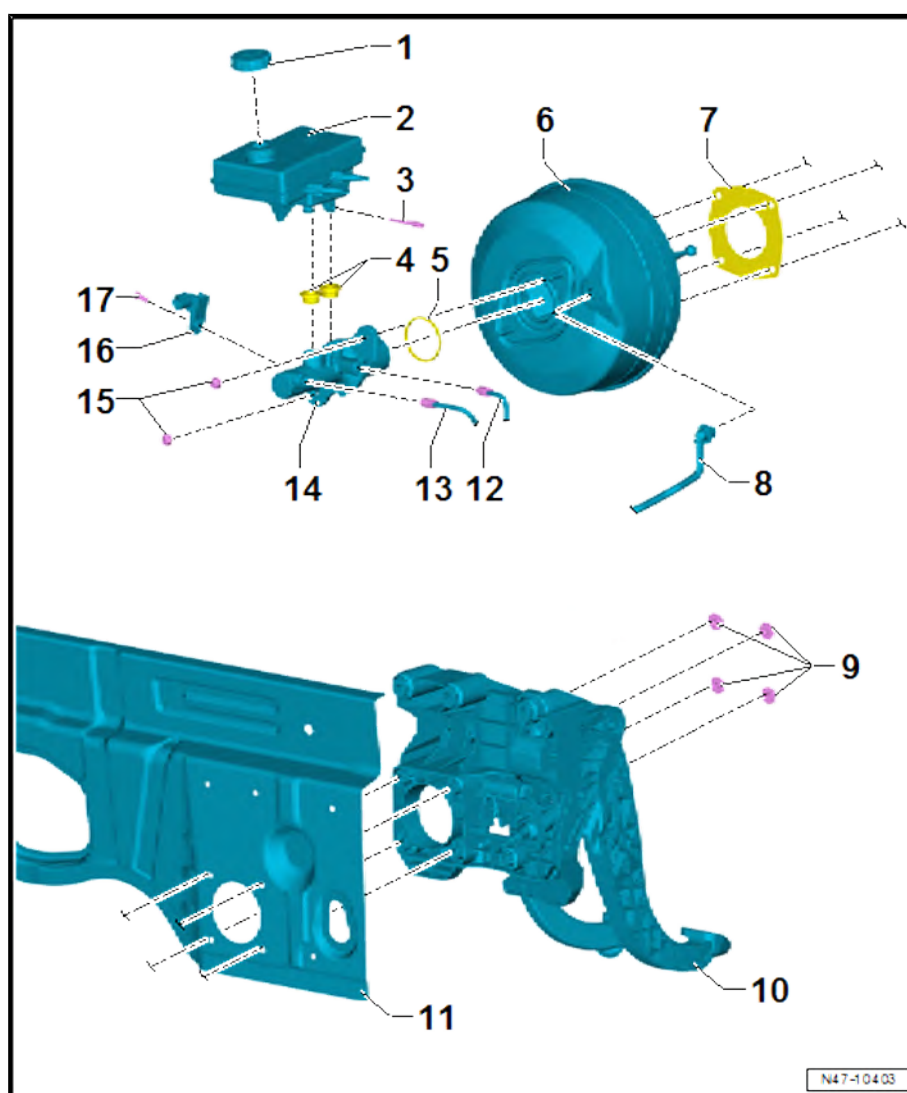
- ❑ Specified torque
⇒ [Item 2 \(page 24\)](#)

14 - Brake master cylinder

- ❑ Removing and installing ⇒ [page 108](#)

15 - Nuts

- ❑ Qty. 2
- ❑ Renew after removal
- ❑ 23 Nm





16 - Brake light switch - F-

- ❑ Removing and installing ➔ [page 107](#)

17 - Bolt

- ❑ 9 Nm

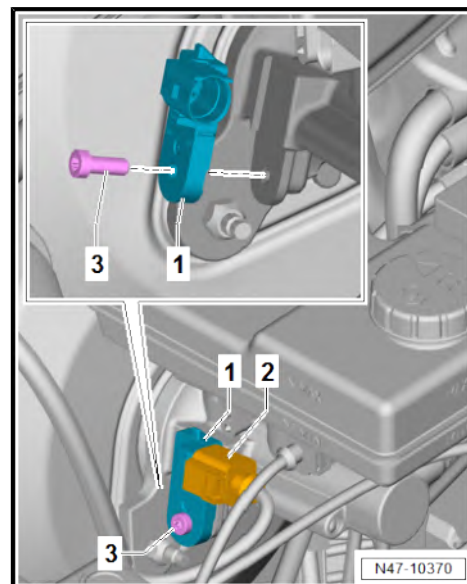
3.2 Removing and installing brake light switch - F-

Removing

- Separate electrical connector -2-.
- Unscrew bolt -3-.
- Remove brake light switch - F- -1-.

Installing

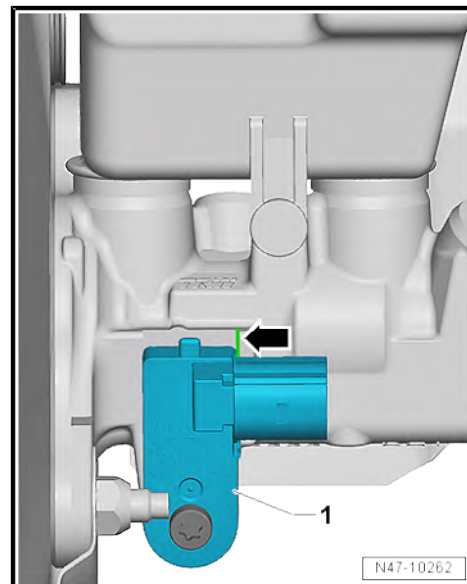
Install in the reverse order of removal, observing the following:



- Ensure that brake light switch - F- -1- seats properly on edge -arrow- of brake master cylinder.

Specified torques

- ◆ ➔ [“3.1 Assembly overview - brake servo/brake master cylinder”, page 106](#)



3.3 Removing and installing brake servo

Special tools and workshop equipment required

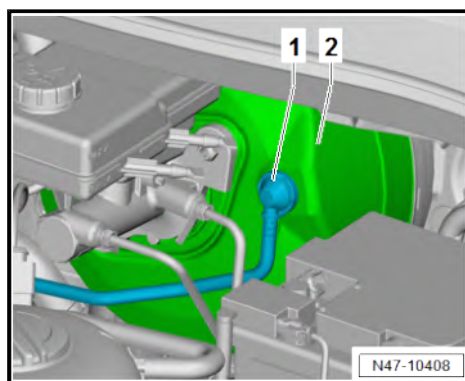


- ◆ Torque wrench - V.A.G 1331-

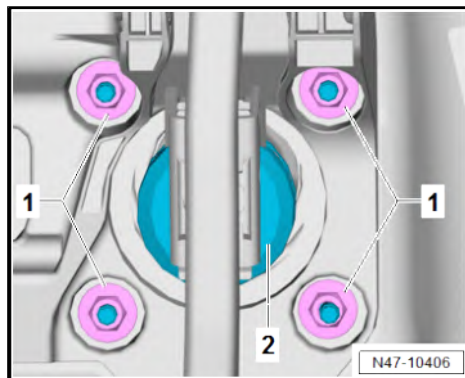


Removing

- Pull vacuum line -1- off brake servo -2-.
- Remove brake master cylinder ➔ [page 108](#) .
- Separate brake pedal from brake servo ➔ [page 92](#) .



- Unscrew nuts -1- from brake servo -2-.



- Remove brake servo -1-.

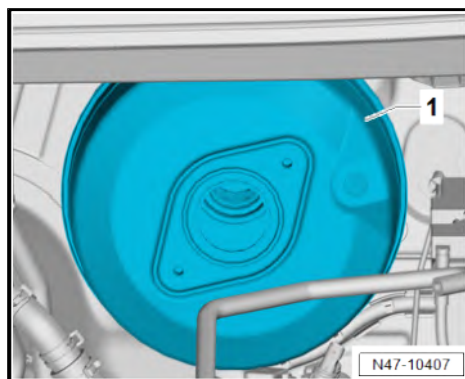
Installing

Install in the reverse order of removal, observing the following:

- Connect brake pedal to brake servo ➔ [page 93](#) .

Specified torques

- ◆ ➔ [“3.1 Assembly overview - brake servo/brake master cylinder”, page 106](#)

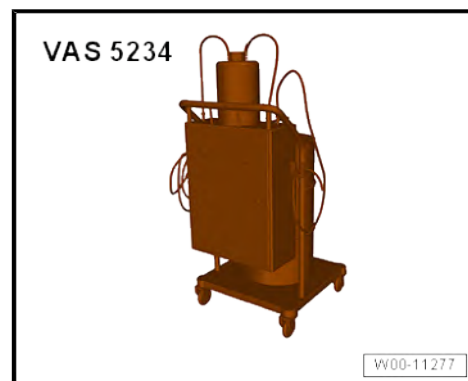


3.4 Removing and installing brake master cylinder

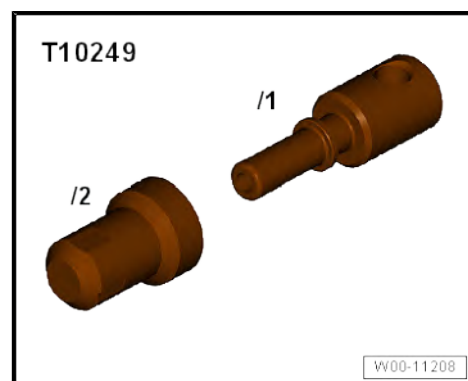
Special tools and workshop equipment required



- ◆ Brake filling and bleeding equipment - VAS 5234-



- ◆ Sealing tool - T10249-



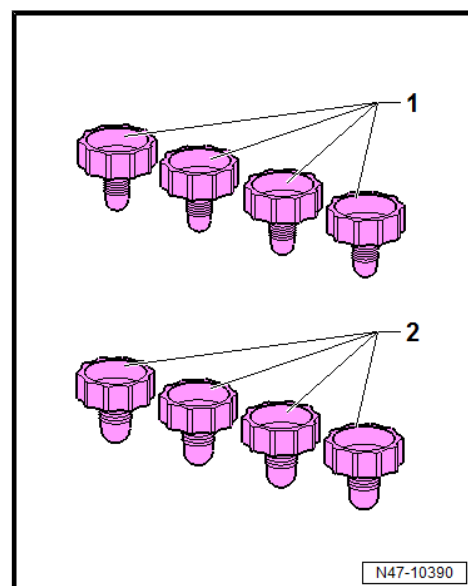
- ◆ Torque wrench - V.A.G 1410-



Sealing plug repair kit

1 - Sealing plug M10

2 - Sealing plug M12





Removing

Vehicles with electric drive

DANGER

Danger to life due to high voltage.

Severe or fatal injury from electric shock.

- The high-voltage system must be de-energised by a suitably qualified technician.

- De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

Continued for all vehicles

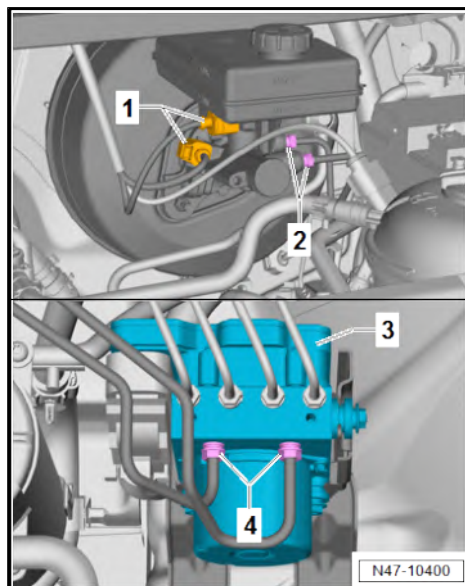
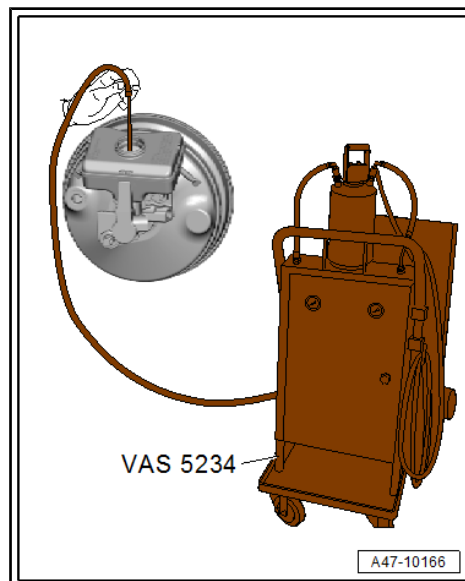
- Lay a sufficient number of lint-free cloths under brake master cylinder.
- Empty brake fluid reservoir with brake filling and bleeding equipment - VAS 5234- .

Vehicles with second battery - A1-

- Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery tray; Removing and installing battery tray .

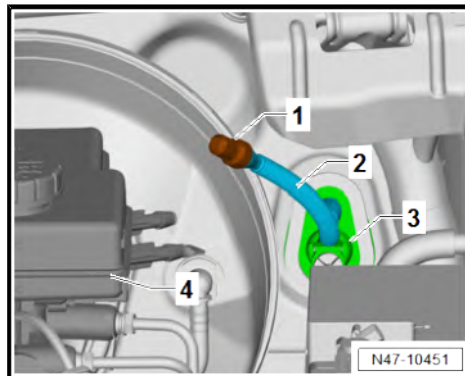
Continued for all vehicles

- Separate connectors -1-.
- Unscrew brake lines -2- and -4-.
- Seal ABS hydraulic unit - N55- -3- using sealing plugs from repair kit.



Vehicles with manual gearbox

- Pull off supply hose -2- for clutch master cylinder from brake fluid reservoir -4-.
- Seal supply hose -2- for clutch master cylinder -3- with sealing tool - T10249- -1-, raise and tie.





Continued for all vehicles

- Unscrew nuts -1-.
- Pull out brake master cylinder -2- in direction of -arrow- from brake servo.

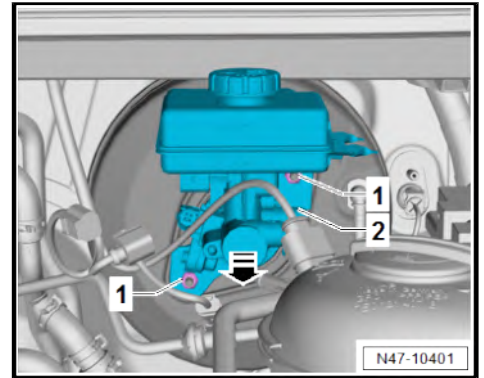
Installing

Install in the reverse order of removal, observing the following:



Note

Make sure seals are correctly seated on brake master cylinder.



Vehicles with manual gearbox



Note

Make sure seals are correctly seated in supply hose.

- Bleed clutch mechanism ⇒ Rep. gr. 30 ; Clutch mechanism; Bleeding clutch mechanism .

Continued for all vehicles

- Bleed brake system ⇒ [page 128](#) .

Vehicles with electric drive



WARNING

Danger to life due to high voltage.

Electrical shocks can cause serious injuries or death.

- Have a qualified technician re-energise the high-voltage system.

- Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

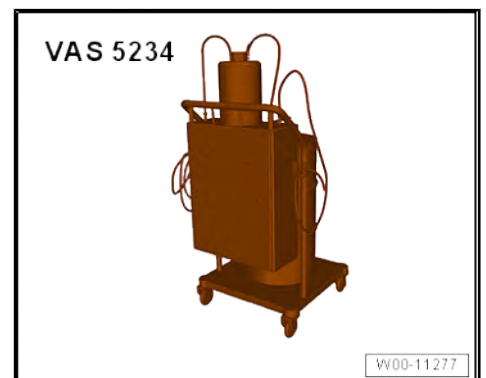
Torque settings

- ◆ ⇒ [“3.1 Assembly overview - brake servo/brake master cylinder”, page 106](#)

3.5 Removing and installing brake fluid reservoir

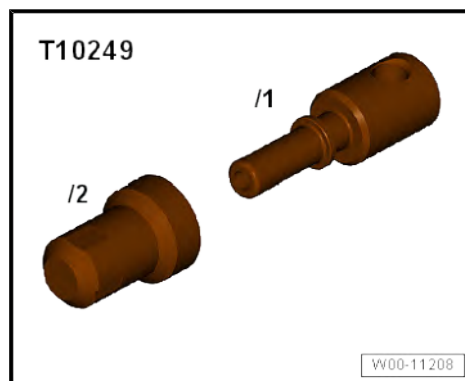
Special tools and workshop equipment required

- ◆ Brake filling and bleeding equipment - VAS 5234-





◆ Sealing tool - T10249-

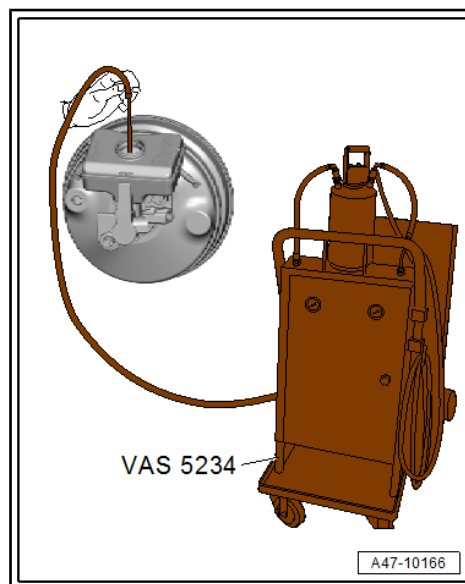


◆ Torque wrench - V.A.G 1410-



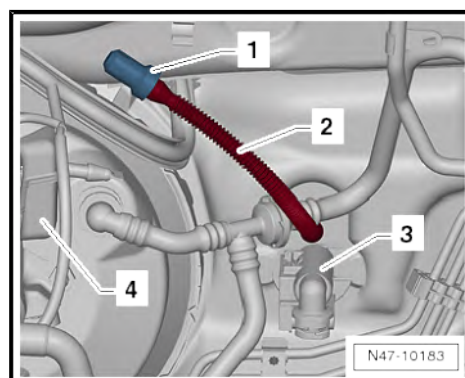
Removing

- Lay a sufficient number of lint-free cloths under brake master cylinder.
- Empty brake fluid reservoir with brake filling and bleeding equipment - VAS 5234- .



Vehicles with manual gearbox

- Pull off supply hose -2- for clutch master cylinder from brake fluid reservoir -4-.
- Seal supply hose -2- for clutch master cylinder -3- with sealing tool - T10249- -1-, raise and tie.





Continued for all vehicles

- Separate electrical connector -2- from brake fluid level warning contact - F34- .
- Unscrew bolt -1-.
- Remove brake fluid reservoir -3- from brake master cylinder in direction of -arrow-.

Installing

Install in the reverse order of removal, observing the following:



Note

Make sure that the sealing plugs are correctly seated in the brake master cylinder.

- Coat sealing plugs with brake fluid before pressing brake fluid reservoir into brake master cylinder.

Vehicles with manual gearbox



Note

Make sure seals are correctly seated in supply hose.

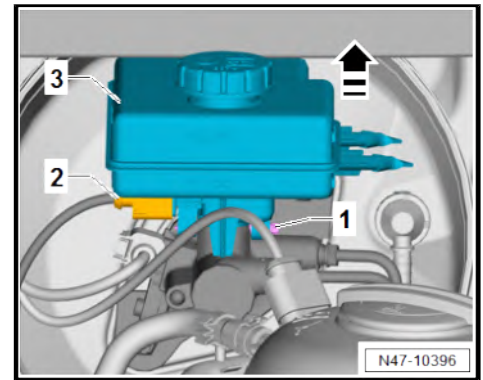
- Bleed clutch mechanism ⇒ Rep. gr. 30 ; Clutch mechanism; Bleeding clutch mechanism .

Continued for all vehicles

- Bleed brake system ⇒ [page 128](#) .

Specified torques

- ◆ ⇒ [“3.1 Assembly overview - brake servo/brake master cylinder”, page 106](#)



4 Vacuum system

⇒ [“4.1 Assembly overview - vacuum pump V22 , e-Crafter and e-TGE”, page 114](#)

⇒ [“4.2 Assembly overview - vacuum pump”, page 115](#)

⇒ [“4.3 Checking non-return valve”, page 115](#)

⇒ [“4.4 Removing and installing non-return valve”, page 115](#)

⇒ [“4.5 Removing and installing vacuum sender G608 ”, page 116](#)

⇒ [“4.6 Checking vacuum system”, page 116](#)

⇒ [“4.7 Removing and installing vacuum pump V22 , e-Crafter or e-TGE”, page 120](#)

⇒ [“4.8 Removing and installing vacuum pump V22 bracket”, page 121](#)

4.1 Assembly overview - vacuum pump - V22- , e-Crafter and e-TGE

1 - Vacuum pump - V22-

- ☐ Removing and installing
⇒ [page 120](#)

2 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

3 - Vacuum pipe

4 - Nuts

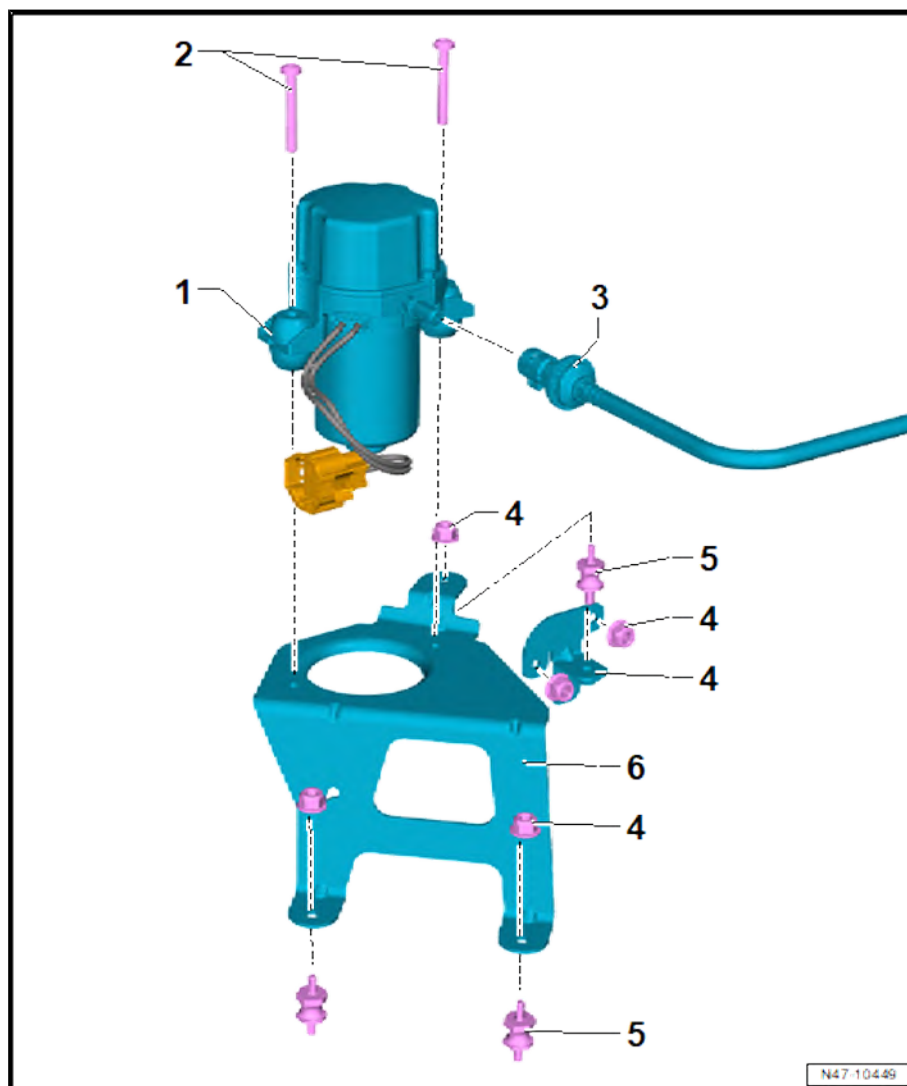
- ☐ Qty. 5
- ☐ 6 Nm

5 - Buffer stop

- ☐ Qty. 3
- ☐ 5 Nm

6 - Bracket

- ☐ For vacuum pump - V22-
- ☐ Removing and installing
⇒ [page 121](#)





4.2 Assembly overview - vacuum pump

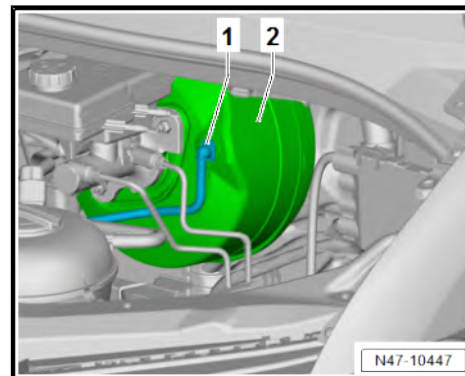
The vacuum pump is integrated into the oil pump ⇒ Rep. gr. 17 ;
Sump/oil pump; Exploded view - sump/oil pump

4.3 Checking non-return valve

Procedure

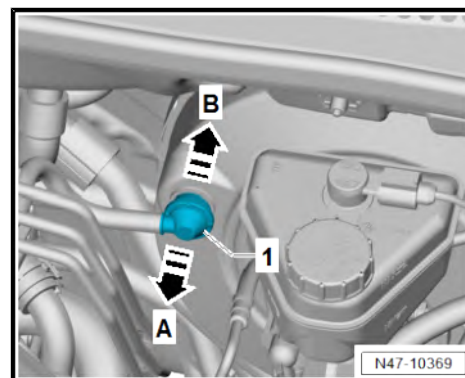
Vehicles with electric drive

- Carefully pull non-return valve together with vacuum line -1- out of brake servo -2-.



Continued for all vehicles

- Carefully pull non-return valve together with vacuum line -1- out of brake servo.
- Blow a little air through non-return valve, first in one direction and then in other.
- Air must pass through non-return valve -1- in direction of -arrow A-.
- Non-return valve -1- must remain closed in direction of -arrow B-. If this is not the case, renew complete vacuum line.



4.4 Removing and installing non-return valve



Note

The non-return valve forms a unit with the vacuum line. In the event of a defect, replace the complete vacuum line with non-return valve.



4.5 Removing and installing vacuum sender - G608-

Removing

- Separate electrical connector -3-.
- Unclip vacuum sender - G608- -1- from vacuum line -2-.

Do not damage plastic locking mechanisms -4-.

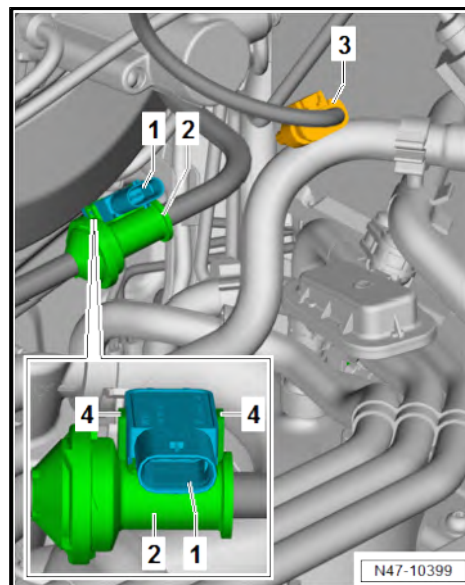
Installing

Install in the reverse order of removal, observing the following:



Note

- ♦ *Make sure to check the seal for damage before installing the vacuum sender - G608- .*
- ♦ *If the seal is damaged, the vacuum sender - G608- must be renewed.*



4.6 Checking vacuum system

⇒ [“4.6.1 Connecting brake servo tester VAS 6721 ”, page 116](#)

⇒ [“4.6.2 Checking vacuum generation”, page 118](#)

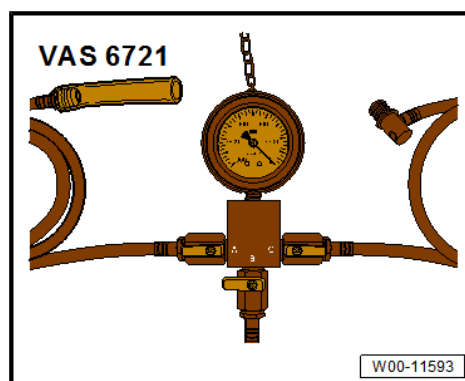
⇒ [“4.6.3 Checking for leaks”, page 118](#)

⇒ [“4.6.4 Vacuum generation with manual vacuum pump VAS 6213 ”, page 119](#)

4.6.1 Connecting brake servo tester - VAS 6721-

Special tools and workshop equipment required

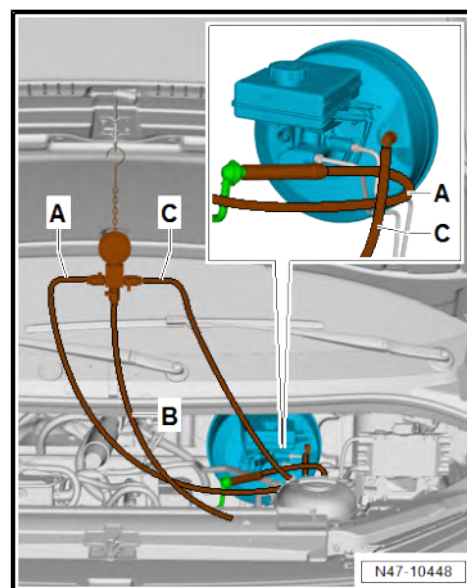
- ♦ Vacuum gauge for brake servo - VAS 6721-





Procedure

Vehicles with electric drive



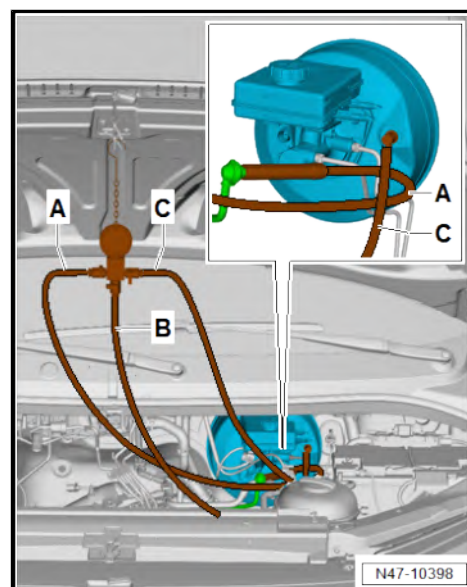
Item	Component	Explanation
A	Shut-off valve	In direction of vacuum line, non-return valve and, if fitted, vacuum pump
B	Shut-off valve	◆ Open to facilitate removal of brake servo tester - VAS 6721- ◆ Open to simulate a fault source ◆ Connection of manual vacuum pump - VAS 6213-
C	Shut-off valve	In direction of brake servo

Continued for all vehicles

- Pull vacuum line out of brake servo.

First press the brake pedal several times in order to facilitate removal of the vacuum line.

- Connect brake servo tester - VAS 6721- as shown.

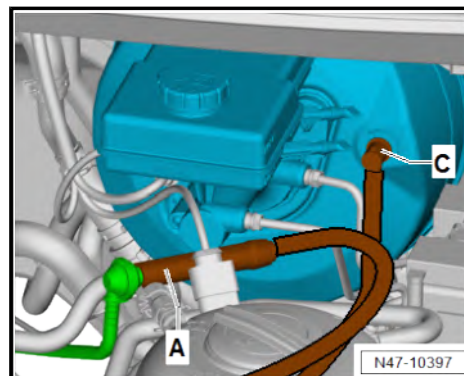


Item	Component	Explanation
A	Shut-off valve	In direction of vacuum line, non-return valve and, if fitted, vacuum pump
B	Shut-off valve	◆ Open to facilitate removal of brake servo tester - VAS 6721- ◆ Open to simulate a fault source ◆ Connection of manual vacuum pump - VAS 6213-



Item	Component	Explanation
C	Shut-off valve	In direction of brake servo

- Connect hose -A- from brake servo tester - VAS 6721- to vacuum line. Press adapter -C- into brake servo.



4.6.2 Checking vacuum generation

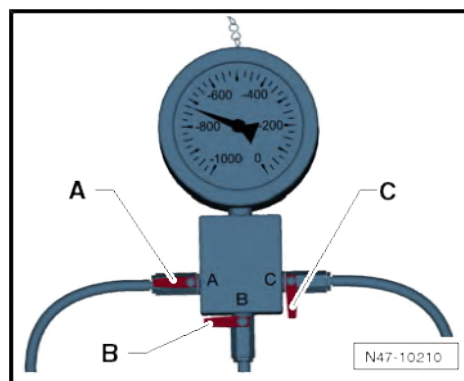
Procedure

- Connect brake servo tester - VAS 6721- ➔ [page 116](#) .
- Open shut-off valve -A-.
- Close shut-off valves -B- and -C-.
- Start warm engine (>60°C), press accelerator briefly (engine speed above 2000 rpm).
- Read measured value shown.

Normally, the vacuum that is generated should be between 600 and 950 mbar (depending on engine size).

If the measured value is not reached even through the preconditions have been met, the vacuum system must first be checked for leaks.

- For comparison purposes, generate the vacuum with the manual vacuum pump - VAS 6213- ➔ [page 119](#) .
- Open cut-off valve -B- to facilitate removal of hose connections and adapter.

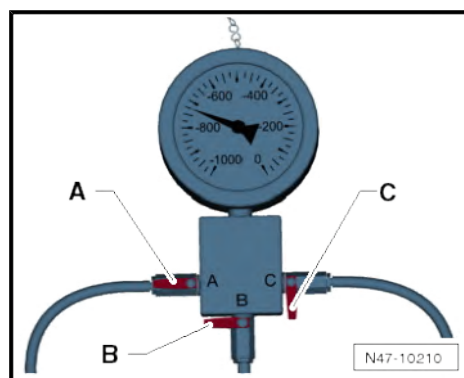


4.6.3 Checking for leaks

Procedure

- Connect brake servo tester - VAS 6721- ➔ [page 116](#) .
- Open shut-off valve -A-.
- Close shut-off valves -B- and -C-.
- Start warm engine (>60°C), press accelerator briefly (engine speed above 2000 rpm).

Normally, the vacuum that is generated should be between 600 and 950 mbar (depending on engine size).





- Open shut-off valve -C- to evacuate brake servo.
- Switch off engine.
- Read and make a note of measured value shown.

A vacuum decrease of 400 mbar in 12 hours is permissible.

If the vacuum decrease is greater, check for leaks in the following areas:

- ◆ Brake servo
- ◆ Non-return valve, vacuum hoses with connectors and vacuum pump/intake manifold

If there are large leaks, the vacuum decreases steeply within a few seconds.

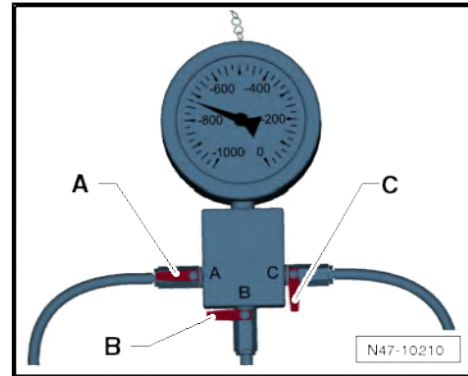
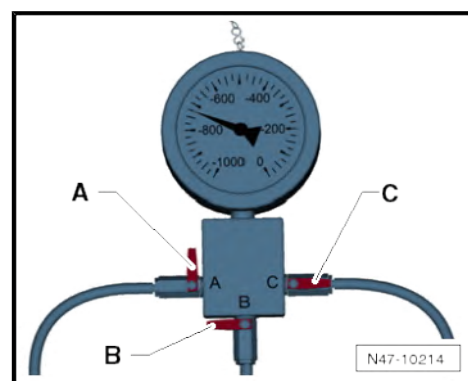
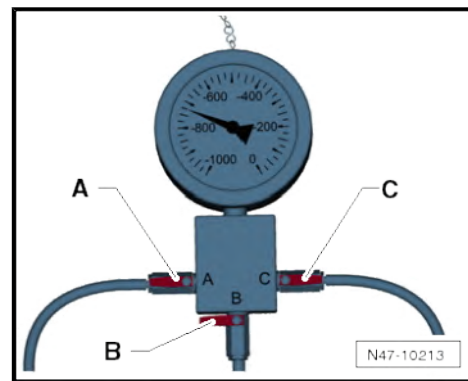
Vacuum check in vicinity of brake servo

- After generation of vacuum, close cut-off valve -A- in order to check brake servo vacuum system.

Vacuum check in vicinity of non-return valve, vacuum hoses with connectors and vacuum pump/intake manifold

- After generating a vacuum, close shut-off valve -C- in order to check vacuum system from brake servo tester - VAS 6721- to intake pipe or to vacuum pump.

Open cut-off valve -B- to facilitate removal of hose connections and adapter.

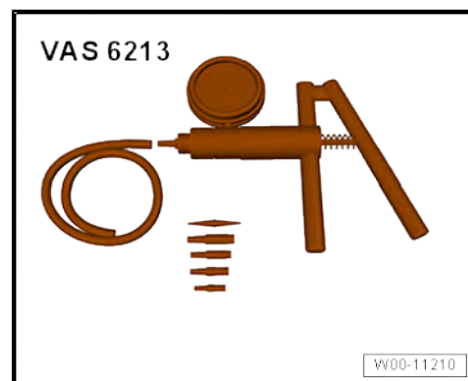


4.6.4 Vacuum generation with manual vacuum pump - VAS 6213-

Procedure

Instead of vacuum generation by means of engine or vacuum pump, the vacuum can be generated with the manual vacuum pump - VAS 6213- in certain cases.

- To do this, connect hand vacuum pump - VAS 6213- to vacuum line from connection -B- on brake servo tester - VAS 6721- .
- Open shut-off valve -B-.
- Generate vacuum with hand vacuum pump - VAS 6213- until between 600 and 950 mbar is shown on brake servo vacuum gauge - VAS 6721- .
- Then perform relevant checks.





4.7 Removing and installing vacuum pump - V22- , e-Crafter or e-TGE

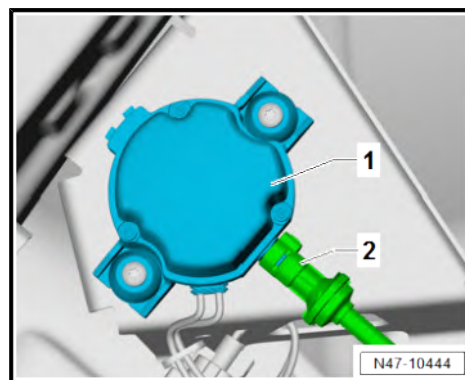
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-

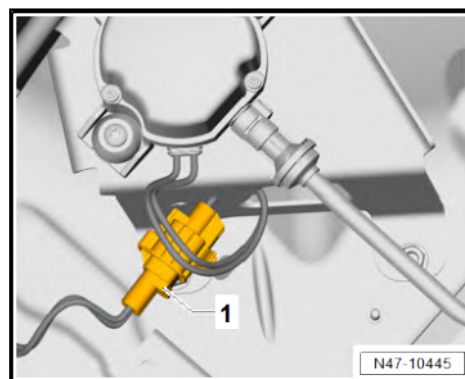


Removing

- Release vacuum line -2-. Pull off from vacuum pump - V22- -1- to do this.



- Disconnect connector -1-.



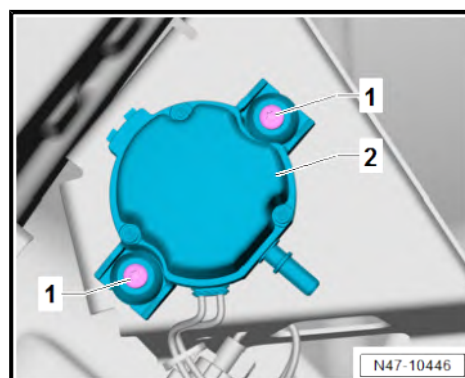
- Unscrew bolts -1-.
- Remove vacuum pump - V22- -2- from receiver.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - vacuum pump V22 , e-Crafter and e-TGE”, page 114](#)

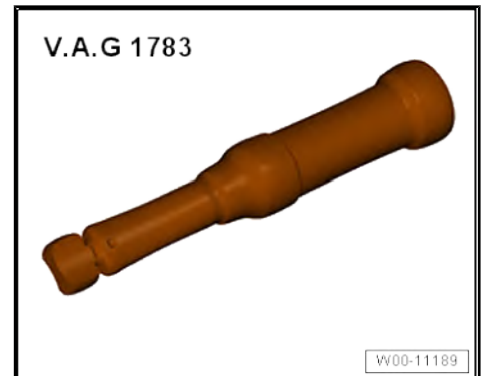




4.8 Removing and installing vacuum pump - V22- bracket

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



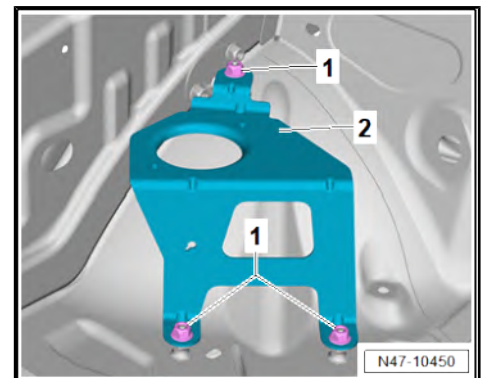
Removing

- Remove vacuum pump - V22- ⇒ [page 120](#) .
- Unscrew nuts -1-.
- Detach bracket -2-.

Installing

Install in the reverse order of removal, observing the following:

- Buffer stops should not twist as nuts are tightened
⇒ [Item 4 \(page 114\)](#) . Tighten nuts when doing this.
- Counterhold buffer stops ⇒ [Item 5 \(page 114\)](#) using a suitable tool.



Torque settings

- ◆ ⇒ [“4.1 Assembly overview - vacuum pump V22 , e-Crafter and e-TGE”, page 114](#)



5 Brake lines

⇒ ["5.1 Repairing brake lines", page 122](#)

5.1 Repairing brake lines

The flanging tool for brake lines - VAS 6056- can be used to flange brake lines with an outer pipe diameter of 5 mm without damaging the coating. In certain cases, this enables partial brake line sections to be renewed at less expense.

The use of flanging tools - V.A.G. 1356- is not permitted due to the coating and the diameter of the black brake lines.

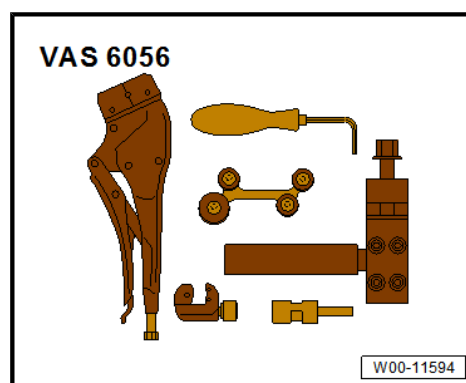


Note

- ◆ *Brake pipes may only be bent up to max. 90°.*
- ◆ *Otherwise they will become kinked or deformed.*
- ◆ *This will result in the cross-section of the lines to be changed in a manner which is not permitted.*
- ◆ *Preferably separate brake lines at underbody.*
- ◆ *The positions of the intermediate pieces must be selected so that they cannot chafe on moving parts.*
- ◆ *Do not grease spindle; merely clean with methylated spirits.*

Special tools and workshop equipment required

- ◆ Flanging tool for brake lines - VAS 6056-



List of individual tools



1 - Flanging tool - VAS 6056/1-

- ❑ Flanging tool - VAS 6056/1- contains flanging jaws - VAS 6056/6-

2 - Pipe cutter - VAS 6056/2-

3 - Brake line scraper - VAS 6056/3-

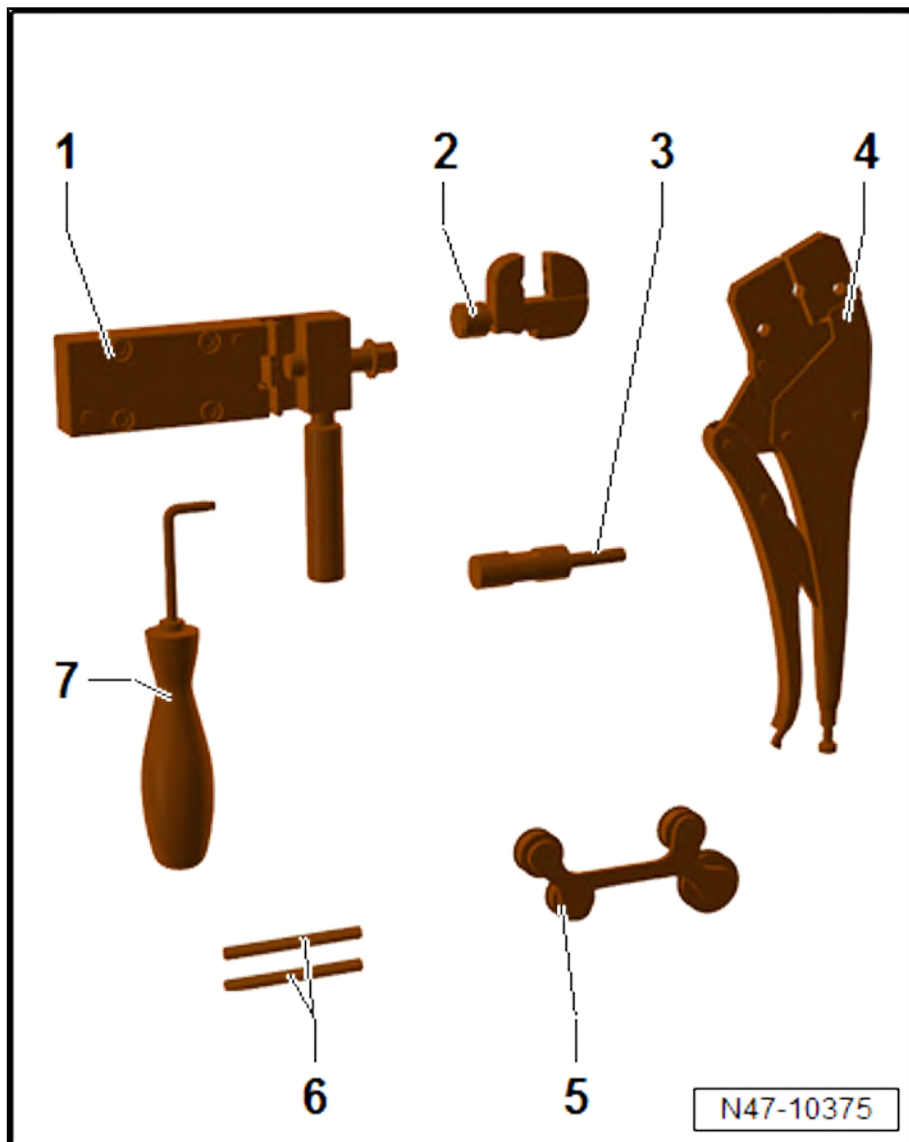
- ❑ Grub screws (in shaft and at side) are adjusted and must not be tampered with!

4 - Set of grips with plastic jaws - VAS 6056/4-

5 - Pipe bending tool - VAS 6056/5-

6 - Flanging jaws - VAS 6056/7-

7 - Special wrench, 6 mm



Assembly overview - flanging tool



1 - Flanging tool upper part

- ❑ Unbolt to change flanging jaws.

2 - Flanging jaws

- ❑ Various.
- ❑ Assembly instructions
⇒ [page 124](#) .

3 - Fastening for handle

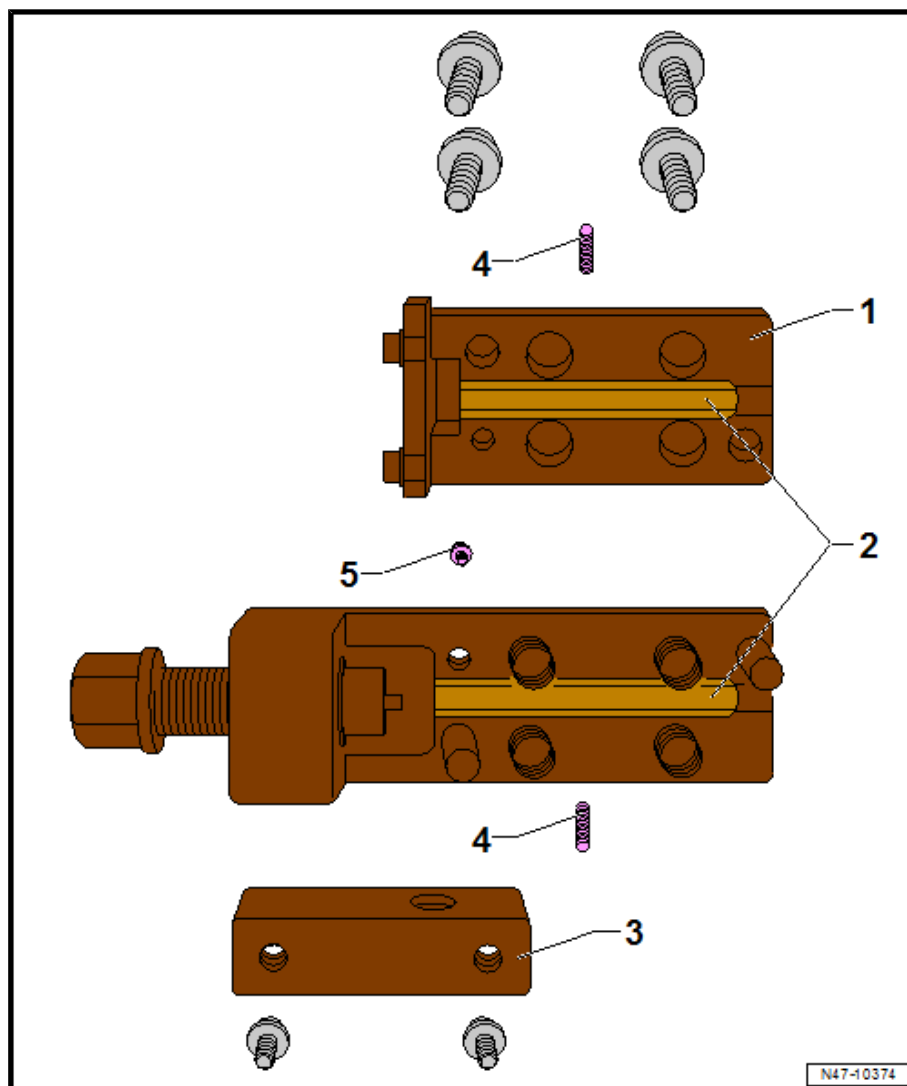
- ❑ Must be unbolted to access securing bolt for upper part

4 - Securing bolt

- ❑ For flanging tool upper part

5 - Grub screws for flanging jaws

- ❑ For centring and holding flanging jaws
- ❑ Allen head bolt, 2 mm



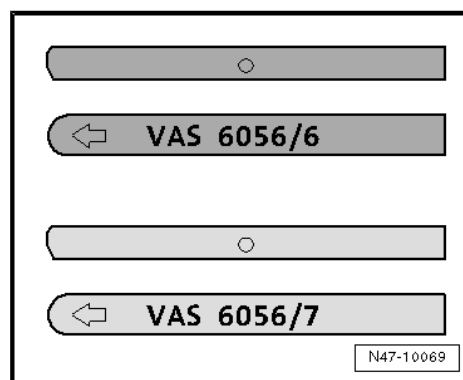
Flanging jaw assembly instructions

- ◆ VAS 6056/6 (dark) for black brake lines
- ◆ VAS 6056/7 (light) for green brake lines



Note

- ◆ *Arrow on rounded side of flanging jaws must point to edge of housing.*
- ◆ *Straight side of flanging jaws must be installed to spindle, as flanging head is not otherwise formed correctly.*

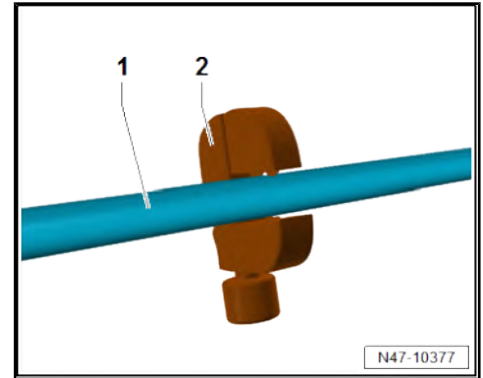


Procedure

- Unbolt respective brake line from brake caliper or wheel brake cylinder.
- Catch escaping brake fluid and dispose of this as per regulations.



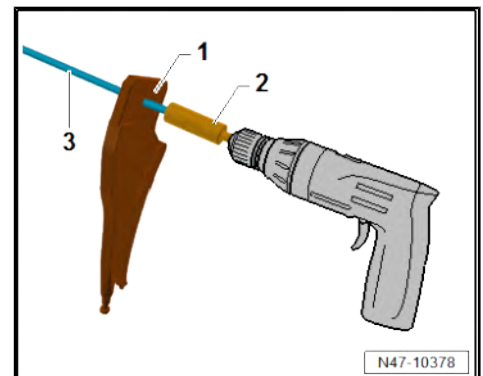
- Cut through brake line -1- at a suitable point (straight, freely accessible section) using pipe cutter -2-.
- Remove section to be renewed.
- Degrease brake line surface.



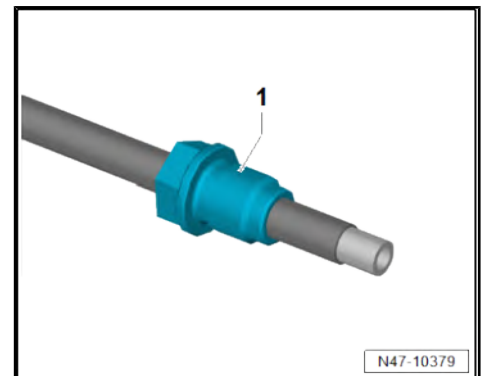
- Clamp brake line -3- in set of grips -1- so that approx. 50 mm project from plastic jaws.
- Clamp scraper -2- into a drill and place it on brake line -3-.
- At slow drill speed and with gentle pressure on brake line, scrape coating off brake line.

Length of scraping is determined by stop in scraper.

- Remove scraper from brake line and remove scraping residues.



- Remove set of grips and slide union bolt -1- onto brake line.



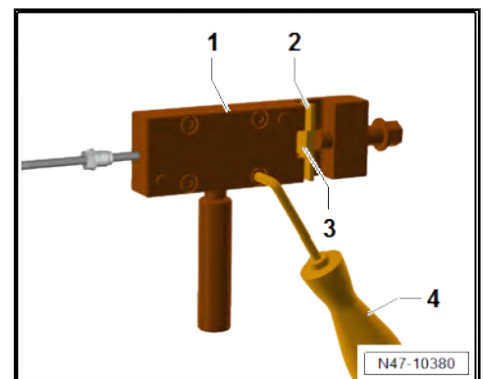
- Slide brake line up to stop -3- in flanging tool -1-.



Note

Brake line must be positioned against stop when the hexagon socket head bolts are tightened, or the flange will not be formed correctly.

- Clamp brake line in flanging tool.
- Push brake line towards front until it can no longer be moved.
- Fold up stop -2-.
- Tighten hexagon socket head bolts diagonally using angle screw driver -4-.

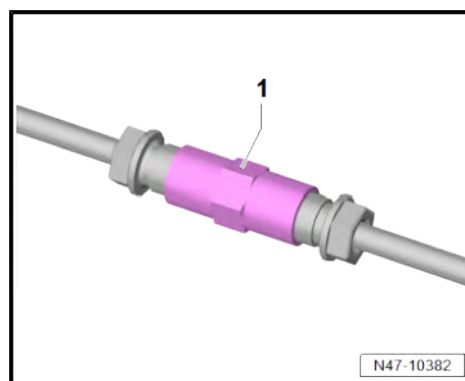
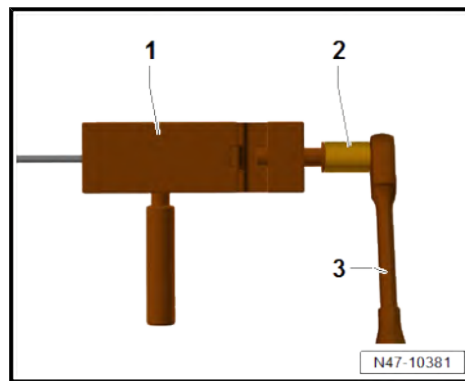




- Turn spindle -2- with ratchet -3- into flanging tool -1- onto stop.
- Turn spindle back again.
- Undo Allen head bolts alternately in diagonal sequence.
- Remove brake line from flanging tool, then clean and inspect brake line and flange.

Briefly flush the section of brake line remaining in the vehicle:

- Connect brake filling and bleeding unit - VAS 5234- , connect bleeder bottle hose to brake line flange and allow brake filling and bleeding unit - VAS 5234- to run briefly until a little brake fluid has run through.
- Blow out new brake line to be inserted with compressed air.
- Join brake lines using connector -1-.
- Install brake line.
- Bleed brake system ➔ [page 128](#) .





6 Hydraulic system

⇒ [“6.1 General notes on brake fluid”, page 127](#)

⇒ [“6.2 Pre-bleeding hydraulic system”, page 128](#)

⇒ [“6.3 Bleeding hydraulic system following standard procedure”, page 128](#)

⇒ [“6.4 Subsequent bleeding of hydraulic system”, page 129](#)

⇒ [“6.5 Leakage test”, page 130](#)

6.1 General notes on brake fluid

⇒ [“6.1.1 Changing brake fluid”, page 127](#)



Note

- ◆ Only use new brake fluid conforming to DOT 4 standard.
- ◆ Brake fluid is poisonous. Also due to its corrosive effect brake fluid must not come into contact with paintwork.
- ◆ Brake fluid is hygroscopic, which means it absorbs moisture from the ambient air and should therefore always be stored in air-tight containers.
- ◆ Wash away any spilt brake fluid using plenty of water.

As the water content in the brake fluid rises, so its boiling point falls. If the brake fluid gets very hot, this can lead to vapour bubbles being formed, and failure of the brake.

The colour of the brake fluid becomes darker over the course of time. A dark brake fluid colour is not an indication of its condition. The discolouration is caused by chemical reactions.



Note

Rubber parts of brake system (sleeves, seals) are not compatible with mineral oil, fuel and aggressive cleaning agents.

Summary of aforementioned points

Always keep brake fluid containers properly sealed. This is the only way of preventing contamination by oil, dirt, cleaning agents or moisture.

Carefully store brake fluid containers separately from oil (including hydraulic oil) and cleaning fluids. This is in order to avoid confusion with other fluids or even filling a brake system with the wrong product.

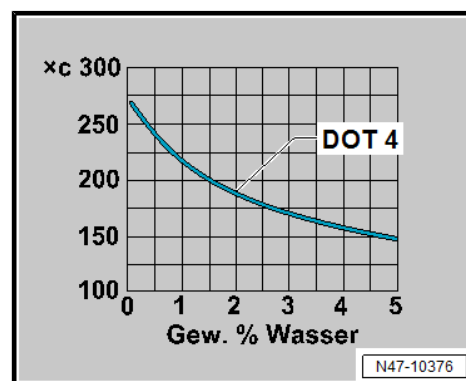
6.1.1 Changing brake fluid



Note

Comply with the brake fluid change intervals as stated in the Maintenance Tables and the Service Schedule.

⇒ Maintenance ; Booklet 10.3 ; Brake and clutch system: Changing brake fluid .

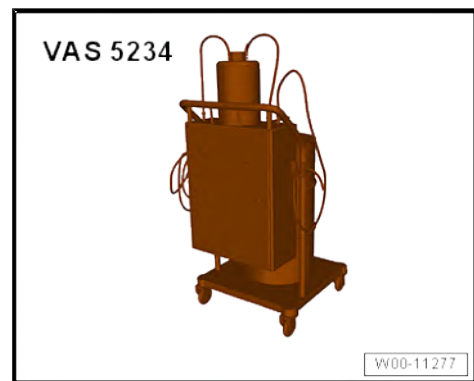




6.2 Pre-bleeding hydraulic system

Special tools and workshop equipment required

- ◆ Brake filling and bleeding unit - VAS 5234-



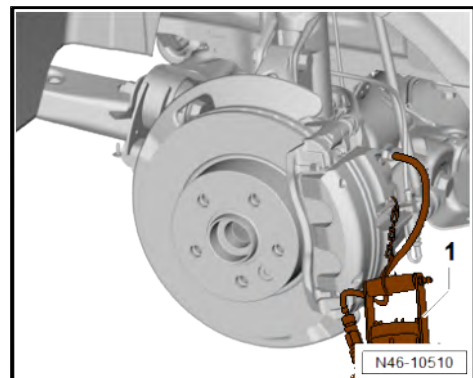
Procedure

- Observe general information on brake fluid ➔ [page 127](#) .

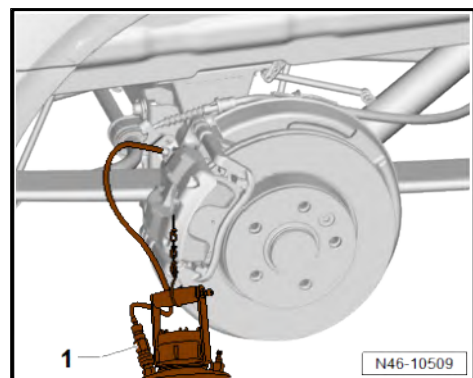
Pre-bleeding is necessary if a chamber of the brake fluid reservoir ran completely empty (e.g. owing to leaks in brake system).

A pre-pressure of 2 bar is required for bleeding the hydraulic unit.

- Connect brake filling and bleeding equipment - VAS 5234- and switch on.
1. Simultaneously bleed both front left and front right brake caliper.



1. Simultaneously bleed both rear left and rear right brake caliper.
- Leave bleeder valves open with brake fluid hoses of bleeder bottle -1- connected until brake fluid free from air bubbles flows out.
 - Connect ➔ Vehicle diagnostic tester and bleed hydraulic unit.
 - Then, bleed brake system normally ➔ [page 129](#) .



6.3 Bleeding hydraulic system following standard procedure

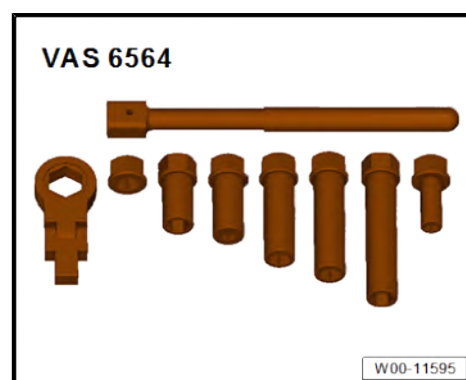
Special tools and workshop equipment required



◆ Brake filling and bleeding equipment - VAS 5234-



◆ Tool set for brake bleeding - VAS 6564-



Adhere strictly to work sequence when bleeding brake system.

- Connect brake filling and bleeding equipment - VAS 5234- and switch on.

A pre-pressure of 2 bar is required to bleed the hydraulic system.

- Open bleeder valves in specified order and bleed brake calipers.
 1. Front left brake caliper
 2. Front right brake caliper
 3. Rear left brake caliper
 4. Rear right brake caliper



Note

Use suitable bleeder hose. It must sit tightly on bleeder valve so that no air can enter brake system.

- Leave bleeder valve of each brake caliper open with bleeder hose fitted until brake fluid discharges free of air bubbles.

6.4 Subsequent bleeding of hydraulic system



Note

- ◆ *This bleed sequence must be carried out 5 times per brake caliper.*
- ◆ *After bleeding, a road test must be carried out, during which at least one ABS intervention must be triggered.*



Repeat bleeding is carried out in connection with excessive brake pedal travel or so-called »spongy« brakes.

A second mechanic is required for this:

- Depress brake pedal firmly and hold.
- Open bleeder valve on brake caliper. Observe bleeding sequence.
- Fully depress brake pedal.
- Close bleeder valve with pedal held down.
- Release brake pedal slowly.

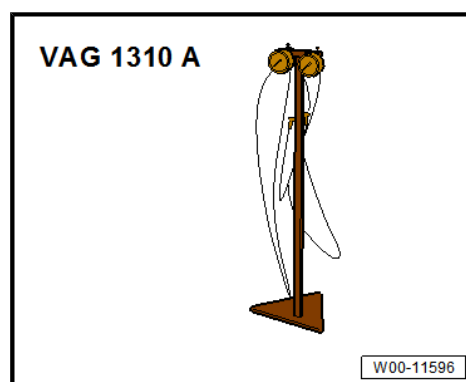
Bleeding sequence:

1. Front left brake caliper
 2. Front right brake caliper
 3. Rear left brake caliper
 4. Rear right brake caliper
- Carry out road test.

6.5 Leakage test

Special tools and workshop equipment required

- ◆ Tester for brake pressure regulator - V.A.G 1310 A-



- ◆ Adapter M10 - V.A.G 1310/6-

Conditions for testing

- Brake system (hydraulic unit, brake hoses, brake lines and brake calipers) operating properly and free of leaks.

Checking

- Remove bleeder valve at one of front brake calipers.
- Connect tester for brake pressure regulator - V.A.G 1310 A- and bleed.
- Apply pressure to brake pedal until gauge indicates a pressure of 50 bar.

The pressure must not drop by more than 4 bar during the test period of 45 seconds. Renew brake master cylinder if drop in pressure exceeds specification.